

Revision Compilation: all seven subjects, one file

READ THE FRAMING PARAGRAPH ON EACH SHEET FIRST, THEN WORK THE TABLES

COMPILED SEPTEMBER 2026

Every sheet in this file is built the same way. A **black strip** at the top says what the whole subject is really about in one paragraph. Then 4 columns of panels, colour coded by lecture. **Yellow boxes** are things to memorise, **red boxes** are traps that cost marks, **blue dashed boxes** are how the question gets asked. Formulas sit in their own grey blocks so you can find them by shape rather than by reading.

01 MAXD 5113 Fundamental of Data Science

Sheet 1. L1 AI and the four types, L2 big data and ethics with the 5 Vs, L3 data understanding with CRISP-DM and EDA, L4 data preparation with every threshold, plus the four scoping canvases and how to attack the paper.

Sheet 2. L5 the analytics ladder and the method families, L6 decision trees with entropy and information gain, fit and overfit, the confusion matrix and ROC, L7 tools, L8 Netflix and Alibaba, L9 model management, plus the worked past year numbers.

02 MAXD 5133 Research Methodology

Sheet 1. What research is, the research process, topic to problem to gap to objective to question to hypothesis, the paradigms, qualitative versus quantitative, validity and reliability, ethics and informed consent, and the literature review end to end.

Sheet 2. Measurement scales, questionnaire development, all the sampling techniques, experimental design and the five design families, simulation design and its twelve steps, data interpretation, and the exact proposal chapter structure with the marking rubric.

03 MPSW 5013 Applied Statistical Method

Sheet 1. Measurement scales, the hypothesis testing frame, the decision rules, **the test chooser**, chi square, the z test and the three t tests, and the whole non parametric family paired with its parametric twin.

Sheet 2. One way and two way ANOVA with the summary table, post hoc tests, correlation and regression with every diagnostic, logistic regression, PCA and factor analysis, time series, statistical process control, and every threshold in the subject on one strip.

04 MAXD 5123 Big Data Management

Sheet 1. Data to information to knowledge, the BDM ecosystem, why NoSQL appeared, ACID versus BASE and CAP, the four NoSQL families, MongoDB with every CRUD command, Neo4j with every Cypher query.

Sheet 2. Hadoop, HDFS with every default number and both data paths, MapReduce with the word count worked, Spark and RDDs, Flume and Sqoop, the comparison tables, and **every final exam question with its recorded answer**.

05 MAXD 5153 Big Data Analytics and Visualization

Sheet 1, text. The NLP pipeline in order, pre-processing with exact function names, Bag of Words and TF-IDF worked, embeddings, classification, sentiment, summarisation, topic modelling, similarity, transformers.

Sheet 2, speech and image. The speech chain, TTS and ASR, image fundamentals, point and histogram processing, spatial filtering, noise matched to its filter, segmentation, morphology, colour, shape description.

Sheet 3, visualization. The grammar of graphics, the chart selection table, storytelling, the six map types, and every exam question with a model answer.

06 MAXD 5143 Applied Machine Learning

Sheet 1, classical. The workflow, the four learning types, features, bias and variance, cross validation, the metrics, decision trees fully worked on PlayTennis, ensembles, KNN, k-means, hierarchical clustering, naive Bayes, association rules.

Sheet 2, deep. The perceptron, activation functions, backpropagation and gradient descent, autoencoders, transfer learning, CNN end to end, reinforcement learning with every formula, scaling, and every exam question.

07 MAXU 5214 Project 1

One sheet. Your own thesis: flood segmentation with cross-modal knowledge distillation. The problem in the order you should say it, the framework drawn, the concepts with formulas, the dataset and its two traps, the method and the loss term by term, the real IoU numbers, and prepared viva answers.

This is the one you will be asked about for twenty minutes. It is last in the file, but revise it first.

HOW How to use this file

- First pass, one hour.** Read only the black strip on each sheet. Seven paragraphs, and you have the shape of the whole degree.
- Second pass.** Read the yellow and red boxes only. Those are the memorise items and the traps.
- Third pass.** Work the tables. Cover the right hand column and try to produce it.
- Before an interview.** Sheet 07 first, then the metrics panel on 5113 sheet 2, then the test chooser on 5013 sheet 1. Those three answer most technical screening questions.

The text in this PDF is real text, not a picture. Ctrl+F works. If you are looking for one formula, search for it rather than scrolling.

Fundamental of Data Science

The whole subject is one arc: **a business question** becomes **data you understand**, becomes **data a model can eat**, becomes **a model**, becomes **a decision someone acts on**. CRISP-DM is the name of that arc. Every lecture is one station on it. If you can say which station a technique belongs to and why that station exists, you can answer most of the paper.

L1 Basic Principle of AI

McCarthy (1955): AI is the science and engineering of making intelligent machines, especially intelligent computer programs.
Turing (1950): if a computer could imitate sentient human behaviour, would that imply the computer itself was sentient.

THE FOUR TYPES OF AI

A 2 by 2: **think vs act** crossed with **human vs rational**.

	LIKE A HUMAN	RATIONALLY
THINK	Introspection, psychological experiments, brain imaging	Laws of Thought. Facts via logic, theorem provers. Cannot handle uncertainty
ACT	Turing Test (the Imitation Game)	Rational Agent. Do the thing that maximises expected return

Exam line: modern AI sits in **Act Rationally**. It is a black box approach, inspired by cognitive science and neuroscience, and it contains correct inference (Think Rationally) as a special case.

7 CAPABILITIES THE TURING TEST DEMANDS

NLP KNOWLEDGE REPRESENTATION AUTOMATED REASONING
MACHINE LEARNING SPEECH RECOGNITION COMPUTER VISION ROBOTICS

8 FOUNDATION DISCIPLINES

Philosophy · Psychology · Mathematics · Computer Engineering · Economics · Control Theory and Cybernetics · Neuroscience · Linguistics

EXECUTIVE VIEW: THE AI CAPABILITY AREAS

AREA	WHAT YOU MUST BE ABLE TO SAY
Image recognition	An image is made of pixels ; the computer sees numerical values, not a picture. Used in manufacturing, medical, agriculture, content moderation, advertising, retail
Speech recognition	Also called ASR , speech to text. Not the same as voice recognition, which identifies who is speaking. Models: acoustic model + language model
Search and knowledge representation	A search problem = state space + start state + goal state (a test on the current state)
NLP	Two halves: NLU (understand) and NLG (produce)
Planning	Realisation of strategies or action sequences for execution by intelligent agents, robots, unmanned vehicles
Reasoning	Deductive: conclusion deduced from known facts. Inductive: hypothesis generating, used at the start. Plus monotonic

THE TWO ORDERED PIPELINES TO MEMORISE

ASR, 4 steps:

Analyze audio ► Break into parts ► Digitize ► Match to text

NLP, 5 steps:

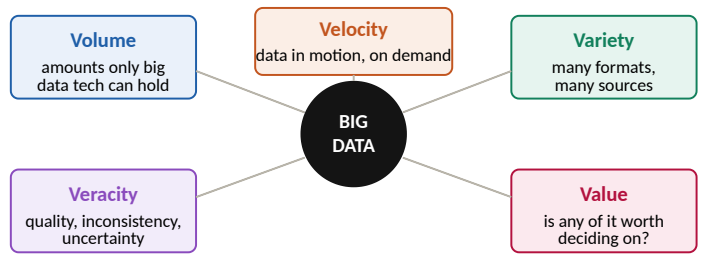
Lexical ► Syntactic (parsing) ► Semantic ► Discourse integration ► Pragmatic

L2 Big Data, Data Science & Ethics

THREE WORDS PEOPLE MIX UP

BIG DATA	DATA SCIENCE	DATA ANALYTIC
Extremely large, complex data sets that need advanced tools to analyse	Multidisciplinary: statistical and mathematical, to extract insight from data	The process of analysing data to uncover patterns that inform a business decision

THE 5 V'S



Gartner names only volume, velocity and variety. Veracity and value were added later, and value is the one that decides whether a project happens.

FOUR TYPES OF DATA

TYPE	DEFINITION	EXAMPLE
Structured	Fixed format, defined type and structure	Transaction table
Semi-structured	Contains both structured and unstructured	JSON, XML
Quasi-structured	Text with irregular formats, formattable with effort and tools	Web clickstream
Unstructured	Unknown form and structure, heterogeneous	Text, images, video

WHERE THE DATA SITS

REPOSITORY	WHO OWNS IT
Spreadsheets and data marts	Low volume, recordkeeping. Analyst depends on a data extract
Data warehouse	Centralised, supports BI and reporting. Analyst depends on IT and DBAs
Analytics sandbox	Assets gathered from many sources, analyst owned, not DBA owned

Warehouse vs lake: the warehouse stores **cleaned, schema-organised** data built for fast querying. The lake stores **raw** data in its original format, mixing structured, semi-structured and unstructured.

THE DATA SCIENTIST, 5 BEHAVIOURS

- Quantitative skill (maths, statistics)
- Technical aptitude (ML, software engineering, programming)
- Skeptical mindset and critical thinking, to find several ways to solve one problem
- Curiosity and creativity, to ask enough questions to actually understand the business problem
- Communicative and collaborative, to articulate business value clearly

FIVE ETHICS IN DATA SCIENCE

- 1 Seek to enhance the value of data science for society
 - 2 Avoid harm
 - 3 Apply and maintain professional competence
 - 4 Seek to preserve or increase trustworthiness
 - 5 Maintain accountability and oversight.
- Case discussed: Crisis Text Line.

BI VS DA, THE QUESTION THAT KEEPS APPEARING

BI ASKS	DA ASKS
What already happened?	Why did it happen, and what happens next?
"How many coffees did we sell last month?"	"Why did coffee sales drop last month?"
"Which branch had the highest sales?"	"Can we predict next month's sales?"

L2+ Project scoping: the four canvases

Asked as a full 20 mark question in both past papers. Learn the field names, not the idea.

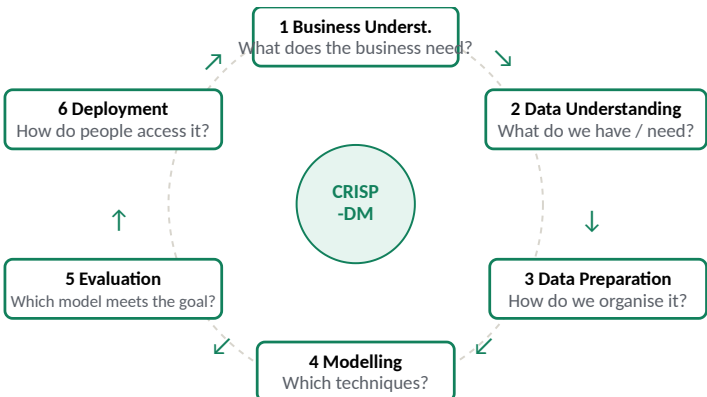
SWOT ► Strategic Thrust ► Capability ► Action Map

CANVAS	FIELDS YOU WRITE IN IT
1 SWOT	Strength (what are you good at), Weakness (what is hard), Opportunity (what can be done), Threats (what might stop you)
2 Strategic Thrust	ST Name, ST Instigator, Stakeholders split into Power / Urgency / Legitimacy, Drivers, Effects, Results, Motivation, Enablers, Resources, Risks, Timings, KPIs
3 Capability	Cap Name, Cap Lead, Overall quality. Do the right things: key monetary value, key non-monetary value, key investment, business requirements, key risk, key service. Do things right: key person, key processes, resource constraints, key information structures, key technologies
4 Action Map	Mission, Vision, Customer value proposition, Business goals, Customer, Internal process, Organisational capability, Core value

How the question is set: you are given a business problem, then asked to fill one canvas and to write **one BI question and one DA question** about it. Write the BI question in the past tense about totals, and the DA question about which attributes drive the outcome.

L3 Data Understanding

CRISP-DM, AND THE QUESTION EACH PHASE ANSWERS



Cross Industry Standard Process for Data Mining. It is a loop, not a line: evaluation can throw you back to business understanding.

DATA UNDERSTANDING: 4 TASKS IN ORDER

Collecting ► Describing ► Exploring ► Verifying

Techniques: data profiling, data visualization, data sampling.

Deliverables, 3: data requirements report, data description report, data quality report.

WHY IT MATTERS, 3 REASONS

- Informed decision making:** insight into characteristics, structure, relationships
- Assessment of data quality:** completeness, consistency, accuracy
- Optimising resource utilisation:** identify potential predictors and reliable variables

VERIFYING DATA QUALITY, 8 CHECKS

Accuracy · Completeness · Consistency and integrity · Timeliness and currency · Standardisation and formatting · Duplicate detection · Audit trails and logging · Profiling and EDA

SUMMARY STATISTICS

MEASURE	USE IT ON
Frequency and mode	Categorical. Mode = most frequent value
Percentiles	Continuous. The 50th percentile is the value below which 50% of values fall
Mean and median	Location. Median is the robust one
Range and variance	Spread. Standard deviation = square root of the variance

LOCATION AND SPREAD
Location = mean Spread = standard deviation

EDA: TUKEY, AND WHAT VISUALIZATION IS

Exploratory data analysis was created by statistician **John Tukey**. Tukey's own focus was visualization; at UTeM the focus is **summary statistics plus visualizations**.

Three aspects of visualization: Representation (map information to a visual format: objects become points, attribute values become positions) · **Arrangement** (placement of visual elements, which changes how easy the data is to read) · **Selection** (choosing a subset of objects).

6 VISUALIZATION TECHNIQUES AND WHEN TO REACH FOR EACH

TECHNIQUE	REACH FOR IT WHEN
Histogram	You want the distribution of one variable. Values go into bins, bar height = count
Box plot	Distribution again, but you also want the outliers named
Scatter plot	Two attributes determine position. Size, shape and colour can carry a third and fourth
Contour plot	A continuous attribute measured on a spatial grid : temperature, rainfall, air pressure
Parallel coordinates	Many variables at once, more than 3 features to compare, and you want to see group differences
Star coordinates	Compare several items across many features. Each item is one star, good for showing a profile of strengths and weaknesses

Principle of graphical excellence, 4: clarity, precision, efficiency, and **maximise ideas, minimise ink**.
Elements of a plot, 3: geometric objects, scale and coordinates, annotations.
Aesthetics, 6: position, shape, size, colour, line width, line type. Colour resource named in the notes: colorbrewer.org

EDA COMPONENTS, 8 IN ORDER

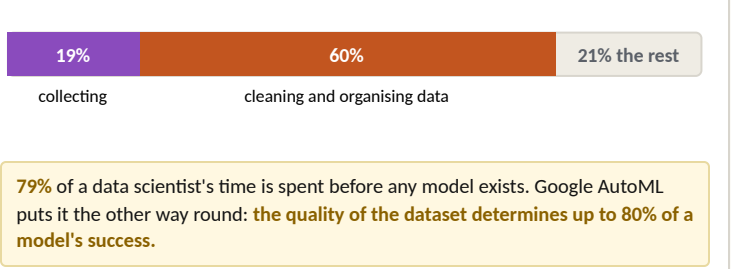
Collection ► Cleaning ► Descriptive stats ► Univariate ► Bivariate ► Multivariate ► Feature engineering ► Visualization

LEVEL	WHAT IT DOES	CHARTS
Univariate	One variable. Understand the distribution, spot outliers, guide the next step	Numeric: histogram, box plot, summary stats. Categorical: bar, pie, frequency table
Bivariate	Two variables. Find relationships, detect trends, spot bias, choose features	Scatter, grouped bar, cross tab
Multivariate	Patterns the first two levels cannot see. How combined variables act together	Parallel coordinates, correlation heatmap, PCA

L4 Data Preparation

Definition: the process of cleaning and transforming raw data before processing and analysis.

WHERE THE TIME ACTUALLY GOES



7 KEY TASKS

1 Data collection · 2 Data cleaning · 3 Data transformation · 4 Data reduction · 5 Handling outliers · 6 Data documentation · 7 Collaboration with domain experts

DATA WRANGLING, 7 STEPS WITH THE ACTUAL THRESHOLDS

#	STEP	THRESHOLD TO QUOTE
1	Drop unnecessary columns	Drop a column with about 75% missing . Drop low variance columns (every row the same value)
2	Remove rows with nulls	Consider it when about 10 to 15% of observations carry nulls
3	Remove noise	Invalid values, outliers, skew. Log transform a skewed variable to make it closer to normal
4	Impute missing values	Numeric: mean, median, mode. Categorical: new category or the most frequent one
5	Non-numeric to numeric	Label and one-hot encoding. Dates are strings by default and must be converted
6	Feature engineering	Binning, standardisation, normalisation. Know which algorithms need scaling
7	Dimensionality reduction	PCA, SVD, LDA, MDS, t-SNE, ICA

Cleaning thresholds asked separately: drop rows when missing is minimal, **under about 5%**. Above **20%**, impute instead. A column that is **70% missing** may be dropped outright. Time series gets **interpolation**, not the mean.

ENCODING CATEGORICAL VARIABLES

ENCODING	WHAT IT DOES	EXAMPLE
Label	One unique integer per category	Male 0, Female 1
One-hot	A new column per unique value	S [1,0,0], Q [0,1,0], C [0,0,1]
Binary	Category as a binary representation	A 000, B 001, C 010

THE LOAN DEFAULT CASE, THE ONE MARKED "SANGAT PENTING"

PROBLEM	FIX, WITH THE CODE
Imbalance: 90% not defaulted, 10% defaulted . Random sampling misses the defaults	Stratified sampling. <code>train_test_split(X, y, test_size=0.2, stratify=y)</code>
Still skewed after splitting: 1800 non-default vs 200 default	Oversample with SMOTE. <code>from imblearn.over_sampling import SMOTE</code>
Loan dates shuffled randomly, so you train on the future and test on the past	That is data leakage . Time based split: train 2020 to 2021, test 2022
Outcome columns left in the training data	Remove any column derived from the answer, and remove IDs before splitting

GARBAGE IN, GARBAGE OUT: 8 OTHER SOURCES OF GARBAGE

- Inaccurate conceptual models or theories, so you model the wrong object
- Inadequate knowledge of causation and causal inference (Maziarz 2020)
- Undocumented or poorly documented code
- Neglecting representative viewpoints, giving biased data (Obermeyer et al. 2020)
- Inadequate research procedures (Gillian et al. 2016)
- Ignoring the **no free lunch** theorem, so you pick a less suitable model
- Disregarding necessary assumptions of a statistical model
- Model assumption bias: modelling something that does not exist in reality

History worth quoting: in 2009 **Mike Driscoll** popularised **data munging**, the painful process of cleaning, parsing and proofing data. In 2013 **Josh Wills** described himself as a **data janitor**.

ETHICS WHEN CHOOSING DATA

The key question the slides put in bold: **"Have we ensured a fair representation of all groups in the dataset?"** Areas to check for bias: age, gender, ethnicity, socioeconomic status, and for environmental data, a balanced representation of conditions.

Lack of diversity, two examples: self driving cars with no night time or snow data and no animal scenarios; predictive maintenance with no failure history and no seasonal variation.

USE How to attack the paper

Both past papers follow the same shape. Recognise the shape and you save fifteen minutes of reading.

WHAT YOU ARE GIVEN	WHAT THEY ARE ACTUALLY TESTING
A business scenario and a blank canvas	Can you name the canvas fields. Write them as labels, then one line each. Do not write an essay
"Write one BI question and one DA question"	BI is past tense and about totals. DA is about why, which drivers, and what next
A small table with a missing value	Impute by category mean , show the arithmetic, then say the disadvantage: it shrinks variability and a small group gives an unrepresentative mean
Counts of predictions	Build the confusion matrix first, then compute. Always finish with the sentence about imbalance
"Should outliers be removed?"	Answer not always . Remove only if they are errors. Real populations really do contain people over 90 and over 110 kg
"Match activities to CRISP-DM phases"	Same currency and combine datasets are preparation . Talk to users is evaluation . Ask how data was collected is understanding . Scale pilot is deployment

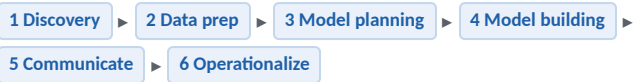
BOX PLOT FENCES, MEMORISE THE NUMBERS
Inner fence = $Q1 - 1.5 \cdot IQR$ and $Q3 + 1.5 \cdot IQR \rightarrow$ suspected outlier
Outer fence = $Q1 - 3.0 \cdot IQR$ and $Q3 + 3.0 \cdot IQR \rightarrow$ extreme outlier
 $Q1 = 25\%$ of data below it $Q3 = 75\%$ of data below it

Two benefits of scaling to [0,1], asked verbatim: it stops one feature such as weight in kg dominating the others, because K-Means, KNN and gradient descent are scale sensitive; and training converges faster.

Sheet 1 ended with clean data. Now: **pick a family of method** (classify, cluster, regress), **build it, judge it with a number that cannot be gamed**, then **keep it alive**. The last part is the one most students skip and it is a whole lecture with its own 50 question test.

L5 Analytical theories, part 1

Where you are: steps 3 and 4 of the lifecycle, **model planning** and **model building**.



What is a model? A mathematical representation of real world patterns learned from data, used to make predictions, classifications, decisions or new information.

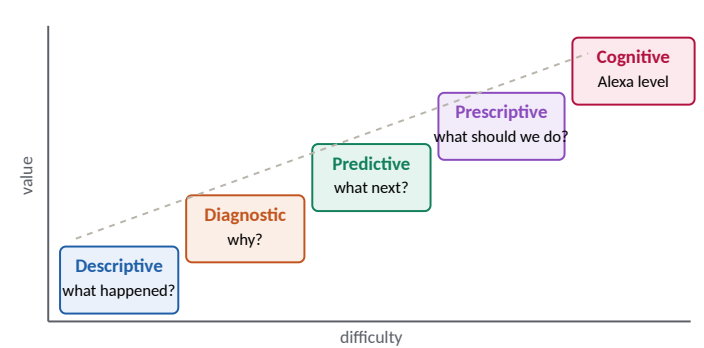
Planning and building overlap and iterate back and forth before you settle on a final model. That sentence is in the slides and is worth a mark.

THE SPLIT, AND THE NUMBERS

SET	SHARE	ANALOGY IN THE SLIDES
Training	70 to 80%	Practice exercises
Validation	carved out of training	Mock exam
Testing	20 to 30%	Final exam

Model tuning means adjusting hyperparameters: learning rate, number of layers, number of trees. **Model selection** means comparing several trained models on the validation set and keeping the best.

THE ANALYTICS LADDER



TYPE	EXAMPLE QUESTION FROM THE SLIDES
Descriptive	What is the highest product sold in January?
Diagnostic	Why did selling product drop this month?
Predictive	What are sales likely to be in 2020?
Prescriptive	How do we increase selling, which influencer do we recommend. Uses what if analysis and optimisation

5 QUESTIONS DATA SCIENCE CAN ANSWER

QUESTION	METHOD
Is it A or B?	Classification
How much, how many?	Regression / forecasting
Which option?	Reinforcement learning
Is it weird?	Anomaly detection
Which groups?	Clustering

Supervised builds a predictive model from **input and output** data. Unsupervised discovers an internal representation **from input only**.

THE THREE METHOD FAMILIES

CLASSIFICATION	CLUSTERING	REGRESSION
Naive Bayes, decision trees, time series classifier, text classifier	K-means, association rules	Linear, logistic

- Time series classifier: Box-Jenkins, ARMA, ARIMA. Accounts for **trend, seasonality, cycles, irregulars**
- Text analysis, 3 steps: parsing, then search or retrieval, then text mining
- K-means: input is numerical and needs a distance metric, normally **Euclidean**. Output is the **centroids**. It is **not predictive**, it finds similarity
- Association rules: also not predictive. Market basket analysis. Three metrics: **support** (how popular an itemset is), **confidence** (how often the rule holds), **lift** (how much more likely Y is when X is bought, versus Y on its own)
- Logistic regression solves **classification**, the outcome is categorical and the output is a probability **between 0 and 1**

L6 Decision trees, entropy, information gain

A decision tree takes a set of properties as input and gives a yes or no as output. It is **supervised**, and specifically **inductive learning**, which means learning from examples.

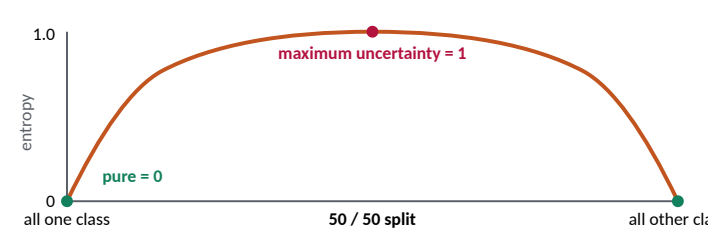
- Each **decision node** is an attribute, each **leaf** is Yes or No, each **branch** is a possible value of that attribute
- Fully expressive within the class of propositional languages
- Aim**: find a **small** tree consistent with the examples. **Idea**: recursively pick the most significant attribute as the root

TWO IMPURITY CRITERIA, AND WHERE EACH LIVES

CRITERION	ALGORITHM
Entropy and information gain, from information theory	ID3 and C4.5
Gini index	CART, classification and regression trees

Remember this one: decision trees in **KNIME** and in **scikit-learn** use **CART** by default, so Gini, not entropy, unless you change it.

ENTROPY



Entropy measures how mixed the classes are. One class only gives 0. An even split gives 1. A value like 0.72 means some uncertainty remains.

ENTROPY OF A BINARY SPLIT, P POSITIVES AND N NEGATIVES
$$E(p/(p+n), n/(p+n)) = - (p/(p+n)) \cdot \log_2(p/(p+n)) - (n/(p+n)) \cdot \log_2(n/(p+n))$$

INFORMATION GAIN
$$IG = \text{Entropy}(\text{before split}) - \text{Remainder}(\text{after split})$$

$$\text{Remainder} = \sum \text{over branches} (\text{rows in branch} / \text{total rows}) \times \text{Entropy}(\text{branch})$$

Information gain measures how much entropy falls when you split on a feature. The algorithm takes the attribute with the **highest** gain.

BUILDING THE TREE, THE 10 STEPS AS EXAMINED

- Compute the entropy of the target using **all** rows
- For each attribute compute the information gain on all rows. These are the **global** gains
- The highest global gain becomes the **root**. In the WillWait example that is **Patrons**
- Split into branches by the root's values. Here: None, Some, Full
- In a branch, if every example has the same label, make a leaf and stop
- Recompute gains using **only the rows inside that branch**. These are the **local** gains
- Split that branch on the highest local gain
- Repeat 5 to 7 until every branch is a leaf or attributes run out
- On a tie, break it arbitrarily, or use gain ratio or pruning
- In the Full branch, raw local gain picks **Type**, but many lecturers take **Hungry** for a simpler tree. Say which you used and why

Named exercises in the slides: WillWait (12 rows), The Simpsons, Play Tennis.

RANDOM FOREST

A supervised classification algorithm that creates a forest of many trees and decides **by vote**. More trees makes the forest more robust and raises accuracy, works for classification and regression, and **with more trees the random forest will not overfit**.

FIT Underfit, good fit, overfit

	UNDERFITTING	APPROPRIATE FIT	OVERFITTING
Meaning	Too simple to learn the pattern	Generalises well	Too complex, memorises the training data
Symptom	Train low, test low	Train high, test high, errors consistent	Train near 100%, test low
Student analogy	Read the notes once and understood nothing	Studied properly, can answer new questions	Memorised past year papers, fails anything else
ML example	Linear model on non-linear data	Pruned tree, tuned random forest	Deep network memorising noise, very deep tree
Cause	Too simple, too few features, high bias	Balanced complexity, enough data, tuned	Too complex, too many features, too little data, low bias high variance
Fix	More complexity, more features, less regularisation, train longer	Validation set, early stopping, cross validation	Regularisation L1 or L2 , prune the tree, drop features, more data, dropout

Three mark question, "how do you avoid overfitting": cross validation, regularisation, and either simplify the model or add more data. Three bullets, three marks.

EVAL Confusion matrix, precision, recall, ROC

A confusion matrix is a table describing the performance of a classification model on test data where the true values are known.

	ACTUAL		
	Yes	No	
PRED Yes	TP has it, you said so	FP false alarm Type 1 error	Precision reads this row
No	FN you missed it Type 2 error	TN correctly left alone	Recall reads the left column

THE FOUR FORMULAS
Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP) when it says yes, how often is it right
Recall = TP / (TP + FN) of the real yeses, how many did it find
F1 = 2 × (P × R) / (P + R) only reaches 1 when both are perfect

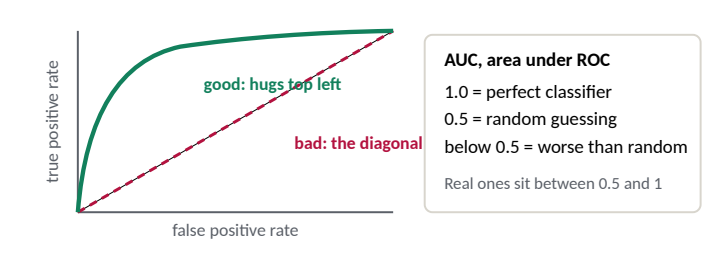
TWO MORE FROM THE STUDY FINAL PAGE
Prevalence = (TP + FN) / Total
Misclassification rate = (FP + FN) / Total

Recall is also called sensitivity, or the true positive rate. Accuracy is only a fair number when the classes are **balanced**.

PRECISION OR RECALL? THE FOUR SCENARIOS ASKED

SCENARIO	OPTIMISE	BECAUSE
Spam filtering	Precision	Sending a real email to spam is worse than letting one spam through
Diagnosing patients	Recall	Missing a sick patient is far worse than a false alarm
YouTube recommendation	Precision	Irrelevant videos kill engagement. Missing a few good ones is fine
Launching a satellite on a weather call	Precision	Positive means "good to launch", so you must be very sure when you say yes

ROC AND AUC



L7 Tools and platforms

PROPRIETARY	OPEN SOURCE
Owned by the individual or company that published it, copyrighted, available only under licence	Anyone can access or change the code, can be free of copyright, usually available to anyone

- MAD Landscape, by Matt Turck and Shivon Zilis, published since 2012. In 2024 MAD stands for **Machine learning, AI and Data**. It groups companies by category so people can grasp a fast growing ecosystem
- Gartner Magic Quadrant, market research reports using **proprietary** qualitative analysis methods, updated every **1 to 2 years**. The slides say there is criticism of it, so read with caution

L8 Communicate and operationalize

Phase 5, communicate result. The team and the major stakeholders decide success or failure **against the criteria set in phase 1**. Assess whether the results are statistically significant and valid, identify surprising results and results in line with the hypotheses, then document the key findings. **This is the most visible part of the process to outside stakeholders and sponsors.**

Phase 6, operationalize. Set up a **pilot project** and deploy in a controlled way. Small scope first, because that is how risk is managed. During the pilot you may need to run the algorithm in the **database rather than with in-memory tools like R** for larger datasets. Monitor accuracy and retrain when needed.

SIX AUDIENCES, AND WHAT EACH ONE WANTS

AUDIENCE	WANTS
Business users	Business benefits and implications
Project sponsor	Business impact, risks, return on investment
Project manager	On time, within budget, goals met
BI analyst	Whether reports and dashboards must change
Data engineer and DBA	The code and the documentation
Data scientist	The code, and the model explained to peers

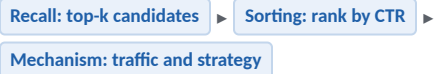
Four deliverables in order: presentation for sponsors (high level), presentation for analysts (process and reporting changes, technical graphs), code, technical specifications.

CASE 1, NETFLIX. THE NUMBERS GET ASKED

Goal	Improve Cinematch , whose accuracy had plateaued
Competition	Netflix Prize, 2007, 1 million US dollars for a 10% improvement, measured by RMSE
Data	100 million anonymised ratings
Entrants	20,000 teams . Won in 2009 by BellKor's Pragmatic Chaos , who submitted 20 minutes before the other team
Methods	Neighborhood models, matrix factorization (the key contributor), regression, restricted Boltzmann machines, temporal effects, and an ensemble of 104 models
Twist	Netflix never used the winning algorithm . Too complex, and the data had changed because of streaming. They used parts of earlier solutions. The contest inspired Kaggle

CASE 2, ALIBABA

1.3 billion active users. Before 2019 the recommender was similarity based, which optimised **relevance** only. After 2019 they introduced **AIRec**, which adds **diversity and discovery**.



Technologies: graph embeddings, transformers, deep learning, knowledge graphs. The load case is **Double 11**, the world's largest shopping event.

L9 Analytics model management

An organisation has hundreds of models built by different analysts in separate units, with **no standardised process** for storing, deploying or managing them.

THE SIX QUESTIONS THE ORGANISATION CANNOT ANSWER

- Who created the models and why?
- What input variables are used to make predictions and decisions?
- How are the models used?
- How are they performing, and when were they last updated?
- Where is the supporting documentation?
- Why does it take so long to put a new model into production?

If you cannot answer those with confidence, you cannot be sure your models deliver real value. The slides describe the state as **hoping for the best**.

THE MODEL LIFE CYCLE



Trap in the MCQ: **auditing is not a life cycle stage**. Also, **model encryption is not an element of model management**, and **Apache Spark is not a model management tool**.

TWO ELEMENTS

MODEL REPOSITORY	MODEL REGISTRATION
Central secure storage for documentation, scoring code and metadata. Includes user and group authentication, version control and audit tracking	The analyst submits a model package : the model, all data transformations, outputs and documentation. Standardises creation, management, deployment and monitoring

TWO ACTIVITIES

GOVERNANCE	VALIDATION
Accountability metrics, version control reports, who changed what and when, movement of models between teams. Gives visibility, auditability and compliance	Every stored model is scored on data and validated for accuracy and reliability, so it is sound, trustworthy and suitable for use. A failed validation sends the model back for further tuning

VERSION CONTROL AND DECAY

Version control needs **event logging**: what changed, who changed it, when. Models **decay** as data patterns change, so you monitor for a performance drop and decide when to refresh or replace. Retired models are archived for **audit, compliance and historical reference**.

THE FOUR NAMED TOOLS

SAS MODEL MANAGER **IBM SPSS MODELER** **KNIME SERVER** **MLFLOW**

MLflow is open source and combines MLOps with model management. Six features: **tracking**, **model registry** with lineage, **packaging**, **deployment**, **projects**, **plugins**. Hardware optimisation is **not** one of them. Neptune AI is also named.

MLOps means automating and streamlining the **deployment, monitoring and maintenance** of ML models. The slides argue the next breakthrough in data analysis is **combining, deploying and maintaining existing algorithms rapidly**, not inventing new ones.

BEST PRACTICE ANSWER, 7 POINTS

1 MLOps with CI/CD for models · 2 version control for models **and** data · 3 automated documentation of algorithm, hyperparameters, variable importance · 4 monitoring and **drift detection** with scheduled retraining · 5 governance and audit trail with approval workflow · 6 controlled deployment, containerised with Docker · 7 a **model retirement plan**

DRILL Worked past year numbers

Do these from memory before the exam. The arithmetic is the easy part, the closing sentence is where the marks are.

SUBSCRIPTION MODEL

Predicted subscribe 5, of which 3 were right. Predicted not subscribe 95, of which 87 were right.

$$TP\ 3\ FP\ 2\ FN\ 8\ TN\ 87$$
$$\text{Accuracy} = (3 + 87) / 100 = 90\%$$
$$\text{Recall} = 3 / (3 + 8) = 27.3\%$$
$$\text{Precision} = 3 / (3 + 2) = 60\%$$

The sentence that earns the marks: accuracy looks high only because most people do not subscribe. Recall of 27% means the model misses most of the real subscribers, so it is a poor classifier for the job it was built for.

FRAUD MODEL

$$TP\ 5\ FP\ 15\ FN\ 5\ TN\ 95$$
$$\text{Accuracy} = 100 / 120 = 83.3\%$$
$$\text{Recall} = 5 / 10 = 50\%$$
$$\text{Precision} = 5 / 20 = 25\%$$

Low recall means half of all fraud goes through, which is the expensive failure. Low precision means many false alarms, which annoys real customers. The model needs improvement on both sides.

CATEGORY MEAN IMPUTATION

Boy ages 9, 7, 7.5, 8. Mean = 31.5 / 4 = **7.875, so about 7.9 years**. Better than the overall mean because boys and girls develop differently, so it keeps the distribution realistic within the group. Disadvantage: a small group gives an unrepresentative mean, and imputation always reduces variability.

FREE TEXT AND CHARTS

A free text Description column becomes features through **TF-IDF**, keyword extraction or product categorisation, which reduces sparsity and standardises the descriptions. For spending by country use a **bar chart**, for category share a **pie chart**, for typical order size a **histogram**.

Retail anomalies: negative quantity means returns or refunds, unit price of zero means free promotional items or replacements. Flag them with a new feature rather than deleting blindly.

Research Methodology

WHAT RESEARCH IS, HOW A STUDY IS FRAMED, AND HOW THE LITERATURE IS READ

The whole course is one chain, and every later chapter is a link in it. A **topic** narrows to a **problem**, the literature exposes a **gap**, the gap becomes **objectives**, **questions** and **hypotheses**, those dictate the **paradigm** and **method**, the method dictates the **instrument**, **sample** and **analysis**, and the whole thing is written up as a **proposal**. The examinable skill is **alignment**: RP to RQ to RO to RM, continuously evaluated and adjusted. Say **why** each piece exists before you define it.

Ch1 What research is

The course opens with definitions because every later term is built on them.

DEFINITION 1, MERIAM-WEBSTER ONLINE DICTIONARY

"studious inquiry or examination; especially: investigation or experimentation aimed at the discovery and interpretation of facts, revision of accepted theories or laws in the light of new facts, or practical application of such new or revised theories or laws."

DEFINITION 2, GROVER, VIJIE (2015)

"searching for knowledge and searching for truth. In a formal sense, it is a systematic study of a problem attacked by a deliberately chosen strategy, which starts with choosing an approach to preparing a blueprint (design) and acting upon it in terms of designing research hypotheses, choosing methods and techniques, selecting or developing data collection tools, processing the data, interpretation, and ending with presenting solution(s) of the problem."

DEFINITION 3, JOHN W. CRESWELL

"research is a process of steps used to collect and analyze information to increase our understanding of a topic or issue."

1 Pose a question

2 Collect data to answer it

3 Present an answer

Week 4 wording:

Research is the systematic investigation of phenomena to answer questions or develop new knowledge."

1.1 FIVE REASONS FOR CONDUCTING RESEARCH

Research needs a motive, and the course names exactly five.

1. **Advancing knowledge.** Answers unanswered questions, validates theories. Neuroscience and memory processing.

2. **Solving problems.** Reducing plastic pollution, biodegradable plastics.

3. **Informing policy and practice.** Smoking research to tobacco warning labels and bans.

4. **Encouraging innovation.** Renewable energy to more efficient solar panels.

5. **Contributing to education.** Active learning to flipped classrooms.

1.2 SIX CHARACTERISTICS OF GOOD RESEARCH

These are the marking criteria for whether a study is worth believing.

1. **Clearly defined objectives.** "To investigate the impact of daily social media use on mental health indicators among adolescents aged 13 to 17."

2. **Methodological rigor.** The randomized controlled trial (RCT) is named "the gold standard in clinical research".

3. **Ethical standards.** Informed consent, honest reporting, privacy.

4. **Reproducibility and transparency.** COVID-19 vaccine development.

5. **Significance and contribution.** Fills knowledge gaps, offers solutions.

6. **Limitations acknowledged.** A remote work study limited to tech professionals, so generalizability is affected.

Conducting scientific research, four principles. The goal is inference, so generalizability - the procedures are public, so replicable - the conclusions are uncertain, "Statistics is never having to say you're certain" - follow the rules of inference.

Ch1 MSc versus PhD research

This exists so you pitch your own proposal at the right level, not above it.

AXIS	MSC RESEARCH	PHD RESEARCH
Scope and depth	Narrow focus, one specific question, limited scope. One factor, rainfall on soil erosion in one region	Broader, comprehensive, multiple perspectives. Climate change plus urbanization plus agriculture across regions
Originality	May build on existing studies, validates known concepts	Must offer original insights and a unique contribution
Duration	1 to 2 years, developing research skills	3 to 5 years, detailed comprehensive process
Design flexibility	Follows established methods	Develops new methodologies or adapts existing ones
Outcome	Thesis demonstrating research competency, no expectation of significant new contribution	Dissertation with substantial original contribution, usually including published papers

CRITICALITY BY LEVEL (WEEK 6)

The same literature is read differently at each level, so know which one you are writing at.

- Undergraduate: backing points with appropriate authoritative sources

- Masters: demonstrating that you understand how knowledge is created

- PhD: demonstrating that you can create new, original knowledge

- Professional academic: peer review, building a coherent body of work, assuring the work of others

BLOOM'S TAXONOMY 1956, REVISED ANDERSON & KRATHWOHL 2001, AS PRINTED

Knowledge → Understanding → Application → Analysis

→ Evaluation → Synthesis / Creation

BASIC VERSUS APPLIED RESEARCH

Supplementary, not in the decks. The lectures never define this pair. The nearest side content is Week 1's Advancing knowledge (a basic motive) against Solving problems and Informing policy and practice (applied motives). Standard distinction, for completeness only; basic seeks knowledge for its own sake and builds theory with no immediate application, applied solves a specific practical problem in a defined context. Do not attribute this pairing to the lectures.

Ch2 The research process

Research is drawn as a process so that you can see which stage feeds which, and where you are.

Definition: "Research is a systematic process designed to explore, analyze, and ultimately answer a research question or solve a problem."

1 Identify a Research Problem
Recognize a specific problem or area of inquiry

2 Literature Review
Examine existing research, what is already known

3 Define Objectives and Questions
What the study aims to achieve, and asks

4 Design the Methodology
Choose collection and analysis: surveys, experiments

5 Data Collection and Analysis
Gather, then make sense of the data

6 Interpretation and Reporting
What the findings mean, and write them up

7 Reflection and Further Research
Recognize limitations, suggest further study

The "What" ladder, 6 report sections (Weeks 4 and 13, shown twice so it is emphasised)

1 Intro, Problems, Objectives, Justification: What? What about? What for?

2 Literature Review: What did prior research say?

3 Methodology: What was done?

4 Results and Discussion: What was found?

5 Implications: So what?

6 Conclusions: What now?

Left ribbon is the Week 2 seven stage process. Dotted ticks align the five step Study Plan. The dashed return arrow is stage 7 feeding a new study.

PARADIGM SPECIFIC CHAINS (WEEK 9)

POSITIVISTIC, DEDUCTIVE
topic → literature review → question / hypothesis
→ design data collection → PILOT STUDY → design data analysis
→ collect data → analyse → interpret → report findings
(Maylor & Blackmon, 2005)

PHENOMENOLOGICAL, INDUCTIVE
topic → literature review → question → design data collection
→ collect data → LITERATURE REVIEW AGAIN → analyse → interpret
→ "Research question answered?" no, loop back
→ report findings

Ch2 Topic, problem, gap, objective, question, hypothesis

These six are examined by asking you to tell them apart, so learn the boundary between each pair.

2.2 RESEARCH TOPIC, FOUR CRITERIA

A topic is chosen first because nothing else can be planned until it is fixed.

- Aligns with interests and expertise

- Has practical or theoretical value

- Is manageable in scope

- Is researchable

Good: "The Effects of Social Media on Teen Self-Esteem." Too broad: "Influence of Technology on Human Psychology."

What do I want to know?

Is it answerable?

Is it researchable?

Is it significant?

Can I do it?

Lecturer rules, typed in the student notes. Topic maximum 15 words. "fix your mind for one topic, please fix to one topic." Use an action verb, investigate rather than understand. "Need to have topics then bar you can scan sketch the plan", the topic comes before the plan.

2.3 PROBLEM VERSUS GAP

Research Problem	"the issue or phenomenon that prompts the research"
Research Gap	"the missing pieces in current knowledge that your research will address"

Studies show social media impacts self-esteem but lack age group insight, so the gap becomes "The Effect of Social Media on Self-Esteem among Preteens."

DEVELOPING THE PROBLEM, 4 STEPS (WEEK 4)

1 Identify a broad area of research interest

2 Review literature to identify gaps

3 Narrow down to a specific issue or question

4 Evaluate for RELEVANCE, SIGNIFICANCE and FEASIBILITY

Ch4 Types of research design

The design is chosen before any method, because it decides what the data can prove.

Definition: "the overall plan for conducting a research project, selecting appropriate research methods and paradigms, as well as determining sampling techniques, data collection methods, and data analysis techniques."

DESIGN	DEFINITION AS GIVEN	IV MANIPULATED	RANDOM ASSIGNMENT
Experimental	"Involves manipulating an independent variable to observe its effect on a dependent variable". Drug versus placebo	yes	yes
Quasi-experimental	"similar to the experimental design, but lacks some of the key features of a true experiment, compare the outcomes of two groups, but without random assignment". Groups already formed naturally, two classrooms, two hospitals	yes	no
Non-experimental	"used when the researcher cannot manipulate an independent variable, often involve collecting data through observation or survey methods". Observe things as they are. An attitude survey	no	n/a
Case study	"in-depth study of a single case or a small number of cases, to explore complex phenomena in real-life contexts"	no	n/a
Longitudinal	"involve studying the same group of individuals over a period of time, used to examine changes in behavior or attitudes over time". Tracking academic performance over several years. Note: "longitudinal design, length of the studies"	no	n/a
Cross-sectional	Named as a positivist research strategy beside experimental and longitudinal studies. SUPPLEMENTARY the 'one point in time versus repeated measures' contrast is only implied, never defined	no	n/a

Ch2 Objectives, questions, hypotheses, rationale, scope

These are the same idea in three grammatical forms, so the exam tests whether you can tell them apart.

2.4 OBJECTIVE, QUESTION, HYPOTHESIS

	DEFINITION	WORKED EXAMPLE, ONE STUDY
Objective	"what the study aims to achieve, providing a roadmap for the research process". A statement	"To examine the relationship between daily social media use and self-esteem levels in teenagers"
Question	"specific questions that the study seeks to answer". Phrased as a question, guides inquiry, gives a framework	"How does the amount of time spent on social media affect teenagers' self-esteem?"
Hypothesis	"testable statements predicting the expected outcome of the study"	"Increased social media use is associated with lower self-esteem in teenagers"

SMART, THE TEST FOR A RESEARCH OBJECTIVE

Specific clear and well defined

Measurable quantitatively or qualitatively

Achievable within the scope of the research

Relevant to the problem and the questions

Time-bound within a specified time frame

CRITERION GOOD POOR

Specific "Impact of social media on brand awareness among millennials in the United States"

Measurable "Compare the effectiveness of two teaching methods on achievement in mathematics"

Achievable "Relationship between exercise and depression in a sample of 500 participants"

Relevant "Factors that influence employee turnover in UK hospitality"

Time-bound "Trends of climate change in the Arctic over the past decade"

WORKED EXAMPLE POOR "To investigate the impact of social media"

"To understand the perceptions of students"

"study the entire population of the United States in three months"

"Impact of climate change on the fashion industry"

"explore the history and development of the internet"

HYPOTHESIS, NOTATION AND QUALITY

Definition: "A Statement of What we believe to be factual."

VARIABLE NOTATION

Independent (X1) and Independent (X2) → Dependent (Y)

Y = f(X1, X2)

Six criteria for a good hypothesis: explanatory power - states the expected relationship and direction if possible - testable - written as simply as possible - relates to a general, not specific phenomenon - plausible.

The fallacy of affirming the consequent. "If A is true then B should follow. We observe B. Therefore A is true." If private schools produce better students (A) then their students get better final marks (B). They do get better marks, therefore A. Invalid, because teaching methods, resources, socioeconomic background and individual differences also drive marks.

2.5 RATIONALE AND 2.6 SCOPE

Rationale answers "Why is this research important?" and "Who will benefit from the findings?" Three parts: Contribution to knowledge - Practical benefits - Personal motivation.

Scope sets four boundaries so the study stays finishable: Population or participants (teenagers 13 to 17 in urban schools) - Geographical scope (three major urban centers) - Timeframe (six months, January to June 2024) - Variables and concepts, stating what is not covered (excludes family influence, academic performance).

RAT, RESEARCH ALIGNMENT TECHNIQUE, 5 STEPS

1 Start with the research problem, the gap or deficiency

2 Develop research objectives, linked, measurable, achievable

3 Choose research methods appropriate to problem, objectives

4 Align RP, RO and RM

5 CONTINUOUSLY EVALUATE AND ADJUST

Mapping chain: RP → RQ → RO → RM "Mapping is important"

Ch4 Research paradigms

A paradigm exists because your beliefs about reality decide your method before you pick one.

Kuhn (1970): "The set of common beliefs and agreements shared between scientists about how problems should be understood and addressed." Also described as a way of looking at the world, a school of thought, a framework for thinking, a set of assumptions. Week 4: "overarching frameworks that guide the design, conduct, and interpretation of research, informed by underlying philosophical assumptions and beliefs about the nature of reality, the role of the researcher, and the purpose of research."

FOUR COMPONENTS OF ANY PARADIGM

Ontology	Epistemology	Methodology	Axiology
"What is reality?" Does God exist?	Inquiry knowledge. "How do we know what we know?"	"How do we go about discovering the answer or reality?" The process of data collection and analysis	"Values and value judgments", literally "rational discourse about values"

THE PARADIGMS, CORE BELIEF AND METHOD

PARADIGM	CORE BELIEF	METHODS	APPROACH
Positivist	"there is a single, objective reality that can be studied through empirical observation and measurement". Objective, verified. Cycle: think of idea, test, report result	Randomized controlled trials, laboratory experiments, surveys	Quantitative
Interpretive / Constructivist	"reality is socially constructed and subjective, and understanding is gained through interpretation and understanding of meanings and experiences"	Ethnographic studies, case studies, phenomenological studies, in-depth interviews	Qualitative
Critical (postmodern)	"research should be used to challenge power structures and social injustices, and to give voice to marginalized groups"	Participatory action research, feminist research, critical race theory, discourse and historical analysis	Both
Pragmatic	"a more flexible and eclectic approach that values the practical outcomes and usefulness of research", studying phenomena in their real-life contexts	Mixed methods, survey plus focus groups or interviews	Mixed
Post-positivist	"seeks to combine the strengths of the positivist and interpretive paradigms, by acknowledging the importance of objective reality while also recognizing the role of interpretation and subjectivity"	Mixed methods, case studies, surveys with open-ended questions	Quantitative and qualitative
Phenomenological	"aims to understand the culturally different 'other' by learning to 'stand in the shoes', look through their eyes' and feel their 'pleasure or pain' (Gearing, 2004)	Action research, case studies, ethnography, grounded theory, hermeneutics	Qualitative

THE PHILOSOPHICAL POSITION TABLE (WEEK 9)

	POSITIVISM	PHENOMENOLOGY
Ontology	Reality is objective and singular, apart from the researcher	Reality is subjective and multiple as seen by participants
Epistemology	Researcher is independent from that being researched	Researcher interacts with that being researched
Axiology	Value free and un-biased	Value-laden and biased
Strategy	Cross-sectional, experimental, longitudinal studies, surveys	Action research, case studies, ethnography, grounded theory, hermeneutics

Features to quote. Positivist: quantitative data, large samples, hypothesis testing, data highly specific and precise, location artificial, reliability high, generalises from sample to population. Phenomenological: qualitative data, small samples, generating theories, data rich and subjective, location natural, reliability low, validity high, generalises from one setting to another. Named framework: the research "onion", Saunders, Lewis & Thornhill (2006).

Student's own one line mapping: positivist to quantitative, interpretive to qualitative, critical to both, post-positivist to quantitative and qualitative. On ethnography: "awak spent time dengan orang community", the researcher lives with the community.

Ch4 Inductive versus deductive

This is the direction of reasoning, and it decides whether theory comes first or last.

DEDUCTIVE, positivistic
theory first, then test it

Define your research topic

Literature review

Define question, i.e. hypothesis

Design data collection

PILOT STUDY

Design data analysis

Collect data

Analyse data

Interpret results

Report your findings

INDUCTIVE, phenomenological
data first, then build theory

Define your research topic

Literature review

Define your research question(s)

Design data collection

Collect data

LITERATURE REVIEW AGAIN

Interpret data

Research question answered?

Report your findings

One chain is a straight line ending in a report. The other loops until the question is actually answered.

	DEDUCTIVE	INDUCTIVE
Logic	"Use existing theory to shape the approach to the qualitative research process and to data analysis"	"Build up a theory adequately grounded in a number of relevant cases", referred to as Interpretative and Grounded Theory
Analytical procedure	Pattern Matching, "predicting a pattern of outcomes based on theoretical propositions to explain what you expect to find". Explanation Building, "attempting to build an explanation while collecting and analysing the data, rather than testing a predicted explanation"	Art of Interpretation, four text stages: Field Text (field notes and documents) to Research Text (notes and interpretations) to Working Interpretive Document (first attempt to make sense) to Public Text (the final tale of the field)
Thematic analysis fit	Coding reliability TA, theory driven	Reflexive TA, data driven
Design versus method, a named distinction. Design = "an outline, sketch, or plan, as of the form and structure of a work of art, an edifice, or a machine to be executed or constructed." Method = "a procedure, technique, or way of doing something, esp. in accordance with a definite plan". Rule: "Failing to distinguish between design and method leads to poor evaluation of designs." Analogy: you cannot order materials or set dates until you know whether you are building a high-rise, a factory, a school, a home or an apartment block.		

COMPONENTS OF RESEARCH DESIGN, 5 ORDERED

The Research Question → The Theory → The Model → The Data → The Use of the Data

Ch4 Qualitative, quantitative, mixed

The three families exist because different questions need different evidence, not because one is better.

#	QUALITATIVE	QUANTITATIVE
1 Aim	Exploration of participants' experiences and life world, understanding, generating theory from data. Exploratory	Search for causal explanations, testing hypothesis, prediction. Confirmatory
2 Approach	Broad focus, process oriented, context-bound, getting close to data	Narrow focus, product oriented, context free, artificial or laboratory setting
3 Sample	Participants and informants, purposeful and theoretical sampling , flexible sampling that develops during research	Respondents, randomised sampling , sample frame fixed before research starts
4 Data collection	In-depth non-standardised interviews, participant observation and fieldwork, documents, photographs, videos	Questionnaire, standardised interviews, tightly structured observation
5 Analysis	Thematic, constant comparative, content analysis, grounded theory, ethnographic analysis	Statistical analysis
6 Outcome	Story, ethnography, theory	Measurable results
7 Relationship	Direct involvement of researcher, relationship close	Limited involvement, relation distant
8 Rigour	Trustworthiness, authenticity, typicality, transferability	Internal validity, external validity, reliability, generalisability

DEFINITIONS TO QUOTE

- Qualitative methods: "methods used to study subjective experiences, meanings, and interpretations of social phenomena. Examples include interviews, focus groups, and observations."

- Qualitative research: "involves collecting and analyzing non-numerical data (e.g., text, video, or audio) to understand concepts, opinions, or experiences." It answers questions about the "why" of human behaviour, and does not start from a hypothesis.

- Quantitative methods: "methods used to study numerical data and relationships between variables. Examples include surveys, experiments, and statistical analysis."

- Quantitative research: "a research strategy that focuses on quantifying the collection and analysis of data. It is formed from a deductive approach where emphasis is placed on the testing of theory, shaped by empiricist and positivist philosophies."

- Mixed methods: "methods that combine qualitative and quantitative methods to study a research question."

Sub-types. Qualitative: narratives, phenomenologies, ethnographies, grounded theory, case studies. Quantitative: experimental designs, non-experimental such as survey. Mixed: sequential, concurrent, transformative.

MIXED METHODS, THREE TYPES

Sequential	One phase, qualitative or quantitative, followed by the other
Concurrent	Both collected and analysed simultaneously. Interview and survey together
Embedded	One data type supports or provides context for the other

FOUR DECISIONS IN SELECTING A MIXED METHODS STRATEGY

Theoretical Perspective explicit / implicit

Integration at data collection / at data analysis / at data interpretation / some combination

Priority equal / qualitative / quantitative

Implementation no sequence, concurrent / sequential

Ch13 Validity and reliability

These two exist to separate "the instrument measures the right thing" from "It does so consistently".

	QUANTITATIVE DEFINITION	QUALITATIVE DEFINITION
Validity	"the extent to which the research accurately measures what it intends to measure"	"the extent to which the research accurately reflects the experiences and perspectives of participants"
Reliability	"the consistency of the research findings over time and across different researchers", the same wording for both	
Assured by	Random sampling, standardization of procedures, use of control groups	Member checking (participants read and confirm the accuracy of the findings), triangulation, peer review (other expert researchers examine the process and conclusions)

Triangulation = "getting information from three different sources, like interviews, observation, and documents, to confirm the same result." Listed again as a qualitative characteristic: **several methods, data collections, investigators.**

THE FOUR TRUSTWORTHINESS CRITERIA

This table exists to translate between the two vocabularies, and it is the single most examinable object in the module.

QUESTION ASKED	ISSUE	QUALITATIVE	QUANTITATIVE
Have we really measured what we set out to measure?	Truth value	Credibility	Internal validity
How applicable are our results to other subjects and other contexts?	Applicability	Transferability	External validity
Would our findings be repeated if the research were replicated in the same context with the same subject?	Consistency	Dependability	Reliability
To what extent are our findings affected by personal interest and biases?	Neutrality	Conformability	Objectivity

Week 9's rigour row also names **authenticity** and **typicality** alongside trustworthiness and transferability.

SIX TYPES OF VALIDITY SUPPLEMENTARY

Honest gap. The decks name internal and external validity only, plus the four trustworthiness criteria. Construct, content, criterion and face validity are not defined anywhere in the course files.

TYPE	THE QUESTION IT ANSWERS
Internal	Did the independent variable, and nothing else, cause the change in the dependent variable? Raised by controlling extraneous variables, the wording the decks use
External	Do the findings generalise to other people, settings and times? The experimental limitation named generalizability
Construct	Does the instrument actually capture the abstract concept it claims to, for example "engagement" rather than mere attendance?
Content	Do the items cover the whole domain of the concept, with nothing important missing? Judged by expert review
Criterion	Does the measure agree with an accepted external benchmark, either at the same time (concurrent) or in the future (predictive)?
Face	Does it simply look like a sensible measure to a lay reader? The weakest form, but it affects response rates

THREE TYPES OF RELIABILITY SUPPLEMENTARY

Test-retest	Same instrument, same people, two time points. Stability over time
Inter-rater	Two or more independent observers or coders score the same material. Agreement between people. This is what coding reliability thematic analysis tests
Internal consistency	Do the items in one scale move together? Split-half, or a coefficient across all items. No coefficient value appears in the course

THREATS TO INTERNAL VALIDITY SUPPLEMENTARY

Each threat is a rival explanation for your result, which is exactly what a control group rules out.

- History. An outside event happens between pretest and posttest

- Maturation. Participants simply grow, tire or learn over the study period

- Testing. Sitting the pretest itself changes the posttest. This is the **pretest effect** the Solomon Four-Group design exists to control

- Instrumentation. The measure or the observer drifts between measurements

- Regression to the mean. Extreme scorers naturally move toward average on retest

- Selection. The groups differed before the treatment. Named in the decks as "pre-existing differences between groups"

- Mortality or attrition. Participants drop out, and not at random

- Diffusion of treatment. The control group finds out about and adopts the intervention

- Hawthorne effect. Being observed changes behaviour

- Placebo effect. Named in the decks: control groups let researchers "rule out alternative explanations for the observed effects, such as placebo effects"

- Extraneous variables. The decks' own umbrella term. "Researchers need to control for as many extraneous variables as possible to increase the internal validity of the study"

Ch3 Managing research, timeline, ethics

Ethics comes before data collection because harm cannot be undone afterwards.

3.1 MANAGING RESOURCES AND TIMELINE, THREE COMPONENTS

Technology and Equipment	Instruments, databases, experimental apparatus, plus logistics preparation for field investigations
Timetable and Milestones	Attainable benchmarks. Tools: Gantt charts, Trellis, Asana. The Gantt chart is required for project section 3.5
Contingency Planning	Extra time in the budget, recognise hazards such as lack of resources or problems accessing data, and prepare contingency measures

3.2 ETHICS OF RESEARCH

Definition: "Ethics in research is a set of norms of conduct in doing research, a set of moral principles and rules concerning the behaviour of the researcher towards the researched, their psycho-physical properties, and the respect of their socio-cultural context." Principles are **universal** (based on respect for human and collective rights) and **specific** (characteristic for a given society, group, nation, minority). Sieber (1993, p. 14): ethics aims to prevent "harming others, promoting goodness, showing respect and being guided by justice." Note: "There is no single guideline or a single list of principles of research ethics."

FIVE CORE PRINCIPLES

1. **Respect for Autonomy.** "Participants must voluntarily agree to be part of the study. They should have the freedom to withdraw at any time without penalty."

2. **Informed Consent.** "Researchers must provide a clear and comprehensive explanation of the study's purpose, procedures, potential risks, and benefits. Consent forms must be written in plain language, and consent should be documented."

3. **Confidentiality and Privacy.** "Personal information must be protected. Use anonymized data wherever possible, and ensure that only authorized personnel have access to sensitive information."

4. **Minimizing Harm.** "Researchers must strive to minimize any physical, psychological, or social risks. This includes avoiding deception unless it's justified and approved by an ethics committee."

5. **Transparency and Integrity.** "Maintain honesty in data collection, analysis, and reporting. Any conflicts of interest must be disclosed, and findings should be shared transparently."

Four types of harm (Pritha Bhandari, Scribbr): psychological, anxiety or negative feelings - social, stigmatization or embarrassment - physical, discomfort or injury - legal, privacy violation from disclosure of private information. "Doing no harm" must be considered prior to the start of the study.

Module 1 Introduction, basic concepts, paradigms, validity, ethics | Module 2 Literature review
Sheet 2 carries Module 3: scales, questionnaires, sampling, Ch11 experiments, Ch12 simulation, Ch13 interpretation, Ch14 proposal.

REVISION SHEET

Ch3 Informed consent and ethics approval

Consent is documented because the participant, not the researcher, owns the decision to take part.

INFORMED CONSENT

Definition: "a process by which a potential adult or child participant voluntarily expresses their willingness to take part in a research study, participants will receive all crucial information prior to the research, including potential risks and how the results will be used, so that the decision to participate is based on complete information about the study. An informed consent form has to be signed." Under 18: parental consent must be obtained, and where possible the consent of the child should also be obtained.

ELEMENTS OF THE CONSENT LETTER, THE SLIDE TEMPLATE

1 Research title

2 Duration of the research

3 Unit, institution or organisation implementing the research

4 Person responsible: name, surname, email address

5 Declaration that the participant has received and understood all the information supplied

6 Freely gives consent, aware they may withdraw at any time without providing a reason

7 Consent to the processing of personal data per the named article / GDPR

8 Consent to processing with anonymity, statements and image, preventing identification by third parties

9 Date and signature

SIX KEY RIGHTS OF PARTICIPANTS

- To reflect before deciding to participate

- To refuse to participate without suffering any consequences

- To refuse to answer certain questions

- To withdraw from the study at any time

- To a guarantee of the security of data obtained, recordings and notes

- To be anonymous. "The general rule is not to collect names"

CONFIDENTIALITY ANONYMITY

"the identity of the participant is known only to us, as researchers", and access to the collected data is given only to persons authorised to analyse them. Sensitive data covers race, religion, ethnicity, sexual orientation, politics, health

"the research participant does not have to reveal their identity, and it remains unknown even to the researcher". It "rarely occurs in qualitative research when the researcher has direct contact". Achieve it by encoding the name or assigning a pseudonym

3.3 SEEKING APPROVAL FROM THE ETHICS COMMITTEE

Approval exists so that an independent body, not the researcher, judges the risk.

- "Ethical board members have to be informed when such a board operates at the university, and its opinion about research ethics is required."

- All crucial information on aim, approach, methodology and data collection must be presented to participants, the research supervisor and ethical board members or reviewers.

- "It has to be included in the research project application submitted to the research supervisor before the start of the project."

- Institutional review boards (IRB) are named for quantitative and for mixed methods research. Ch14 requires the proposal to "discuss any ethical review processes or approvals that need to be obtained".

Vulnerable groups and images. Children need written guardian consent covering purposes, methods, data protection and use, and the child's own consent should also be sought. For Indigenous Peoples, First Nations and ethnic minorities, guidelines are reviewed and updated by universities and the communities themselves. Never publish images without consent: "every time we are going to take a photograph of a child, we must ask the child for permission." Do not publish uncoded images showing participants negatively, exposing emotional state or unclothed.

M2 Literature review, purpose and search

The literature review exists to prove the gap is real before you spend a year filling it.

THREE DEFINITIONS

1. "A written systematic summary of journal articles, books and other documents that describes the past and current states of information."

2. "the presentation, classification and evaluation of what other researchers have written on a particular subject."

3. Turton (2005): "A research that reviews, critiques and synthesizes representative literature on a topic in an integrated way such that new frameworks or perspectives are generated."

Process as well as outcome. Outcome: It appears in the final draft as part of the introduction or as a separate chapter. Process: "critical engagement (thinking, reading and writing) with relevant research on your topic."

HART'S (1998) EIGHT PURPOSES

- Distinguishing what has been done from what needs to be done

- Discovering important variables relevant to the topic

- Synthesizing and gaining a new perspective

- Identifying relationships between ideas and practice

- Establishing the context of the topic or problem

- Understanding the structure of the subject

- Identifying the main methodologies and research techniques that have been used

- Placing the research in a historical context

Seven functions. It establishes the terms and context - presents a survey of preceding literature - explores ways others have solved similar questions - outlines the relationship of these texts to each other - evaluates the quality and relevance of the literature - establishes the gaps or inadequacies - demonstrates your scholarly rigor.

THE FIVE STAGE SEARCH PROCESS

PLANNING A LITERATURE REVIEW

1 Problem formulation which topic or field, component issues

2 Literature Search finding materials, use keywords

3 Evaluation, Select and Manage which literature makes a significant contribution

4 Analysis and Interpretation discussing findings and conclusions of pertinent work

5 Writing and Referencing

SEARCH STRATEGIES

Databases: IEEE Xplore, ScienceDirect, Springer, Elsevier, Scopus, ACM Digital Library, Google Scholar. Off campus you need a proxy; ProQuest Dissertations and Theses sits in the library online database.

BOOLEAN AND SYNTAX RULES, EXACTLY AS TAUGHT

AND OR NOT the logical connectors that combine words

"flexible learning" quotation marks find a specific multi-word phrase

+e-government +taxes the addition sign means the word MUST be included

PRIMARY VERSUS SECONDARY SOURCES

PRIMARY	SECONDARY
Empirical research reports - Theoretical articles - Autobiographies - Conference papers - Theses	Review articles - Newspapers - Magazines - Textbooks - Encyclopedias

5.1 MANAGING THE LITERATURE

Reference managers exist so that citations and the reference list can never drift apart.

Mendeley

Free reference manager, builds the library, inserts citations and generates the reference list. Also named as a productivity tool in Ch14

EndNote

The other named manager, same role, used with the word processor

5.2 Note taking and the summary table. The tool named is the literature review matrix, a summary table with one row per article and columns for author, year, method, sample, findings and gap, plus a mind map to connect concepts and arguments back to the research purpose.

5.3 Analysing and evaluating, four tasks: summarizing, synthesizing, critiquing and comparing. "Analysis: you see how one part relates to another. Synthesis: you bring things together."

Other tools named: scite ai - Connected Papers - Elicit.org for related articles - Pop AI - ChatGPT - Jenni.ai and Copilot for paraphrasing. "SLR, do the systematic literature review."

M2 Writing the review, citation, plagiarism

Writing is separated from searching because a pile of summaries is not a review.

6.1 STRUCTURE

Organise the review chronologically, thematically, by trends, approaches or techniques, or by major debates and controversies, and "probably a combination of these".

PARAGRAPH SHAPE

Topic sentence → discussion of the topic → closing sentence with thematic and grammatical links, and logical progression

6.2 SUMMARISING, SYNTHESISING, PARAPHRASING

Summarising

Restating the main points of one source in shorter form, in your own words, still cited

Synthesising

Bringing several sources together so they speak to each other and a new perspective emerges. This is what earns marks

Paraphrasing

Restating a specific passage in your own words and sentence structure, same length, always cited

The take-home assignment states the standard: at least 5 articles, at most 5 pages, synthesis not summary, a coherent argument, an introduction, discussion and conclusion, in-text documentation and a reference list.

CONNECTIVES THAT SIGNAL THE MOVE

PURPOSE	CONNECTORS
Agreement	Moreover, Further, Furthermore, Additionally, also, and
One idea leads to another	Hence, as a result, therefore, accordingly, consequently
Difference	However, but, in spite of this, Nevertheless, on the other hand, Although, yet

Exam cue from the lecturer: "word however show arguments." The word however is where the gap is announced in a literature review paragraph. Find it, and you have found the writer's contribution claim.

6.3 IN-TEXT CITATION AND REFERENCING

The style is APA, and there are exactly two in-text forms.

FORM A, EMPHASIS ON THE AUTHORS

Muhammad and Grundy (2015) compared between ...

FORM B, EMPHASIS ON THE INFORMATION

The black seeds need to be fermented (Grundy, 2015).

Tense: "When reporting related works, it is suggested that past-tense or present perfect tense verb is used." Rule from the lecturer: "Whatever you cite needs to be in the reference list."

6.4 ISSUES OF PLAGIARISM

Instruction, verbatim: "Ensure that you have enclosed in quotation marks and cited all works and ideas that are not yours. Avoid plagiarism."

Six reporting rules. Be honest in reporting your data - be transparent - do not plagiarise other authors - do not self-plagiarise - do not falsify or fabricate data or results - do not present unreliable or misrepresented data.

Named misconduct cases from the ethics lecture: A technician computer scientist accused of plagiarism in at least eight papers - a mathematician banned from several journals for a word-for-word copy - IEEE's 2012 "Notice of Violation of IEEE Publication Principles" - a University of Florida professor who fabricated results and pressured his PhD student,

Research Methodology

HOW YOU MEASURE, WHO YOU ASK, WHAT YOU CAN CLAIM, AND HOW YOU WRITE IT UP

REVISION SHEET

Everything on this sheet answers one question: **can I believe this number?** The **scale** decides which statistic is even legal. The **questionnaire** decides whether the answer means what you think. The **sample** decides who the answer applies to. **Validity and reliability**, on sheet 1, decide whether it is the right answer, consistently. **Experiments** buy causality with randomisation and a control group, **simulation** buys it with a model when the real world is too costly or unethical, and **Ch14** is the document where you promise all of it in advance. **Honesty marker**: items tagged **SUPPLEMENTARY** are standard textbook content the lecture decks never covered. Learn them, but do not cite them as course material.

Ch13

Measurement scales

The scale exists because it fixes, in advance, which statistics you are allowed to compute.

SCALE	DEFINITION AS STATED	EXAMPLES	LEGAL DESCRIPTIVES
Nominal	"Categories without order"	Gender, ethnicity (both named as demographic variables) - spam or not spam - department	Frequency distribution, Mode, cross-tabulations
Ordinal	"Categories with a meaningful order"	Likert agreement levels, ranks, education level, promoter or passive or detractor	Frequency distribution, Median, Percentile, Mode, cross-tabulations
Interval	"Numeric data with equal intervals"	Temperature in Celsius, calendar year, standardised test scores	Mean, Median, Mode, Variance, SD, Quartiles, Correlation, Regression
Ratio	Same, and "ratio also has a true zero"	Age, Income, reaction time, website response time, count of samples	All of the above, plus meaningful ratios such as "twice as long"

Know this exam trap. Ch13 groups **interval and ratio as one row**, so the course's operative distinction is three-way: **nominal, ordinal, interval-ratio**. That is why the big matrix below has three columns, not four. The only thing separating interval from ratio is the **true zero**.

MEMORY HOOK

nominal → counts, chi-square

ordinal → ranks, non-parametric tests

interval-ratio → means, parametric tests

LIKERT GUIDANCE SUPPLEMENTARY

Likert items sit awkwardly between ordinal and interval, so the rules below decide how you may analyse them. No deck in this set gives Likert construction guidance.

- Use **5 or 7 point balanced scales** with a labelled midpoint
- Label every point**, not just the two anchors
- Keep each item **unidimensional**, avoid double-barrelled and leading wording
- Reverse-code** some items to detect acquiescence, agreeing with everything
- A single Likert **item is ordinal**: median, mode, non-parametric tests per the matrix
- A summated multi-item **scale** is treated as **interval** for parametric analysis
- Report **internal consistency** for any summated scale

Ch9

Questionnaire development

The questionnaire is the standardised quantitative instrument, so it must be built before any data exists.

What the decks do state. Quantitative data collection is by **questionnaire**, **standardised interviews** and **tightly structured observation**, with the **sample frame fixed before research starts**. Qualitative data instead come from **open-ended**, **semi-structured**, **in-depth interviews**. Warning given in Week 4: "the inclusion of a section for comments on a quantitative survey does **not** constitute qualitative research."

Honest gap. Syllabus items **9.2 questionnaire development steps** and **9.3 types of questions** are listed in the teaching plan but **no surviving slides cover them**. The steps and types below are the standard version, marked **SUPPLEMENTARY**.

EIGHT DEVELOPMENT STEPS SUPPLEMENTARY

- Define the construct.** Work back from the research objectives and the conceptual framework, one construct per objective
- Specify the target respondent** and the sampling frame, since wording depends on who reads it
- Generate the item pool**, usually adapted from validated instruments found in the literature review
- Choose question types and the response format**, including the scale points
- Word and sequence the items.** Easy and general first, sensitive and demographic last, related items grouped
- Expert review for content validity**, then revise
- Pilot study.** The positivistic chain on sheet 1 names **pilot study** as an explicit stage. Check comprehension, timing and internal consistency
- Finalise, administer, and document the instrument** in proposal section 3.4

TYPE OF QUESTION SUPPLEMENTARY

TYPES	WHAT IT IS, AND WHEN TO USE IT
Open-ended	Respondent writes freely. Use when the answer set is unknown. Rich but slow to code
Closed-ended	Fixed options. Fast, comparable, and the default for quantitative surveys
Dichotomous	Two options only, yes or no, male or female. Nominal
Multiple choice	One answer from a list. Nominal unless the list is ordered
Checklist, multiple response	Select all that apply. Each option is a separate binary variable
Likert or rating scale	Degree of agreement, satisfaction or frequency. Ordinal per item
Ranking	Put options in order of preference. Ordinal, and tiring beyond about five items
Semantic differential	A scale between two opposite adjectives, for example slow to fast
Matrix or grid	Many Items sharing one response scale. Saves space, risks straight-lining
Contingency or filter	Asked only if a previous answer qualifies the respondent
Demographic	Age, gender, ethnicity, socioeconomic status. Ch13 uses these for the demographic characteristics analysis

Wording faults to avoid: double-barrelled (two questions in one) - leading or loaded wording - jargon and abbreviations - double negatives - overlapping response categories - missing categories with no "other" - asking for recall the respondent cannot have.

Ch9

Sampling

Sampling exists because you cannot measure the whole population, and it decides who your findings apply to.

VOCABULARY

Population every unit you want to say something about

Sample frame the actual list you can draw from

Sample the units you measure

Sampling the rule that turns the frame into the sample

WHAT THE DECKS ACTUALLY STATE

- Quantitative**: "randomised sampling, sample frame fixed before research starts" - "Participants should be selected **randomly** to ensure that the sample is representative of the population" - random sampling is named as a measure to ensure validity and reliability
- Qualitative**: "**purposive and theoretical** sampling, flexible sampling that develops during research" - "Sample sizes in qualitative research are typically small, **10 to 20 subjects**, and the participants or sites are usually **purposely sampled**, selected on the basis of their knowledge or experience of the research phenomenon"
- A worked telecom abstract names **random sampling, purposive survey and multistage sampling**
- Syllabus item **11.8 stratification, also known as blocking** is named for experiments

Honest gap.

No deck in this set enumerates sampling techniques. The full taxonomy below is the standard one, **SUPPLEMENTARY** except for **random, purposive, theoretical, multistage and stratification**, which are named in the materials.

PROBABILITY SAMPLING

Every unit has a known, non-zero chance of selection, so sampling error can be estimated and results can be **generalised**. This is the quantitative and positivist family.

TECHNIQUE	HOW IT WORKS	USE IT WHEN
Simple random	Every unit has an equal chance, drawn by lottery or random numbers	You have a complete frame and the population is fairly homogeneous
Systematic	Pick a random start, then take every k-th unit, where k = population divided by sample size	The frame is a long ordered list and simple random is impractical. Avoid if the list has a hidden cycle
Stratified (blocking)	Split the population into strata that differ, then sample randomly within each. Proportionate or disproportionate	A known characteristic such as faculty, gender or region strongly affects the outcome and must be represented
Cluster	Split into naturally occurring groups, randomly choose whole clusters, measure everyone inside	The population is geographically spread and listing every individual is impossible
Multistage	Cluster sampling repeated in stages, for example state, then district, then school, then pupils	Large national studies. Named in the Week 5 worked example

NON-PROBABILITY SAMPLING

Selection chance is unknown, so statistical generalisation is not claimed. This is the qualitative and interpretive family.

TECHNIQUE	HOW IT WORKS	USE IT WHEN
Convenience	Whoever is available and willing	Pilot testing and exploratory work where speed matters more than representativeness
Purposive or judgmental	The researcher deliberately selects units "on the basis of their knowledge or experience of the research phenomenon"	Qualitative studies needing informed participants. The default qualitative technique in this course
Theoretical	Successive cases are chosen by what the emerging theory needs next, until saturation	Grounded theory . Named alongside purposive in the Week 9 table
Quota	Fill fixed counts per category, but choose within the category non-randomly	You need category balance without a usable sampling frame
Snowball	Existing participants recruit further participants	Hidden or hard to reach populations, the "difficult to access populations" qualitative research is said to target
Self-selection	Participants volunteer in response to an invitation	Open online surveys. Carries strong volunteer bias

Sample size cues in the materials: qualitative **10 to 20 subjects** - the "achievable objective" example uses **50 participants** - true experimental step 2 says the sample must be **"large enough to detect a meaningful difference"**. No numeric formula, power calculation or alpha level appears anywhere in the course.

Ch10

Qualitative method, instrument and sampling

The qualitative chapter exists because some questions have no number that could answer them.

Nelson et al. (1992, p. 4): "Qualitative research is an interdisciplinary, transdisciplinary, and sometimes counter disciplinary field. It crosses the humanities and the social and physical sciences. It is multiparadigmatic in focus. Its practitioners are sensitive to the value of the multimethod approach. They are committed to the naturalistic perspective, and to the interpretative understanding of human experience."

WHAT IT IS

"an exploratory method of enquiry that typically involves investigating unknown subject matter, sensitive topics, or difficult to access populations". Researchers do **not start out with a hypothesis to test**

WHAT IT IS NOT

"not descriptive research using a qualitative method, and the inclusion of a section for comments on a quantitative survey does not constitute qualitative research"

10.3 Instrument design.

The instrument is the interview guide, not a fixed form. Data come from **open-ended, semi-structured, in-depth interviews**, focus groups, participant observation, ethnographic study and **projective techniques**. Sampling is **purposive and theoretical**, and it **develops during the research** rather than being fixed in advance.

QUALITATIVE RESEARCH AS A PROCESS

Theory to Method to Analysis all three interconnect

WHY IT IS POPULAR	ITS LIMITATIONS
Cheaper than quantitative - no better way to understand in-depth motivations and feelings - improves the efficiency and effectiveness of quantitative research	Does not distinguish small differences as well as quantitative can - not representative of the population of interest - the multitude of individuals who without formal training profess to be experts

Aims by researcher type: qualitative researchers explore **behaviour, perspectives, feelings and experiences** - **ethnographers** focus on culture and customs - **grounded theorists** investigate social processes and interaction - **phenomenologists** consider the meanings of experience and describe the life world.

Ch11

Experimental design, variables and components

Experiments exist for one reason: they are the only design that can test cause and effect.

Definition: "Experimental research design is a type of research design in which a researcher **manipulates one or more variables to observe the effects on an outcome variable, while controlling for other variables** that might affect the outcome. This design is commonly used in natural and social sciences to test cause-and-effect relationships."

FIVE KEY COMPONENTS

- Independent Variable (IV)**, "the variable that is manipulated by the researcher". **Dependent Variable (DV)**, "the variable that is measured to observe the effects of the manipulation". Caffeine is the *IV*, attention is the *DV*.
- Control Group (CG)**, "a group of participants who are similar to the experimental group in every way **except that they do not receive the experimental manipulation**. The purpose of the control group is to **rule out alternative explanations** for the observed effects of the manipulation."
- Random Assignment**, "the process of assigning participants to either the experimental or control group at random. This helps to ensure that any differences between the groups are **due to chance rather than systematic differences** between the groups."
- Experimental Manipulation**, "the specific intervention that the researcher is interested in testing", either **physical** (administering a drug) or **psychological** (providing feedback).
- Outcome Measures**, "the specific measures used to assess the effects of the experimental manipulation on the DV". A computer-based reaction-time task.

Three worked examples: new medication to blood pressure, control gets a placebo - new teaching method to test scores, control gets traditional teaching - social media to self-esteem, control does not use social media.

THE FOUR VARIABLE ROLES

Worked example from the lecture: firm performance is dependent, technology adoption is independent, firm size and training are mediator or moderator. The tell for a mediator is the word through.

Confounding variables are never named as such in these decks.

The term used throughout is **extraneous variables**: "Researchers need to control for as many extraneous variables as possible to **increase the internal validity** of the study." A confounder is an extraneous variable that varies with the IV, so it offers a rival explanation for the result. **Randomisation** is the general defence: it distributes unknown extraneous variables evenly across groups, so the only systematic difference left is the treatment.

NINE STEPS IN SETTING UP EXPERIMENTAL RESEARCH

WORKED THROUGHOUT ON A MALWARE TAXONOMY EXAMPLE

- Research problem identification
- Hypothesis development
- Selection of participants
- Design of the experiment
- Randomization and control
- Data collection
- Data analysis
- Interpretation of results
- comparing control group with experimental group
- Conclusion

Hypothesis used:

"there is a significant relationship between the type of malware and the method of propagation." Participants: cybersecurity experts. Extended example: signature-based versus behavior-based malware detection.

STEP LISTS PER FAMILY, MEMORISE THE DIFFERENCES

FAMILY	STEPS, WITH WHAT CHANGES MARKED
Pre-experimental, 5	RQ and hypothesis, select participants and sample size, select the treatment , measure DV before and after, analyze and draw conclusions
True experimental, 7	RQ and hypothesis, select participants (randomly, representative, large enough to detect a meaningful difference), randomly assign to treatment or control, administer treatment (standardized and consistent) , measure DV before and after, analyze, draw conclusions
Quasi-experimental, 7	As true, but participants are selected on pre-existing characteristics or group membership , and it adds control for extraneous variables , because quasi designs are "more susceptible to extraneous variables"
Field experiment, 8	RQ and hypothesis, choose the field site , select participants, administer treatment, measure DV, control extraneous variables, analyze, draw conclusions
Laboratory experiment, 7	RQ and hypothesis, choose the sample , manipulate the IV , measure DV, control extraneous variables, analyze, draw conclusions

Three limitations of experimental research. **Artificiality**, manipulations may not reflect real-world situations. **Ethics**, may involve manipulating variables potentially harmful to participants. **Generalizability**, findings may not generalize to other populations or situations.

Ch11

The five design families

The families are ranked by what they have, because what they have is exactly what they can claim.

FAMILY	DEFINING FEATURE	NAMED SUB-TYPES, IN ORDER
Pre-experimental	Lacks one or more key components: no control group and/or no random assignment . Simplest	1 One-Shot Case Study - 2 One-Group Pretest-Posttest - 3 Static Group Comparison
True experimental	All key components: control group, random assignment , IV manipulation. "the gold standard for experimental research because they include all the key components necessary for establishing causality"	1 Pretest-Posttest Control Group - 2 Posttest-Only Control Group - 3 Solomon Four-Group
Quasi-experimental	IV manipulation plus a comparison group but no random assignment , uses existing groups such as two classrooms or two hospitals. "less rigorous than true experimental designs due to the lack of random assignment"	1 Non-equivalent Control Group - 2 Time Series - 3 Interrupted Time Series
Field experiments	Real-world setting, "more ecologically valid"	1 Natural Experiments - 2 Quasi-Field Experiments
Laboratory experiments	Controlled environment, precise manipulation	1 Pretest-Posttest Control Group - 2 Posttest-Only Control Group - 3 Solomon Four-Group

Choice rule, stated:

"the choice of design will depend on the **research question**, the **resources available**, and the **ethical considerations** involved."

EVERY DESIGN, DEFINED AND DISCRIMINATED

DESIGN	R	CG	PRE	WHAT IT IS
One-Shot Case Study	no	no	no	"a single group is studied after they have been exposed to a treatment". Exploratory, or when control groups are impossible. Give one group new software, then measure performance
One-Group Pretest-Posttest	no	no	yes	"taking measurements before and after a treatment to a single group. There is no control group". Problem-solving pre-test, computer game, post-test
Static Group Comparison	no	yes	no	"two groups are compared, but the groups are not randomly assigned. One group receives the treatment, the other does not. There is no pre-test". Lecture versus interactive programming instruction
Pretest-Posttest Control Group	yes	yes	yes	Two groups, both measured before and after, only the experimental group receives the treatment
Posttest-Only Control Group	yes	yes	no	Two groups, both measured only after, "used when it is not feasible to measure the outcome before the treatment"
Solomon Four-Group	yes	yes	half	"combines both the pretest-posttest control group design and the posttest-only control group design. There are four groups: two experimental and two control." One experimental and one control get a pretest, the others do not. Purpose: to control for the effects of the pretest on the outcomes
Non-equivalent Control Group	no	yes	yes	Groups not randomly assigned, one receives treatment, the other is control. Two groups taught different programming languages
Time Series	no	no	many	"multiple measurements are taken over time, both before and after the intervention", task completion time before and after a new tool
Interrupted Time Series	no	no	many	"includes a discrete intervention at a specific point in time , followed by multiple measurements before and after". Mandatory two-factor authentication, track breach frequency
Natural Experiment	no	yes	-	"the treatment or intervention is not directly manipulated by the experimenter, but rather occurs naturally". A new traffic signal, a new search engine algorithm
Quasi-Field Experiment	no	yes	-	"the treatment is manipulated by the experimenter, but the participants are not randomly assigned ". A new program in one school district, volunteer programmers

PRE-TEST AND POST-TEST NOTATION SUPPLEMENTARY

The slides describe every design in prose only. This shorthand is the conventional Campbell and Stanley notation, and it makes the discriminators visible at a glance.

R = RANDOMLY ASSIGNED, O = OBSERVATION, X = TREATMENT, HYPHEN = NO TREATMENT

One-Shot Case Study X O

One-Group Pretest-Posttest O1 X O2

Static Group Comparison X O1

Pretest-Posttest Control R O1 X O2

R O3 - O4

Posttest-Only Control R X O1

R - O2

Solomon Four-Group R O1 X O2

R O3 - O4

R - X O5

R - - O6

Non-equivalent Control Group O1 X O2 (no R)

O3 - O4

Interrupted Time Series O1 O2 O3 X O4 O5 O6

Randomisation is what turns the second block into the fourth: same structure, but R removes selection bias, so the groups are equivalent at baseline and the only systematic difference left is X.

Syllabus items named without slide content: 11.2 purpose of control - 11.3 formulate the question or goal in advance - 11.4 comparison and control - 11.5 replication - 11.7 randomization - 11.8 stratification, also known as blocking, 11.9 factorial experiments, which cross two or more IVs so that main effects and interactions can both be estimated.

Ch12

Simulation research design

Simulation exists for the experiments you are not allowed, cannot afford, or cannot wait for.

Definition: "Simulation in research design refers to the use of **computer models or software to imitate real-world phenomena or systems**. It involves creating a virtual environment or model that represents a particular system or situation and then conducting experiments or observations within that virtual environment." It lets researchers study systems "difficult or impractical to study directly in the real world" and is "particularly valuable when conducting experiments or studies that involve **high costs, risks, ethical concerns, or time constraints**". Fields: engineering, economics, social sciences, healthcare, computer science.

FIVE TYPES OF SIMULATION-BASED RESEARCH

TYPE	DEFINITION	TOOLS, IN THE ORDER GIVEN
Agent-based	"creating individual agents with specified characteristics and observing how they interact with each other and with their environment"	NetLogo, Repast, AnyLogic, MASON, GAMA, Swarm
Discrete event	"modelling the behavior of a system over time, taking into account events that occur at discrete points in time "	SimPy, Arena, Simio, AnyLogic, ExtendSim, GPSS World
Monte Carlo	"using random numbers to simulate the behavior of a system or process"	MATLAB, R, Python, Crystal Ball, Palisade @RISK, Crystal Ball
System dynamics	"modeling the behavior of a complex system over time, taking into account the interrelationships between different components "	AnyLogic, Vensim, Stella, Powersim Studio, Simio
Gaming	"creating a game-like environment in which participants can interact with each other and with the simulation to explore different scenarios and outcomes"	Unity, Unreal Engine, Blender, CryEngine, Simio, Construct

Applications by type.

Agent-based: traffic, social network analysis, economic modelling, epidemiology, AI and swarm robotics. Discrete event: manufacturing and supply chain, queuing systems, computer networks, project management, healthcare patient flows. Monte Carlo: risk analysis, optimization, performance evaluation, sensitivity analysis, option pricing. System dynamics: IT infrastructure planning, software development processes, supply chain, IT project management, technology adoption and diffusion. Gaining: training and education, VR and AR, serious games, game-based learning.

Memory fact: AnyLogic appears in three types and Simio in four. Six network simulators: NS-3 · OPNET, now Riverbed Modeler · OMNeT++ · QualNet, wireless focus · GloMoSim · J-Sim. PhD exemplar: "Optimizing The Search Of Access Control Lists For IP Packet Filtering Applications", Seyed Mohamed Buhari, MMU, 2003, using MATLAB, OPNET, NS and Kappa-PC.

Ch12

Simulation steps, validation, verdict

The 12 steps exist because a model that was never validated is an opinion with a graph.

THE 12 STEPS OF A SIMULATION STUDY

ORDERED, HIGH PROBABILITY EXAM LIST

- Define the research problem
- Determine the research objectives
- Select the appropriate simulation method
- DEVELOP A CONCEPTUAL MODEL, key components and relationships of the system
- DEFINE THE SIMULATION MODEL, detailed structure and parameters: variables, rules, constraints, interactions
- Gather input data
- Implement the simulation model
- VALIDATE THE SIMULATION MODEL
- Conduct simulation experiments
- Analyze and interpret the results
- Draw conclusions and implications
- Communicate and report the research

VERIFICATION VERSUS VALIDATION

Validation, verbatim: "Validate the simulation model by comparing its outputs to real-world observations or other established benchmarks. This step ensures that the model behaves as expected and accurately represents the system being simulated." Also: "comparing its behavior and outputs with real-world data, empirical evidence, or expert knowledge. **Calibration and sensitivity analysis** can be conducted to ensure the accuracy and reliability of the model."

Validation, verbatim: "Validate the simulation model by comparing its outputs to real-world observations or other established benchmarks. This step ensures that the model behaves as expected and accurately represents the system being simulated." Also: "comparing its behavior and outputs with real-world data, empirical evidence, or expert knowledge. **Calibration and sensitivity analysis** can be conducted to ensure the accuracy and reliability of the model."

STRENGTHS, FIVE ADVANTAGES

- Control and repeatability.** Repeat experiments with identical conditions to validate results
- Cost and time efficiency.** A fraction of the cost and time of real-world experiments
- Manipulation of variables.** Easily manipulate variables to observe impact
- Risk reduction.** Test hypotheses without real-world risk
- Complex system analysis.** Numerous interconnected variables, observe emergent behavior

WEAKNESSES, SIX LIMITATIONS

- Simplified representation of reality.** Accuracy relies on the quality of the simplifications and assumptions
- Uncertainty and validation.** Validating against real-world data is challenging and time-consuming
- Limited scope of application.** Most effective for **well-defined** systems, hard for poorly understood ones
- Incomplete knowledge representation.** Gaps lead to overlooked factors and **misleading or biased conclusions**
- Computation and resource requirements.** Time, power, expertise
- Ethical and practical constraints.** May not capture all considerations, such as human decision-making

Ten key challenges: model validity · data collection · calibration and parameter estimation · computational resources · model complexity · interpretation of results · replicability and reproducibility · validation with real-world data · ethical considerations · generalizability.

Twelve tips: clearly define objectives · conduct a literature review · develop a conceptual model · validate the model · document assumptions and limitations · use realistic input data · **conduct sensitivity analysis** · **ensure sufficient replications**, to account for randomness and variability · analyze and interpret results · communicate findings effectively · document and share the model · seek peer feedback.

Ch13

Data interpretation

Interpretation exists because raw output is not a finding until someone says what it means.

Tool: SPSS, "a widely used tool for statistical analysis", supporting categorical and continuous data.

13.1 THREE TYPES OF ANALYSIS COVERED

Demographic characteristics	Variables such as age, gender, ethnicity, socioeconomic status. Helps understand the composition of the sample population
Descriptive	"summarizes key features of the dataset", using measures of central tendency, variability, and distribution
Inferential, hypothesis testing	"used to test relationships and differences in data; determines whether results are statistically significant "

13.2 TO 13.4 THE THREE PART CYCLE

Data reduction > Data display > Conclusion drawing

- Data reduction:** "Involves cleaning, sorting, and filtering data; removes irrelevant or redundant information; improves clarity and accuracy of analysis"
- Data display:** "uses tables, charts, and graphs to present data visually; enhances communication and understanding of findings"
- Conclusion drawing:** the third move. Qualitative analysis "focuses on non-numeric data, involves data reduction, data display, and drawing conclusions, helps identify **themes and patterns**"

SEVEN TYPES OF DATA ANALYSIS

TYPE	DESCRIPTION	EXAMPLES
Descriptive	Summarizes main features using mean, median, mode, standard deviation	Average website response time, summarizing respondent demographics
Inferential	Draws conclusions about a population from sample data , via hypothesis testing and statistical significance	Differences in user satisfaction between two software versions
Machine learning	Identify patterns and make predictions without explicit programming instructions	Spam classification, stock price prediction
Network analysis	Analyzes relational data : structure, behavior, dynamics of interconnected entities	Communication patterns in social networks, network vulnerabilities
Text mining and NLP	Extracts insights from unstructured text	Sentiment analysis of reviews, topic modeling
Data visualization	Presents data through charts, graphs, maps	Bar charts for sales trends, heatmaps
Time series	Data over time for trends, patterns, seasonal effects , forecasts	Forecasting monthly sales, daily stock price trends

Ch13

The scale by technique matrix

This grid is the single densest examinable object in the course, so learn it as a grid, not as a list.

THE BIG MATRIX, TECHNIQUE BY SCALE

This grid exists so that the scale you chose at design time tells you exactly which test to run. Reading left to right, more scale information buys more powerful tests.

	NOMINAL	ORDINAL	INTERVAL / RATIO
Descriptive	Frequency Distribution; Mode; Cross-tabulations	Frequency Distribution; Median, Percentile; Mode; Cross-tabulations	Mean, Median, Mode, Variance, SD, Quartiles; Correlation; Regression Analysis
Inferential tests	Chi-Square; McNemar's; Fisher's Exact	Mann-Whitney U; Wilcoxon Signed-Rank; Kruskal-Wallis; Spearman's Rank Correlation	Independent Samples t-test; Paired Samples t-test; One-way ANOVA; Pearson Correlation
Machine learning	Decision Trees; Naive Bayes; Support Vector Machines	Ordinal Regression; Ordinal Logistic Regression; Ordinal Random Forest	Linear Regression; Multiple Regression; Generalized Linear Models; ANOVA with post-hoc tests
Network analysis	Node Degree Distribution; Modularity	Rank Centrality; Kendall's Tau	Edge Betweenness Centrality; PageRank
Text mining, NLP	Word Frequency; Topic Modeling	Ordinal Text Sentiment Analysis	Text Similarity; Named Entity Recognition; Latent Dirichlet Allocation
Visualization	Bar Charts; Pie Charts	Ordered Bar Charts; Cumulative Frequency Diagrams	Histograms; Box Plots; Scatter Plots

Ch13

Qualitative analysis

Qualitative data have no arithmetic, so the analysis is a disciplined route from raw text to themes.

QUALITATIVE ANALYSIS, FILLING 13.4 TO 13.7

APPROACH TO ANALYSING QUALITATIVE DATA (WEEK 5)

Categorisation → Uniting data → Recognising relationships and developing the categories → Developing and testing hypotheses to reach conclusion

The process is INTERACTIVE: collection, analysis, and the development and verification of relationships and conclusions are all interrelated.

Tools: Summaries, Self Memos, Researcher Diary.

Thematic analysis (Braun & Clarke, 2006, p. 79):

"a method for identifying, analysing and reporting patterns (themes) within data. It minimally organises and describes your data in (rich) detail. However, frequently it goes further than this, and interprets various aspects of the research topic."

SIX PHASES OF THEMATIC ANALYSIS

- Familiarisation with the data
- Coding the data
- Generating initial themes
- Reviewing and developing themes
- Refining, defining and naming themes
- Producing the report

Clarke et al. (2019), three approaches: coding reliability (small q) · reflexive (Big Q) · codebook (medium q, template analysis). **Semantic codes** capture surface meaning, **latent codes** capture the assumptions underpinning it. **Themes as topic summaries** (a bucket unified by shared topic) versus **fully realised themes**, "patterns of shared meaning underpinned by a central concept".

13.6 and 13.7 fall back to the validity box on this sheet. Quantitative validity is accuracy of measurement assured by random sampling, standardization and control groups. Qualitative validity is faithfulness to participants assured by member checking, triangulation and peer review. Triangulation closes the chapter. **No p-value cut-off, alpha level, sample-size rule or reliability coefficient appears anywhere in this course.**

Ch14

Proposal writing and presentation

The proposal exists to prove, before you spend the time, that the study is worth doing and can be done.

Definition: "A research proposal is a document that outlines the key elements and components of a research study. It serves as a **blueprint or a roadmap** for the research project and provides a detailed plan for conducting the study." **Purpose:** to present the question to be researched and its importance

Applied Statistical Method

MEASUREMENT SCALE, THE HYPOTHESIS FRAME, AND PICKING EXACTLY ONE TEST

REVISION SHEET

Every question on this paper is the same three questions in disguise. **What scale is the variable on** (nominal, ordinal, interval, ratio), **what shape is the design** (one group, two groups paired or independent, three or more, or a relationship), and **do the assumptions hold** (normality, equal variance, independence). Those three answers name exactly one test, and the rest is arithmetic: **statistic, degrees of freedom, critical value, decision**. The decision line never changes: **$p < \alpha$ reject H_0 , $p > \alpha$ do not reject H_0** .

L8 Measurement scales, the gate on every test

The scale limits which arithmetic is meaningful on the numbers, so it decides the test before any data is seen. "The level of measurement is critical in determining the appropriate multivariate technique to use."

SCALE	WHAT THE NUMBER MEANS	LEGAL SUMMARY	TEST FAMILY
Nominal nonmetric	Label only. The size of the number is unrelated to the amount of the characteristic. 1 = male, 2 = female	Counts, mode, proportions	Chi square
Ordinal nonmetric	Larger numbers mean more or less, but not how much more. Ranks, Likert positions	Median, ranks, percentiles	Nonparametric, rank based
Interval metric	Ordinal plus equal differences between scale points. Zero is arbitrary. Temperature in C	Mean, standard deviation	z, t, ANOVA, Pearson r
Ratio metric	Interval plus a natural zero , so ratios are meaningful. Income, weight, time	Everything, plus ratios	z, t, ANOVA, regression

The gate in two words: **nonmetric** (nominal, ordinal) means you may only count or rank, so nonparametric. **Metric** (interval, ratio) makes means and variances meaningful, so parametric becomes legal.

THE VARIATE, AND MEASUREMENT ERROR

A technique never works on one variable, it works on one weighted combination of them, so the weights are what it actually estimates.

THE VARIATE
 $Y' = W_1X_1 + W_2X_2 + \dots + W_nX_n$
Weights W are chosen empirically to best achieve the objective of the technique. Each respondent has one Y'.

Observed score is true score plus error, so error blurs real relationships and costs power. The remedy taught is the **summed scale**, checked on two things: **validity**, the degree to which a measure accurately represents what it is supposed to measure, and **reliability**, the degree to which the observed variable measures the "true" value and is free of error.

L1 The hypothesis testing frame

A test is a rule for deciding under uncertainty, so both ways of being wrong are named and priced before you look at the data.

H0 AND H1

- **H0, the null**: no difference, no association, or the claimed value holds. It is the statement you can compute a distribution for
- **H1, the alternative**: the research claim. It never gets "accepted", H0 only gets rejected or not rejected
- **One tailed**: the wording says **greater than** (right tail) or **less than** (left tail), one direction only
- **Two tailed**: the wording says **not equal to**, or "is there any difference". **No direction mentioned means two tailed**

THE TWO ERRORS, IN ONE TABLE

	H0 IS TRUE	H0 IS FALSE
FAIL TO REJECT H0	1 - α Correct decision	β Type II error
REJECT H0	α Type I error	1 - β Correct decision, POWER

DEFINITIONS
 α = P(reject H_0 | H_0 true) Type I error
 β = P(fail to reject | H_0 false) Type II error
Power = 1 - β = P(reject H_0 | H_0 false)

POWER AND ITS THREE DRIVERS

Power is the chance of finding an effect that really is there, so every design choice is a power choice.

- Effect size**. The actual magnitude of the effect, a difference between means or a correlation. **Smaller effect sizes always require larger samples**
- Alpha**. As α is set smaller, power decreases. Typical $\alpha = .05$. A more stringent $\alpha = .01$ needs a **larger sample** for the same power; a less stringent $\alpha = .10$ raises power
- Sample size**. As n increases, power increases. **Any increase in power is most likely achieved by increased sample size**

Rules of Thumb 1-1: always design for **power = .80**. **Effect sizes**: r 0.10 small, 0.30 medium, 0.50 large · Cohen d 0.20, 0.50, 0.80 · eta squared = SSB/SST, 0.01, 0.06, 0.14.

Practical vs statistical significance: with very large n even a trivial effect turns significant, so report effect size, not only p.

L1 The decision rules, and reading the tables

Every test on both sheets ends at one of these two lines, so learn them once and reuse them everywhere.

THE DECISION RULES, ALL OF THEM

UNIVERSAL
p-value < α -> REJECT H_0
p-value > α -> DO NOT REJECT H_0
Critical value route, two tailed, $\alpha = .05$:
 $|z| > 1.96$ -> reject ($\alpha/2 = 0.025$ per tail)
 $|z| > 2.58$ -> reject at $\alpha = .01$
One tailed, $\alpha = .05$: $z > 1.645$ (or < -1.645)
Chi square and F: RIGHT TAILED ONLY, reject if statistic > critical value. Example chi-square critical = 7.815 at df = 3, $\alpha = .05$

Reading Table E, the standard normal table: if the critical value you want is **negative**, just look up the alpha value directly. If it is **positive**, look up 1 - α . Two tailed with $\alpha = .05$ gives 1 - 0.025 = 0.9750.

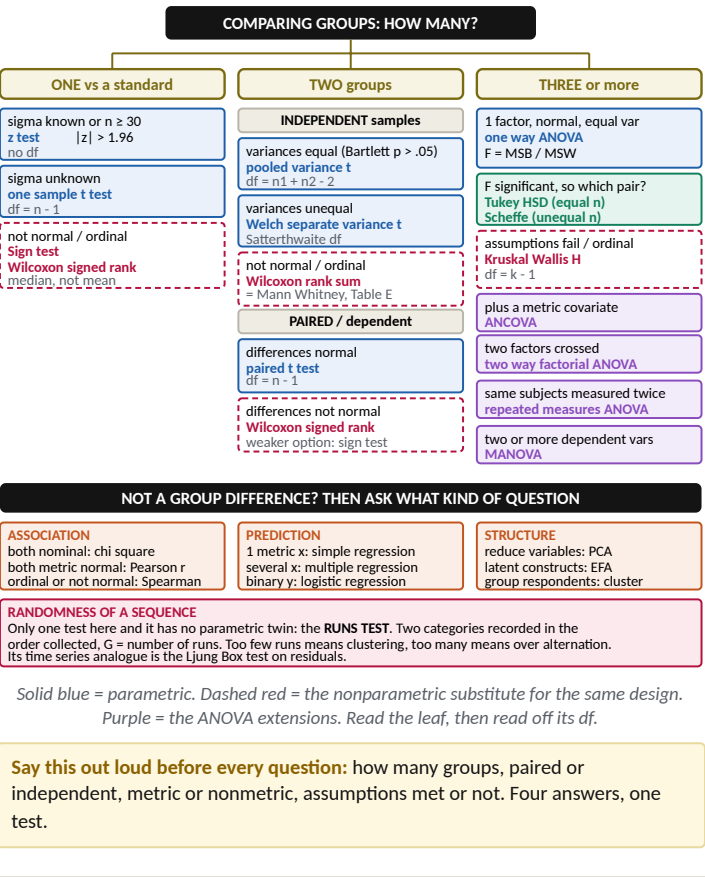
R SIGNIFICANCE STARS, READ THEM OFF EVERY OUTPUT

STARS	P RANGE	WORDING TO USE
***	p < 0.001	Very highly significant
**	p < 0.01	Highly significant
*	p < 0.05	Statistically significant
.	0.05 to 0.10	Marginally significant
blank	p $\geq$ 0.10	Not significant

Both routes always agree. A Notion worked case checks them side by side: **chi square 4.36 < critical 9.488 and p = 0.3595 > 0.05**, so both say do not reject H_0 . Quote both in an exam answer.

KEY The test chooser

One diagram answers "which test". Walk it top down: how many groups, paired or independent, then do the assumptions holds.



L1 Chi square: goodness of fit and independence

Frequencies cannot be averaged, so when the data are counts in categories the only comparison available is observed count against expected count. Both tests use the same statistic and Table G, and both are **right tailed only**.

1. GOODNESS OF FIT, "SAME OR NOT SAME"

Q: does the observed frequency distribution across k categories match a claimed or theoretical distribution? Shoe style preference, accidents by day of week, call volume by hour for staffing. **Data**: one categorical variable, **frequency counts**, never percentages. Actual frequency = Observed O, calculation frequency = Expected E.

Assumptions: random sample, observations independent, categories mutually exclusive, **expected frequency E $\geq$ 5 in every cell**.

STATISTIC AND EXPECTED FREQUENCIES
chi-square = SUM over all cells of (O - E)² / E
Equal expected proportions: E = n / k
Unequal claimed proportions: E = n * p_i
n = total observations, k = number of categories, p_i = claimed proportion for category i.
df = k - 1

H0: the population follows the specified distribution. H1: it does not. **Decision**: reject if chi-square_calc > chi-square_crit(α , df), or p < α .

R: `chisq.test(x = observed, p = expected)`, where expected is a vector of proportions summing to 1.

2. TEST OF INDEPENDENCE, "INDEPENDENT OR DEPENDENT"

Q: are two categorical variables associated, or independent? Data are cross tabulated into an r by c **contingency table** of counts. **Assumptions**: random sample, independent observations, expected cell frequency $\geq$ 5 in at least 80% of cells and none below 1.

STATISTIC, EXPECTED CELL, DF
chi-square = SUM over all r*c cells of (O_{ij} - E_{ij})² / E_{ij}
E_{ij} = (row_i total * column_j total) / grand total
df = (r - 1) * (c - 1)
2 x 4 table -> df = 3 3 x 3 table -> df = 4

H0: the two variables are independent, no association. H1: they are dependent.

R: `matrix(..., ncol=, byrow=TRUE)`, then `as.table()`, then `chisq.test(data)`. Read **X-squared, df, p-value**.

The wording trap from the notes: a chi square question **never mentions independent or dependent samples**. A Wilcoxon question always does. If the scenario says "before and after the same people", it is not chi square.

WORKED EXAMPLES, WITH THEIR REAL NUMBERS

DATA	EXPECTED	X-SQ	DF	P	DECISION
13, 10, 16, 22, 24	17 each	8.2353	4	0.0833	Do not reject
Blood A 12, B 8, O 24, AB 6	10, 14, 18, 8	5.4714	3	0.1404	Do not reject
Year by Race, 724/335/174/107 vs 370/292/152/140	row x col / N	60.144	3	5.48e-13	Reject , associated
Industry by unemployment duration, 3 x 3	row x col / N	4.9743	4	0.2899	Do not reject

CORE z test and the three t tests

A sample mean wanders around the population mean, so you need a yardstick for "how far is too far". That yardstick is the standard error.

ONE SAMPLE Z TEST

Q: does one sample mean differ from a known population mean? **Assumptions**: random sample, normal population or n $\geq$ 30, σ known. **df**: none, it is the standard normal.

Z FOR A MEAN AND FOR A PROPORTION
 $z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$
 $z = (\text{phat} - p_0) / \sqrt{p_0 * (1 - p_0) / n}$
Reject two tailed if $|z| > 1.96$ at $\alpha = .05$, $|z| > 2.58$ at $\alpha = .01$, one tailed 1.645.

ONE SAMPLE T TEST

Q: does a sample mean differ from a hypothesised value when σ is unknown? **Assumptions**: random sample, approximately normal population.

T, DF = N - 1
 $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$
df = n - 1
Reject if $|t| > t_{\text{crit}}(\alpha/2, n - 1)$

Nonparametric twin: sign test, or Wilcoxon signed rank.

INDEPENDENT SAMPLES T TEST

Q: do two independent groups differ on a metric outcome? One metric DV, one dichotomous grouping variable. **Assumptions**: independence, normality within groups, and **homogeneity of variance**, tested with **Bartlett's** or Levene's test. Check that first, it picks which formula you use.

POOLED VARIANCE, EQUAL VARIANCES ASSUMED
 $s_p^2 = [(n1-1)s1^2 + (n2-1)s2^2] / (n1 + n2 - 2)$
 $t = [(\bar{x}1 - \bar{x}2) - (\mu_1 - \mu_2)] / \sqrt{s_p^2 * (1/n1 + 1/n2)}$
df = n1 + n2 - 2

SEPARATE VARIANCE, WELCH, VARIANCES NOT EQUAL
 $t = [(\bar{x}1 - \bar{x}2) - (\mu_1 - \mu_2)] / \sqrt{s1^2/n1 + s2^2/n2}$
Welch-Satterthwaite df =
 $(s1^2/n1 + s2^2/n2)^2 / [(s1^2/n1)^2/(n1-1) + (s2^2/n2)^2/(n2-1)]$
Conservative alternative: df = smaller of n1 - 1 and n2 - 1

Nonparametric twin: Wilcoxon rank sum, that is, Mann Whitney.

PAIRED SAMPLES T TEST

Q: do two **related** measurements differ, before and after or matched pairs? **Assumptions**: the differences are approximately normal, pairs independent of one another.

WORK ON THE DIFFERENCES ONLY
d_i = x1_i - x2_i dbar = SUM(d_i) / n
s_d = sqrt([SUM(d_i²) - (SUM d_i)² / n] / (n - 1))
t = (dbar - μ_d) / (s_d / sqrt(n))
df = n - 1

Nonparametric twin: Wilcoxon signed rank, or the cruder sign test.

SPSS reading rule: read **Levene's Test** first. Sig > .05 use the "Equal variances assumed" row, Sig $\leq$.05 use "Equal variances not assumed". Then read t, df, Sig. (2 tailed), Mean Difference.

L2 The nonparametric family, paired with its parametric twin

Parametric tests are tests about **population parameters** (mean μ , standard deviation σ , proportion p) and they assume a normal population. Nonparametric tests exist for when that assumption cannot be met, so they test hypotheses that do not involve those parameters and work on **ranks and signs** instead.

THE TWIN LADDER

PARAMETRIC (needs normality)	NONPARAMETRIC (ranks, signs)
one sample t test / z test	Sign test · Wilcoxon signed rank
independent samples t test	Wilcoxon rank sum = Mann Whitney
paired samples t test	Wilcoxon signed rank (sign test)
one way ANOVA, 3+ groups	Kruskal Wallis H
Pearson correlation r	Spearman rank correlation r _s

NO TWIN: Runs test (randomness of a sequence) and chi square (frequency data) stand alone on the nonparametric side. Nothing parametric does their job.

ADVANTAGES AND DISADVANTAGES

Advantages: work on nominal and ordinal data · fewer and weaker assumptions, no normality · usable with very small samples · simpler arithmetic · robust to outliers.
Disadvantages, as the notes state them: **less sensitive**, so larger differences are needed before H_0 can be rejected · **less information**, the sign test knows only above or below the median, not by how much · **less efficient**, so larger samples are needed: **the sign test is about 60% as efficient as the z test, so n = 100 is needed where the z test needs n = 60**.

L2 Sign test and Wilcoxon rank sum

SIGN TEST

Q: does the median differ from a hypothesised value, or do paired observations shift in one direction? **Data**: ordinal or better, only the **direction** (+ or -) of each difference is used, zeros are discarded and n is reduced. **Assumptions**: random sample, continuous underlying distribution. No normality, no symmetry.

STATISTIC, SMALL AND LARGE SAMPLE
Small sample: test statistic = smaller of (number of + signs , number of - signs)
Reject H_0 if that count $\leq$ the sign-test table value.
Large sample, n $\geq$ 26:
 $z = [(X + 0.5) - (n/2)] / (\sqrt{nt} / 2)$
X = smaller of the plus/minus counts
0.5 = continuity correction. df: none

Twin: one sample or paired t test.

WILCOXON RANK SUM, THAT IS, MANN WHITNEY

Q: "uses ranks to determine if **two independent samples** were selected from populations that have the same distributions." **Table E, standard normal**. **Assumptions**: independent random samples, at least ordinal data, similarly shaped distributions if you read it as a shift in medians, which R calls "true location shift". **Steps**: 1 combine and rank, 2 sum the ranks, 3 use the formula.

STATISTIC, DF NONE, NORMAL APPROXIMATION
Rank all n1 + n2 values together, smallest to largest, averaging tied ranks.
R = sum of ranks of the SMALLER sample (size n1).
 $\mu_R = n1 * (n1 + n2 + 1) / 2$
 $\sigma_R = \sqrt{n1*n2*(n1 + n2 + 1) / 12}$
 $z = (R - \mu_R) / \sigma_R$
R's statistic: W = R - n1*(n1 + 1) / 2

Decision: two tailed at $\alpha/2 = 0.025$, boundaries -1.96 and +1.96. The worked slide shows **z = -2.41**, beyond -1.96, so reject H_0 . R: `wilcox.test(A, B)`, read W and p.

Worked R results: Males vs Females **W = 113, p = 0.1425** do not reject · Community college vs four year credit transfer **W = 43, p = 0.6228** do not reject · Handheld game lifetimes Brand A vs B **W = 47, p = 0.3922** do not reject.

L2 Wilcoxon signed rank and Kruskal Wallis

WILCOXON SIGNED RANK

Q: "whether **two dependent samples** have been selected from two populations having the same distributions", used for matched samples and before and after tests on the same subjects. **Table K. Assumptions**: pairs random and independent, differences from a continuous and **symmetric** distribution, at least ordinal.

STATISTIC W_s, DF NONE
1. D = X1 - X2 ; discard D = 0 and reduce n.
2. Rank |D| smallest to largest, average ties.
3. Reattach the sign of D to each rank.
4. w_s = smaller of (sum of positive ranks , sum of negative ranks)
R reports V = sum of POSITIVE ranks.
Large sample, n $\geq$ 30:
 $z = [w_s - n(n+1)/4] / \sqrt{n(n+1)(2n+1) / 24}$

Direction trap: for small n you reject when **w_s is less than or equal to** the table value. That is the opposite direction to almost every other test on this sheet.

Worked R results: school district legal costs year 1 vs year 2 **V = 2.5, p = 0.03547** reject, costs differ · drug prices US vs Canada **V = 52, p = 0.01437** reject, Canada cheaper · bowling scores day 1 vs day 2 **V = 21, p = 0.7422** do not reject, scores did not improve.

KRUSKAL WALLIS

Q: "whether **three or more** samples came from populations with the same distributions." One ordinal or metric DV, one factor with k $\geq$ 3 levels, independent samples, **chi square table**. **Assumptions**: independent random samples, at least ordinal data, each group n $\geq$ 5 for the chi square approximation, similar shapes for a median reading. **Steps**: 1 arrange lowest to highest, 2 rank sum per group, 3 substitute.

H STATISTIC, DF = K - 1
Rank ALL N observations together,
N = n1 + n2 + ... + nk, R_i = rank sum of group i.
H = [12 / (N * (N + 1))] * SUM_i (R_i² / n_i) - 3 * (N + 1)
Tie correction, divide H by:
1 - [SUM(t³ - t) / (N³ - N)]
df = k - 1. Right tailed: reject if
H > chi-square_crit(alpha, k-1), or p < alpha.

R labels H as "**Kruskal-Wallis chi-squared**". Post hoc: Dunn's test or pairwise Wilcoxon with Bonferroni. Twin: one way ANOVA.

Worked R results: generic 3 group **H = 4.9062, df 2, p = 0.08603** do not reject · sugar content of 3 candy bar brands **H = 10.405, df 2, p = 0.005504** reject, brands differ · job offers at 3 colleges **H = 8.6014, df 2, p = 0.01356** reject.

L2 Spearman and the runs test

SPEARMAN RANK CORRELATION

Why it exists, verbatim: testing p = 0 for the Pearson coefficient assumes the populations are normally distributed. If that cannot be met, the nonparametric equivalent r_s is used on ranked data. **Table L. Assumptions**: random paired sample, both variables at least ordinal, **monotonic** but not necessarily linear, no normality. Range -1 to +1. **Steps**: 1 rank each set, 2 subtract the rankings pairwise, 3 square the differences, 4 sum, 5 substitute.

R_s AND ITS SIGNIFICANCE TEST
 $r_s = 1 - [6 * \text{SUM}(d^2)] / [n * (n^2 - 1)]$
d = difference between the two ranks of a pair
n = number of pairs
R reports S = SUM(d²), so r_s = 1 - 6S/(n³ - n)
Significance: t = r_s * sqrt((n - 2) / (1 - r_s²))
df = n - 2

Decision, stated twice on the slides: r > critical value, reject H_0 . r < critical value, do not reject H_0 . The worked instance is **0.857 > 0.738** reject, and -0.8 < -0.738 reject.

Worked R results: rho = 0.857, S = 12, **p = 0.01071** reject · subway vs rail rho = 0.6, **p = 0.2417** do not reject · hospitals vs nursing homes rho = 0.8929, **p = 0.0123** reject · calories vs cholesterol rho = 0.036, **p = 0.9323** do not reject.

RUNS TEST

Why it exists, verbatim: "when samples are selected, you assume that they are selected at random. How do you know if the data obtained from a sample are truly random? One way to answer this question is to use the runs test." **Table M plus Table E. Data**: a sequence of **two category** outcomes in the order observed, True/False, M/F, Up/Down, above or below the median. **Assumptions**: recorded in collection order, each observation in exactly one of two categories.

G, ITS MEAN AND STANDARD DEVIATION, DF NONE
A RUN = a maximal succession of identical symbols.
G = observed number of runs
n1 = count of category 1, n2 = count of category 2
 $\mu_G = (2 * n1 * n2) / (n1 + n2) + 1$
 $\sigma_G = \sqrt{ [2*n1*n2 * (2*n1*n2 - n1 - n2)] / [(n1 + n2)^2 * (n1 + n2 - 1)] }$
 $z = (G - \mu_G) / \sigma_G$

Decision: if the number of runs falls **between** the two critical values, do not reject H_0 . Too few runs means clustering, too many means over alternation, and **both** reject randomness. Large sample: reject if $|z| > 1.96$. The worked slide shows **z = -3.96** against ± 1.96 , so reject randomness.

Scenarios used: True/False exam answers, twenty shoppers by gender at $\alpha = 0.05$, absences over 30 days, market Up/Down over 30 days.

Exam warning from the notes, verbatim in spirit: for every test you must know which table to read, which formula to use, and exactly what the question scenario is. Table E for rank sum and runs, Table K for signed rank, Table L for Spearman, Table M for runs counts, Table G for chi square.

Big Data Management

WHAT BIG DATA IS, AND THE DATABASES INVENTED TO HOLD IT

The arc of this half: **raw facts** become **information**, become **knowledge** once an ontology says how the pieces relate. Relational storage then breaks on the **scale, variety and speed** of that data, so the ecosystem splits into layers and the database layer splits into **four NoSQL families**. MongoDB is the document family, Neo4j is the graph family. Know why each family exists before you quote any syntax.

Ch1

Introduction to BDM

This chapter exists to fix the vocabulary every later chapter reuses.

TERM	EXACT DEFINITION
Data	Raw facts that were taken, observed, recorded, agreed . Words, numbers, observations, surveys
Information	The fact taken, observed, recorded, agreed. Information is processed data
Knowledge	The set of relationships between information elements following an ontology . "Mapped information": how they are related, what is compound in what, where
Ontology	The definition of meanings that frames the set of information

Reproduced as written, the two labels look swapped versus convention.

Qualitative, "Valuated facts", collected by measurement tools, analysed by statistical and numerical methods. **Quantitative**, "Described facts", collected by observation, analysed by adapted methods of classification and quantification.

Why data matters: Availability of massive data at unprecedented scale, formats and domains, and **Exploitability**, turning it into information, a long standing IT problem worth facing because massive data allows results never achieved before. Nine uses are listed, from targeting customers and optimising processes to healthcare, sports, science, security, cities and financial trading.

BIG DATA, NO SINGLE STANDARD DEFINITION

1. Data whose **scale, diversity, complexity** require new **architecture, techniques, algorithms, analytics** to manage it and extract hidden knowledge.

2. Data referring to a rapidly growing amount which traditional database mechanisms handle inefficiently in **storage, processing and analysis**.

Size: **terabytes (TB), petabytes (PB), exabytes (EB) and beyond**. **Four types**: **UNSTRUCTURED** **QUASI-STRUCTURED** **SEMI-STRUCTURED** **STRUCTURED**

THE VS, IN THE WORDING THAT EARNS MARKS

V	WHAT TO WRITE
Scale (Volume)	"Big means extremely large, more than terabyte or petabyte." Data must necessarily be distributed , and the number of sources can be huge
Complexity (Variety)	Text, numerical, images, audio, video, sequences, time series, social media. Static versus streaming data . All types must be linked to extract knowledge. Entity resolution resolves or identifies entities and detects relationships
Speed (Velocity)	Generated fast, processed fast, late decisions miss opportunities. Speeding the feedback loop is the advantage. Stream processing consumes data on the fly instead of storing it for later batch processing
Veracity	Widely different quality, often little or no schema controls structure. Uncertainty from multichannel sources under four headings: Archive, Multiple profiles, Delegation, Forecasted by nature

3Vs = Volume, Variety, Velocity. 4Vs adds Veracity. The 5th V appears in the notes only as an image, never spelled out in the text.

Challenges: standard tools are not designed for such volumes, infrastructures cannot yet meet the distributed computational need, and traditional techniques cannot process or analyse it.

Storage

Preprocessing

Processing

Security

Big Data Management is the organization, administration and governance of large volumes of structured and unstructured data, covering **policies, procedures, technology** for collection, storage, governance, organization, administration and delivery. It ensures data quality and accessibility for BI, and helps companies find value in **call detail records, system logs and social media sites**.

Ch2

BDM Ecosystems

The ecosystem chapter exists to show big data is a pipeline of layers, not one product.

Acquisition

Storage

Preprocessing

Quality

Analytics

Security

Three cited definitions. **Harvey 2017**: putting the right **people, policies, technologies** in place to ensure the **accuracy, security, quality** of large stores of data. **Wessler 2016**: the **capture, integration, administration, governance** of big data for an organization's data related applications. **Baker 2014**: a complex process, especially when abundant heterogeneous data feeds BI and decision making.

Layer 1, Acquisition exists to collect, clean and filter data **before** storage. Five channels: **Data Streaming**, IoT devices from smart fridges to logistics - **Batch and Events sourcing** from cloud and on premise infrastructures - **Real Time Data Flows** - **Data transactions**, structured, mainly on premise - **Cloud or Grid Computing**, resources anywhere on the Internet processing in distributed mode, giving sharing, archiving and high scale.

Layer 2, Storage saves, protects, manages, recovers, archives, alters, deletes and retrieves records. Databases are SQL plus NoSQL such as MongoDB and Cassandra.

	DATA WAREHOUSE	DATA LAKE
Scope	Connected storages for one organization	Repositories not related to one organization
Feeds	Data Marts, then BI reporting, dashboards, decisions	Data Analytics, for processing
Sources	Transactional: ERP, CRM, SCM	Multiple
Data	Structured, formatted models	Structured and unstructured, unformatted models

Layer 3, Preprocessing. **Spark** is open source, suits a wide assortment of workloads and is more powerful in **streaming mode**. **Hadoop MapReduce** processes data in **HDFS** with the MapReduce algorithm plus **HBase**, and is more powerful in **batch mode** for wide volumes. **Data Quality** cleanses for **truthfulness, consistency, compliance, significance**, plus **Master Data Management**. **Security** protects sensitive data during collection, integration and aggregation using **Data Masking and Data Encryption**, which also reads the organization for auditing.

The 7 analytics roles: Business User (benefits from results) · Project Sponsor (genesis of the project) · Project Manager (milestones on time and at quality) · BI Analyst (domain expertise, **KPIs**, reporting) · DBA (provisions the database environment) · Data Engineer (extractions and manipulation) · Data Scientist (techniques, modelling, meets the objectives).

6 ANALYTICS LIFECYCLE PHASES

1 Discovery learn domains, past history, frame the problem

2 Data Preparation execute ELT (extract, load, transform) and ETL (extract, transform, load) into the sandbox

3 Model Planning method, techniques, workflow, select key variables

4 Model Building testing, training, production datasets. build, execute

5 Communicate Results success or failure, key findings, business value

6 Operationalize final reports, briefings, code, technical document

DATA LIFE CYCLE, 8 STEPS

Plan → Collect → Assure → Describe → Preserve → Discover → Integrate → Analyze

Four named models. **Jimmy Nolan**: Data Acquisition, Data Normalization, Data Mining and Analytics, Storage. **IBM**: 3 core phases Test data management, Data Masking, Archiving, and 7 sub-phases Create, Use, Share, Update, Archive, Store or Retain, Dispose. **Demchenko, De Laat and Membrey**: architecture of data models, storage, data source, applications. **Kurt Cagle**: Data Acquisition, Data Awareness, Data Analytics, with **Data Governance** as the key process.

5 BDM TECHNOLOGIES

– **MPP**: two or more processors collaborate on the same program, each with its own OS and dedicated memory, for concurrent ingest, loading and analysis

– **MapReduce Technology**: processing distributed across nodes in server clusters, results assembled. Core Hadoop technology beside **HDFS**

– **Storage and File System**: distributed file storage, to address the limitation of shared disk file systems

– **Network**: higher speed, coverage and bandwidth. Most recent advance is **5G**, supporting Internet of Things connection

– **Database**: beyond relational, for unstructured data. This motivates the advent of **NoSQL**

How to succeed with BDM, 4, reduce the amount of data to be managed - develop talents for data management - set a clear big data management plan and policy - reduce the amount of problematic data.

Ch6

Why NoSQL appeared

NoSQL exists because relational databases hit four walls at big data scale.

- **Not suitable for all types of problems**
- **Do not scale well for very large datasets**
- **Struggle to handle unstructured data**
- Large scale solutions become **very expensive**

Name and date: coined in 1998 by Carlo Strozzi, meaning "Not Only SQL". A design supporting structured, semi-structured and unstructured data, suited to large scale distributed data, especially in cloud environments.

8 characteristics: distributed computing across servers - flexible, **schemasless** model - asynchronous inserts and updates with weaker transactional support - follows **BASE** and **CAP** instead of **ACID** - does not strictly rely on SQL - **no joins**, data is denormalized - lower cost to scale - easy to implement and scale.

ACID, THE RELATIONAL PROMISE	BASE, THE NOSQL TRADE
Every transaction is all or nothing and leaves the database consistent immediately. Neo4j keeps this	The system stays available and partition tolerant, and consistency arrives later. This is eventual consistency

Honest gap: the notes name ACID and BASE but never expand the letters. Only CAP is unpacked, and it is unpacked inside Ch7 as MongoDB's choice. Quote the note definitions in the triangle below, they are the ones you were taught.

all replicas contain the same version of the data

Pick any two

a distributed system cannot hold all three

MongoDB chooses C and P

Availability operational on failing nodes

Partition tolerance multiple entry points, operations on system split

AP: stay up, go eventual · CA: one machine only

The three definitions are the exact wording the MongoDB chapter gives for C, A, and P.

STRENGTHS	WEAKNESSES
High scalability, schema flexibility	Too many products, above 150 , which to pick
Distributed computing: reliability, scalability, sharing of resources, speed	No common standardized query language as in SQL
No complicated relationships, lower hardware cost	Limited query capabilities so far, no ACID
Open source, all except Amazon S3 (Amazon Dynamo), a low cost entry point	Eventual consistency is not intuitive for strict scenarios like banking. Some have no secure authentication, failover or backup

Use NoSQL when the RDBMS model is too restrictive and you want a flexible schema - ACID is not "really" needed - Object to Relational (O/R) impedance - RDBMS is neither distributed nor scalable by nature - logging from distributed sources - events and temporal data - temporary data such as shopping carts, wish lists, session data - **Polyglot Persistence**, the best store for the nature of the data.

Do not use NoSQL for financial data, data requiring strict ACID compliance, or business critical data.

Ch6

The four NoSQL families

Each family exists to make one access pattern cheap. Name the pattern, then the product.

FAMILY	PRODUCT	USE CASE
1 Document	MongoDB	Content management systems, mobile application data handling
2 Graph	Neo4j	Walmart, personalized product recommendations and promotions
3 Wide Column	Cassandra	Facebook Search, rewritten off MySQL
4 Key-Value	Redis	Session data, user profiles, preferences, shopping carts

1 Document. Stores semi-structured data along with its description, in **JSON, BSON, XML**. Pairs each key with a complex data structure, and **indexes are done via B-Trees**. Documents hold key-value pairs, key-array pairs, or nested documents. Also: Couchbase, DynamoDB, CouchDB, RethinkDB, Terrastore, RavenDB, MarkLogic, OrientDB.

2 Graph. Neubauer (2010): graph databases "model the whole data set as one big dense network structure". Nodes are real world entities, a person, object, place or concept. A **relationship** is how two nodes are associated. Both have different properties.

u1 is connected to u2 by FRIENDS, u1 is connected to u3 by FRIENDS, u2 is connected to p1 by LIKES, u2 is connected to p1 by AUTHOR

Circle = Node (User or Post) Line = Relationship

Only 2 entities exist, Users and Posts. u1 is AUTHOR of p1, u2 LIKES p1, u1 and u2 are FRIENDS, u1 and u3 are FRIENDS.

THE ONE GRAPH QUERY THE NOTES STATE, IN WORDS

Find all of the Friends of any User who has Liked one of my Posts,

in alphabetical order of username.

Use a graph database when the data is highly relational - it has many **Many-To-Many** relationships - the schema is not absolutely rigid - you want complex analysis or expensive queries spanning multiple types of data.

Walmart	uses Neo4j to provide customers personalized, relevant product recommendations and promotions
Medium	uses Neo4j to build their social graph to enhance content personalization
Cisco	uses Neo4j to mine customer support cases to anticipate bugs
HP	uses Neo4j for its own network and IT operations

Also: OrientDB, Titan, ArangoDB, Apache Giraph, InfiniteGraph.

3 Wide Column. Data sits in cells grouped in **columns** rather than rows, all cells of a column stored as a continuous disk entry, read and written by column.

ROW STORE	COLUMN STORE
read pages containing entire rows	read only columns needed
one row = 212 columns! is this typical? (it depends)	In this example: 7 columns caveats: "select *" not any faster, clever disk prefetching, clever tuple reconstruction

FACEBOOK SEARCH, QUOTE THESE NUMBERS EXACTLY

MySQL > 50 GB Data Writes Average ~300 ms Reads Average ~350 ms

Cassandra > 50 GB Data Writes Average 0.12 ms Reads Average 15 ms

"What about vertical partitioning? (it does not work with ad-hoc queries)." Also: HBase, Accumulo, Hypertable, Google Bigtable.

4 Key-Value. The simplest family, built on a **hash table**. Values are reached by strings called **keys**, data has no required format, the model is **(key, value) pairs**.

BASIC OPERATIONS, EXACT

Insert(key, value) Fetch(key) Update(key) Delete(key)

Do not use when there are relationships among entities, when queries need the content of the value part, or for set operations across multiple pairs. Also: Memcached, Riak KV, Hazelcast, Ehcache, Aerospike, Amazon SimpleDB, Oracle Berkeley DB, ArangoDB.

Ch7

MongoDB: every CRUD command

The query language is based on **JavaScript**. Reads return a **cursor** for iteration, and each command is evaluated on a **single collection**.

THE FOUR CRUD OPERATIONS

db.collection.insert() db.collection.update() db.collection.remove() db.collection.find()

INSERT

Parameters: 1 Document, one or more; 2 Options. A given `_id` must be unique, and when missing it is generated automatically as an ObjectId. Collections are created automatically when they do not yet exist.

db.actors.insert({
 {
 firstname: "Jennifer",
 lastname: "Lawrence"
 }
})

Result: { _id: ObjectId("8"), firstname: "Jennifer", lastname: "Lawrence" }

db.movies.insert({
 {
 _id: ObjectId("9"),
 title: "The Hunger Games",
 year: 2003,
 actors: [ObjectId("4"), ObjectId("8")]
 },
 {
 title: "Passengers",
 year: 2016,
 actors: [ObjectId("8")]
 }
})

UPDATE

Parameters: 1 Query; 2 Update; 3 Options. **At most one document is updated by default, unless { multi: true } is specified.**

db.actors.update(
 { _id: ObjectId("8") },
 {
 firstname: "Ana",
 lastname: "Lawrence"
 }
)

db.movies.update(
 { year: { \$gt: 2015 } },
 {
 \$set: { new: true },
 \$inc: { rating: 3 }
 },
 { multi: true }
)

Replace mode applies when the update parameter has no update operators, so the whole document is replaced and `_id` is preserved. **Update mode** applies when it has only update operators, each usable at most once.

FIELD OPERATORS	ARRAY OPERATORS
\$set sets a field or fields	\$push adds items to the end of an array
\$unset removes a field or fields	\$addToSet adds items, duplicates ignored
\$rename renames a field or fields	\$pop removes first or last item
\$inc increments, \$mul multiplies	\$pull removes all items matching a condition
\$currentDate stores current date, time or timestamp	

Upsert: with { `upsert: true` } and no match, a new document is inserted. Replace mode inserts all fields from the update parameter, Update mode inserts all value fields from the query parameter plus the results of all update operators, and `_id` is preserved or generated automatically.

db.movies.update(
 { title: "Joy", year: { \$gt: 2000 } },
 {
 \$set: {
 director: { firstname: "David", lastname: "Russell" },
 year: 2001
 },
 \$inc: { rating: 2 }
 },
 { upsert: true }
)

Result: { _id: ObjectId("11"), title: "Joy",
 director: { firstname: "David", lastname: "Russell" },
 year: 2001, rating: 2 }

REMOVE

Parameters: Query, Options. All matching documents are removed unless { `justOne: true` } is specified, then only one is removed.

FIND

Parameters: 1 Query; 2 Projection. Matching documents come back on an iterable cursor, allowing **sort, skip, limit** to be chained.

db.movies.find()
db.movies.find({})
db.movies.find({ _id: ObjectId("2") })
db.movies.find({ year: 2000, rating: { \$gt: 1 } })
db.movies.find({ year: { \$gte: 2005, \$lte: 2015 } })

TITLES FILMED 2005 AND LATER, SORTED DESCENDING
db.movies.find(
 { year: { \$gt: 2005 } },
 { _id: false, title: true }
)
.sort({ title: -1 })

Result: { title: "The Double" } { title: "The Social Network" }

The pairs below are NOT equivalent. Value equality on an embedded document or an array compares the whole thing, order included.

db.movies.find({ director: { firstname: "David", lastname: "Russell" } })
db.movies.find({ director: { lastname: "Russell", firstname: "David" } })
db.movies.find({ actors: [ObjectId("7"), ObjectId("5")] })
db.movies.find({ actors: [ObjectId("5"), ObjectId("7")] })

Dot notation: embedded documents use "field.subfield" such as "director.firstname", arrays use "field.index" such as "actors.2", index starting at 0.

db.movies.find(
 {
 "director.firstname": "David",
 "director.lastname": "Russell"
 }
)

db.movies.find({ actors: { \$all: [ObjectId("5"), ObjectId("7")] } })

GROUP QUERY OPERATORS

Comparison \$eq \$ne \$lt \$lte \$gt \$gte, \$in matches any value in a list, \$nin not in list. Comparisons consider **BSON data types**, some numeric conversions are automatic

Element \$exists

Evaluation \$regex, \$text which requires a text index

Array \$all contains all specified items, \$size, \$elemMatch at least one element satisfies all conditions

Logical \$and, \$or need at least two expressions at top level; \$not negates and cannot be used at top level

Querying arrays, the trap: a value equality condition is true when the whole field equals the value OR at least one element does. A query operators condition is true when the whole field satisfies the operators OR at least one element does, and that element may not be the same for all operators. Use \$elemMatch when one element must satisfy all conditions.

Projection: `true` or `1` includes, `false` or `0` excludes, and include and exclude cannot be mixed except for `_id`. `$elemMatch` returns the first matching array item, and if there is none the field is not returned. `$slice` positive gives the first N, negative the last N, and can skip items.

db.movies.find(
 { _id: ObjectId("2") },
 { title: true, actors: true }
)

db.movies.find(
 { year: 2005 },
 { title: 1, _id: 0, actors: { \$slice: -2 } }
)

sort, limit, skip are optional, chainable in any order, and must be applied before retrieving documents. Order is undefined unless sorted, use indexes on large datasets, and **sorting happens before projection**.

Slide typo: the notes write "1 → descending". The working values are 1 ascending and -1 descending, which is what the sorted find above uses.

AGGREGATION AND MAPREDUCE

db.orders.aggregate(
 { \$group: {
 _id: type,
 totalQuantity: { \$sum: quantity }
 }
)

NEEDS A MAP FUNCTION, A REDUCE FUNCTION, AND A RESULT SET NAME
db.collection.mapReduce(
 <mapfunction>,
 <reducefunction>,
 {
 out: <collection>, query: <document>,
 sort: <document>, limit: <number>,
 finalize: <function>, scope: <document>,
 jsMode: <boolean>, verbose: <boolean>
 }
)

INDEXES, REPLICATION, CONSISTENCY

– **B+ tree indexes**, one created automatically on `_id`. Users add indexes for performance or to enforce unique values

– **Single field and compound**, where field order matters as in SQL. Indexing an array creates one index entry per element

– **A sparse index** only indexes documents containing the field. **Unique plus sparse** rejects duplicates but allows documents without it

– **Only one index is used per query**, chosen by the optimizer. An **index covering query** is answered from the index alone

Replication gives redundancy, backup and automatic failover. The **Primary** handles writes, the **Secondary** replicates. Secondary types: Secondary-only, Hidden, Delayed, Arbiters, Non-voting. If the primary fails, secondaries **elect a new primary**.

Consistency: reads from the primary give strong consistency, reads from a secondary give eventual consistency and may return older data. Failure before full replication may need a rollback, applied manually. Memory mapped files use mmap(). Geospatial: Spherical using longitude and latitude, and Flat, 2D points on a plane, with geospatial indexes.

Ch1 Introduction | Ch2 BDM Ecosystems | Ch6 NoSQL | Ch7 MongoDB | Ch8 Neo4j
Sheet 2 carries the Hadoop stack: HDFS, MapReduce, Spark, Flume, Sqoop, and the final exam questions.

REVISION SHEET

■ Ch1 Introduction ■ Ch2 Ecosystems ■ Ch6 NoSQL and CAP ■ Ch4 Four families ■ Ch7 MongoDB ■ Ch8 Neo4j

Built from the MAXD 5123 chapter notes. Every command and number reproduced as given

One sentence holds the stack together: **Hadoop moves computation to the data instead of data to the computation**. HDFS decides where the data sits, so replica placement and rack awareness exist. MapReduce runs beside the block, so data locality exists. Spark keeps the intermediate result in **memory** instead of on disk, so it wins on iterative work. Flume and Sqoop exist only to get data into HDFS, one for streams, one for tables.

Ch10.1 Hadoop

Hadoop exists because one machine cannot store or process petabyte scale data, and a bigger machine stops being an answer long before the data stops growing.

Open source software from the **Apache Software Foundation**, a framework that lets you store, access, process big data in a distributed environment across clusters of computers, managing resources across the computing and network resources.

Hadoop is NOT big data itself, it is a framework to work with big data and part of a much larger ecosystem - not an operating system or packaged software application - not a brand name, it is an open source project.

WHY IT WAS INVENTED

WHAT TO WRITE

1 Big volume	RDBMS cannot store huge amounts. Hadoop stores data distributed across many nodes in a cluster , used in the field at petabyte scale
2 Variety	Real data is unstructured, semi-structured and structured. Hadoop digests all formats because it does not enforce a schema
3 Speed	RDBMS fails at real-time speed. Hadoop moves computation to data, not data to computation , which is Data Locality
4 Cost	Traditional storage is expensive. Hadoop runs on commodity hardware , so large storage costs a reasonable amount

History, honestly, The Hadoop timeline appears in the notes as an image only, with no dates in the text. The dated facts this subject gives are: MapReduce created by Google engineers **Jeffrey Dean** and **Sanjay Ghemawat** in 2004, and **HDFS** derived from the **Google File System**.

THE CLUSTER, PHYSICALLY

Data centre → Racks → Nodes

- A **node** is a process on a virtual or physical machine that actually executes the jobs
- Every **rack is interconnected through 1 gigabyte of Ethernet, 1 GigE**
- Topology: **Server → Rack Switch → Cluster Switch → external network**
- Nothing is shared between nodes except the network that connects them**, a shared-nothing architecture

Master and slave. Masters: one machine as **NameNode**, another as the **JobTracker**. Slaves: the rest, acting as **DataNode** and **TaskTracker**.
5 design principles: data distributed around the network with no centralized data server and replication for fault tolerance · scaling out from one server to thousands · computation sent to data, results aggregated · master and slaves, the master or **JobNode** launches the application and **SlaveNodes** do the bulk of computation · hardware failure.

3 operation modes: **Standalone**, one process, for development and debugging · **Pseudo-distributed**, separate processes on one machine, to analyse performance bottlenecks · **Fully Distributed**, real cluster mode, for final job runs.

THE 4 CORE COMPONENTS

HDFS	Manages storage of and access to data distributed across the various nodes of a Hadoop cluster
YARN	Cluster resource manager and job scheduler, responsible for allocating system resources to applications
MapReduce	A YARN-based programming framework and processing engine for large scale batch applications
Common	Utilities and libraries supporting the other three core modules

YARN, exactly as far as the notes go. It is described only as cluster resource manager, scheduling jobs and allocating system resources, plus Sqoop's line that parallel import and export use multiple mappers via YARN. No **ResourceManager**, **NodeManager** or **ApplicationMaster** appear anywhere. The daemons you were taught are **NameNode** and **JobTracker** on the master side, **DataNode** and **TaskTracker** on the slave side.

ECOSYSTEM, 7 COMPONENTS

HBase	Column-oriented data warehouse for extremely large data
Hive	Data warehouse software for querying and managing large datasets in distributed storage
Sqoop	Transfers bulk data between Hadoop and structured datastores such as relational databases
Pig	High-level data-flow language and execution framework for parallel computation. The language is PigLatin
Mahout	Scalable machine learning and data mining library
Oozie	Workflow scheduler to manage Apache Hadoop jobs
Zookeeper	Operational services for a Hadoop cluster group services

Together they give **absorption, analysis, storage and maintenance of data**.

9 characteristics: **OPEN SOURCE** · **DISTRIBUTED PROCESSING** ·

FAULT TOLERANCE · **RELIABILITY** · **HIGH AVAILABILITY** · **SCALABILITY** ·

ECONOMIC · **EASY TO USE** · **DATA LOCALITY**

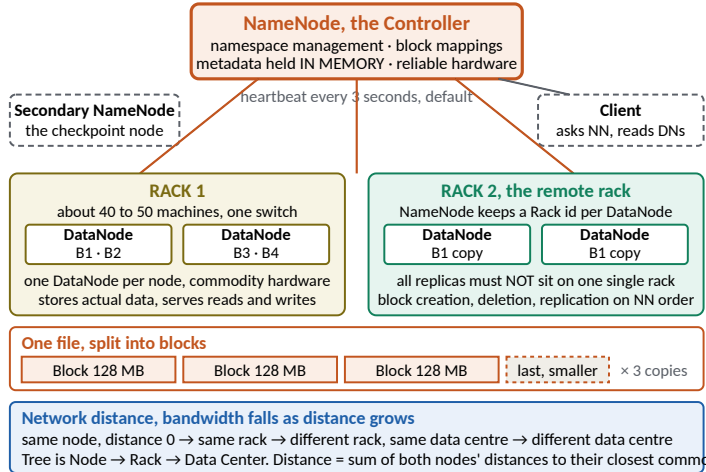
THE NUMBER THAT SITS INSIDE FAULT TOLERANCE
By default, 3 replicas of each data block are stored across the cluster.
The replication factor can be adjusted.

4 weaknesses: **security**, hard to manage and Hadoop lacks built-in encryption at **storage and network levels** · **vulnerable by nature**, mostly written in **Java**, often targeted by cybercriminals · **not fit for small data**, no efficient random reading of small files · **stability**, many contributors, so use the latest stable version.

Flavours. Free: Apache Hadoop, <https://www.hadoop.apache.org/>. Commercial: **Cloudera**, **Hortonworks**, **MapR**, **IBM**, **Microsoft**.

Ch10.2 HDFS

HDFS exists to store and manage very large data on commodity hardware that is expected to fail. **Source: the Google File System**.



EVERY DEFAULT NUMBER IN THE HDFS CHAPTER

- Block size, default **128 MB** (adjustable)
- Replication factor, default **3 copies** (adjustable)
- DataNode heartbeat frequency **3 seconds**, by default
- Machines in one rack **about 40 to 50**, one network switch
- Rack interconnect **1 gigabyte of Ethernet, 1 GigE**
- Distance, same node to itself **0**
- MongoDB shell port, **Ch7 28015** (the only port in the whole subject)

No port numbers appear in any Hadoop chapter. If a question demands one, say the notes do not specify it rather than invent a value. Likewise **no HDFS shell commands** such as `hdfs dfs -put` appear. The only shell level command referenced is HDFS's **put** command, named in the Flume chapter as the naive way to upload data.

THE 6 STEP PROCESS

- Client sends request**, contacting the **NameNode** to ask where the data is stored, a metadata request
- NameNode responds**, checking metadata, file name, location, replicas, and naming the **DataNodes** to use
- Client reads or writes** directly with the selected DataNodes
- Data is split into blocks** across multiple DataNodes
- Replication happens**, each block copied to other DataNodes, for example across Rack 1 and Rack 2
- Fault tolerance ensured**, if one DataNode fails the data is still on the replicas

NAMENODE, 4 TASKS	DATANODE TASKS
1 Manage the file system namespace	Store the actual data
2 Regulate the client's access to files	Serve read and write requests from clients
3 Execute file system execution such as naming, closing, opening files and directories	Block creation, deletion and replication on instruction from the NameNode
4 Take care of the Replication Factor of all the blocks	Send a heartbeat reporting HDFS health, default every 3 seconds

NameNode metadata: filename, number of blocks, number of replicas, location of blocks, block IDs, held in **memory** for faster retrieval, which is why it needs **reliable hardware** as the centerpiece of HDFS. The DataNode is typically **one per node** and runs on commodity hardware.

DATA STORAGE, THE 7 NUMBERED RULES

- HDFS splits huge files into small chunks called **blocks**
- A block is the **smallest unit of data** in the file system
- Default block size is **128 MB**, adjustable based on requirements
- All blocks are the same size **except the last block**, which can be smaller
- If the file is smaller than the block size, the block uses only the **actual data size**, no space wasted
- Each block is replicated, with a **default of 3 copies**
- All blocks are stored in the Hadoop file system, distributed across multiple **DataNodes**

REPLICA PLACEMENT, LEARN VERBATIM, IT IS WORTH 18 MARKS

- One replica on the **LOCAL NODE**
- Second replica on a **REMOTE RACK**
- Third replica on the **SAME REMOTE RACK**
- Additional replicas are **RANDOMLY PLACED**
- Clients read from the **NEAREST REPLICA**

Rack awareness exists because one switch failure takes out a whole rack of **about 40 to 50 machines**. The NameNode keeps **Rack ids** for each DataNode, chooses a DataNode in the same or a nearby rack, and **ensures all replicas are not stored on a single rack**. It is needed for high availability and reliability, cluster performance, and network bandwidth.

READ PATH AND WRITE PATH

HDFS is **write-once, read-many**. Files **cannot be edited**, but data can be **appended**. In both paths the client talks to the NameNode first for metadata, then directly to the DataNodes.

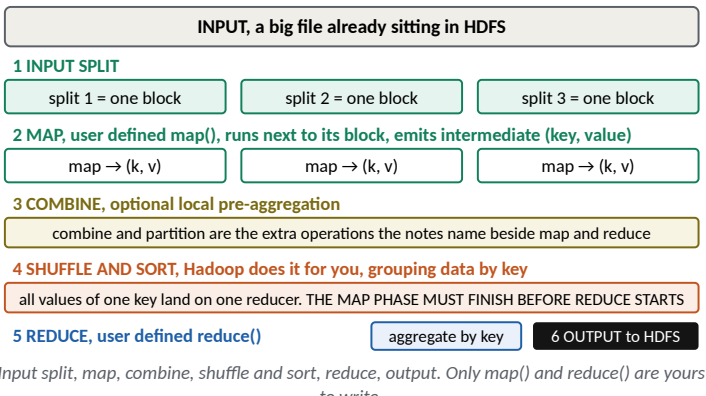
#	READ	WRITE
1	Client connects to NameNode to read data	Client connects to NameNode to write data
2	NameNode tells client where to find the data blocks	NameNode provides the list of suitable data nodes
3	Client reads blocks directly from the closest DataNodes, without going through the NameNode	Client writes blocks directly to these data nodes with the desired replication factor
4	On node failure the client connects to another node holding the missing block	On node failure the NameNode figures it out and replicates the missing blocks

Q. Why not read through the NameNode? To stop it becoming a **bottleneck**, to scale for **many concurrent users**, and to distribute data traffic across the cluster.
Q. Which replica does the NameNode pick? It uses **rack awareness based on network topology**, and the client reads from the **nearest DataNode**.
Q. Where do the three replicas go, and what trade-offs? Reliability, data safety · **Write bandwidth**, speed when saving · **Read bandwidth**, speed when accessing.

Ch11.1 MapReduce

MapReduce exists so a programmer can parallelise across thousands of CPUs by writing only two functions.

A **parallel programming framework** for large data sets, **terabytes and petabytes**, stored in **HDFS** across nodes in a network cluster. Created by **Google engineers Jeffrey Dean** and **Sanjay Ghemawat** in 2004. Motivations: process lots of data, **greater than 1 TB** · parallelize across hundreds or thousands of CPUs · make this easy for users.



MAP	REDUCE
Collects relevant data from the source. The user-defined map() is applied to each data chunk and produces intermediate (key, value) pairs . Data is split into blocks, each block processed separately	Takes the mappers' output, groups all values by the same key, and the user-defined reduce() processes each key with its list of values, producing the final aggregated result and storing it in HDFS

YOU WRITE TWO FUNCTIONS, HADOOP HANDLES FIVE THINGS

You define: map() and reduce()

Hadoop automatically handles:
parallel processing · data distribution · load balancing
fault tolerance · shuffling, which is grouping data by key

WORD COUNT, THE WORKED EXAMPLE

MAPPER OUTPUT A, INPUT "MELAKA IS A NICE PLACE TO VISIT" (Melaka,1) (is,1) (a,1) (nice,1) (place,1) (to,1) (visit,1)
MAPPER OUTPUT B, INPUT "TOURISTS ALWAYS VISIT BANDA HILIR IN MELAKA" (Tourists,1) (always,1) (visit,1) (Banda,1) (Hilir,1) (in,1) (Melaka,1)
REDUCER OUTPUT (Melaka,2) (is,1) (a,1) (nice,1) (place,1) (to,1) (visit,2) (Tourists,1) (always,1) (Banda,1) (Hilir,1) (in,1)

"Melaka" appears in both sentences so its count is 2, "visit" appears in both so its count is 2, all other words count 1.

Three implementations: **Google MapReduce**, the original proprietary one · **Apache Hadoop MapReduce**, the common open source one built on Google's specifications · **Amazon EMR**, running it on **AWS EC2**, with similar services on **Microsoft Azure HDInsight** and **Google Cloud**.
Problems it solves: a benchmark for big data processing such as sorting · Google's **word count**, **ads**, **PageRank**, **indexing** · simple algorithms like **search (grep)** and text indexing · **data mining** such as classification · Facebook demographics · **financial analytics** · **scientific research** such as astronomy · **semantic web** and **Web 3.0**.

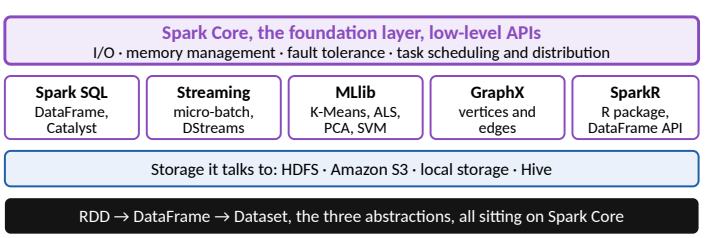
Summary points worth a mark each: handles petabytes and exabytes · simple Map and Reduce operations · supports **combine and partition** · the **Map phase must finish before Reduce starts** · tasks can run on the same physical machine · the number of map and reduce tasks is configurable · processing happens close to the data · low cost commodity hardware · splitting and movement are automatic · works with distributed storage like HDFS.

Ch11.2 Spark

Spark exists because MapReduce writes every intermediate result to disk, which is fatal for iterative work.

An open source, wide-range **data processing engine** for large scale analytics on clusters or a single node: **machine learning**, **SQL** and **graph analytics**. Written in **Scala**, and it can be part of the Hadoop ecosystem.

These notes contain no driver, cluster manager and executor topology. Spark is taught here through features, ecosystem and data abstractions, not runtime topology. If a question demands the three-part answer, name **driver**, **cluster manager** and **executor** from standard reading, then anchor it in what the notes do say: Spark has its own **cluster manager**, runs **independently or on top of a Hadoop cluster**, and uses **DAG** execution.



Spark Core is the foundation, SQL, Streaming, MLlib, GraphX and SparkR all depend on it.

WHY SPARK BEATS MAPREDUCE, THE 3 SHORTCOMINGS IT FIXES

- Hadoop is written in Java → high-level APIs for Java, Scala, Python and R
- MapReduce is batch only → batch AND stream, real-time data
- MapReduce is disk-based → in-memory, faster performance

With Hadoop: Spark has its own cluster manager, can run independently or on a Hadoop cluster, needs no replacement of Hadoop, improves traditional MapReduce, supports **iterative** and **real-time stream** processing, is much faster by **in-memory computation**, and reaches Hadoop data sources such as HDFS and Hive.

2009	Academic project at UC Berkeley, AMP Lab , led by Dr. Matei Zaharia
2010	First paper, "Spark: Cluster Computing with Working Sets"
2012	RDD introduced in "Resilient Distributed Datasets: A Fault-Tolerant Abstraction for In-Memory Cluster Computing"
2014	Adopted by Apache, Spark 1.0 released , May 2014. Databricks founded, Zaharia became CTO

8 features: **Speed**, up to **100x faster in memory**, **10x faster on disk** · **Dynamic in nature**, many high-level operators · **In-Memory Computation** with **DAG**, **Directed Acyclic Graph** execution · **Reusability**, one codebase for batch and streaming · **Fault Tolerance** through **RDD** · **Real-Time Stream Processing** via **Spark Streaming** · **Lazy Evaluation**, plan first then run · **Multiple Language Support**, Java, Scala, Python, R, while Hadoop is Java only.

RDD, THE FUNDAMENTAL UNIT OF DATA

Resilient	Fault tolerant. A partition lost to node failure is recomputed automatically , because Spark tracks transformations as lineage
Distributed	Spread across nodes, divided into partitions processed in parallel
Dataset	Records from JSON, CSV, text files or databases via JDBC. No strict schema required

THREE WAYS TO CREATE AN RDD

- Parallelized Collections `parallelize()` converts a driver collection into an RDD
- External Datasets `textFile()` takes a file path or URL, reads lines
- Existing RDDs apply a transformation to an existing RDD

TRANSFORMATION	ACTION
Creates a new RDD from an existing one. Lazy , executed only when an action is called, which lets Spark optimize. Does not modify the existing RDD	Performs computation on RDDs and returns a value to the driver program after the computation

RDD TRANSFORMATIONS, EXACT SYNTAX

<code>map(func)</code>	each element through a function
<code>filter(func)</code>	select where the function returns true
<code>flatMap(func)</code>	like map, each input can give many outputs
<code>sample(withReplacement, fraction, seed)</code>	sample a fraction, with or without replacement
<code>union(otherDataset)</code>	elements from both datasets
<code>distinct([numTasks])</code>	distinct, unique elements
<code>groupByKey([numTasks])</code>	group values by key into (K, Seq[V]) pairs
<code>reduceByKey(func, [numTasks])</code>	aggregate values for each key
<code>sortByKey([ascending], [numTasks])</code>	sort by key, ascending or descending
<code>join(otherDataset, [numTasks])</code>	join two datasets with the same key
<code>cogroup(otherDataset, [numTasks])</code>	group multiple datasets by key
<code>cartesian(otherDataset)</code>	all possible pairs between two datasets

RDD ACTIONS, EXACT SYNTAX

<code>reduce(func)</code>	aggregate, must be commutative and associative
<code>collect()</code>	all elements as an array to the driver
<code>count()</code>	number of elements
<code>first()</code>	the first element
<code>take(n)</code>	array with the first n elements
<code>takeSample(withReplacement, fraction, seed)</code>	random sample, with or without replacement

Pair RDDs hold elements that are key-value tuples (key, value), the key identifying and the value carrying the data, both of any type. Create them from a list of pairs with `parallelize()`, or from an existing RDD with `map()`.

PAIR RDD TRANSFORMATIONS AND ACTIONS, THE FULL LISTS

1. aggregateByKey · combineByKey · combineByKeyWithClassTag · flatMapValues · foldByKey · groupByKey · mapValues · reduceByKey · reduceByKeyLocally · sampleByKey · subtractByKey · keys · values · partitionBy · fullOuterJoin · join · leftOuterJoin · rightOuterJoin
A: collectAsMap · countByKey · countByKeyApprox · lookup · reduceByKeyLocally · saveAsHadoopDataset · saveAsHadoopFile · saveAsNewAPIHadoopDataset · saveAsNewAPIHadoopFile

– **DataFrame**: a distributed collection in **named columns**, like a relational table or an R data frame. Better optimization than RDD, supports CSV, JSON, AVRO, HDFS, Hive tables. Created from RDDs, SQL queries, or external data

– **Dataset**: an **extension of DataFrame**, combining RDD strong typing and lambdas with Spark SQL's optimized engine. **Scala and Java only**, **Python does not support it**

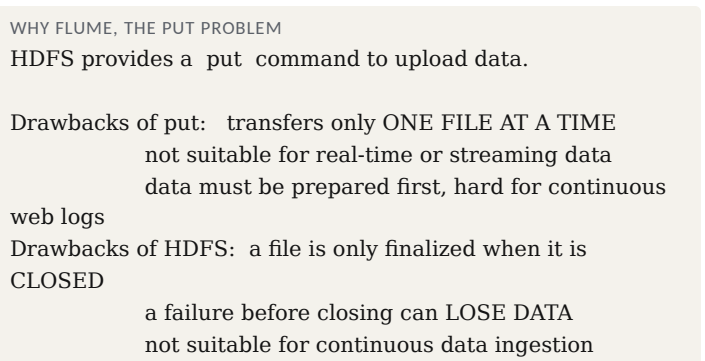
– **GraphFrames**: graph data in two **DataFrames**, vertices and edges. Breadth-first search, shortest path, PageRank

– **Tungsten**: an optimized memory engine using **off-heap memory** and **Kryo serialization**. Cached data uses **4 to 5 times less space**

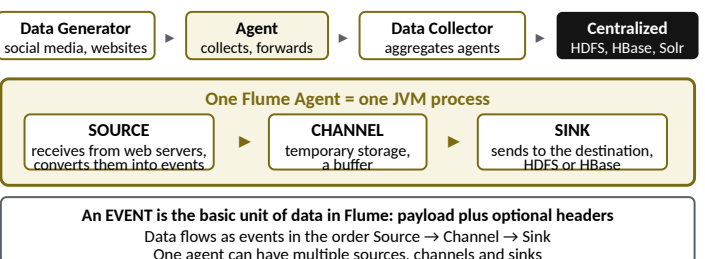
5 weaknesses. No **file management system** of its own, so it integrates with Hadoop or a cloud platform · in-memory capability becomes a **bottleneck** for cost-efficient processing · very high memory consumption · it requires large data · MLlib lacks algorithms, for example **Tanimoto distance**.

Ch12 Flume and Sqoop

Both exist for one reason: **HDFS** stores data but **cannot ingest it**. Flume brings streams in, Sqoop brings tables in.



Flume is the solution which can transfer the "streaming data" from servers to HDFS with little delay. It **collects, aggregates and transfers** large amounts of data to **HDFS or HBase**. Streaming sources are system logs and events from **network traffic, social media, emails**. Batch sources are tail, reading local files and sending them to HDFS, and **Apache log4j**, letting applications send log data to HDFS via Flume.



THE FLOW TO DRAW IN THE EXAM

Data Source → Source → Channel → Sink → HDFS

Features: flexible design for streaming flows · fault tolerant with failover and recovery · two delivery modes, **best-effort delivery** with no guarantee on failure and **end-to-end delivery** which guarantees it · continuous or batch transfer · processing or transformation before sending · multiple destinations.

FLUME ADVANTAGES	FLUME DISADVANTAGES
Stores streaming data into centralized systems, HDFS and HBase	Weaker message ordering guarantees
Steady data flow between producer and consumer	Messages are not guaranteed to be 100% unique
Supports contextual routing, ensures reliable message delivery	Complex topology and difficult reconfiguration
Reliable, scalable, extensible, fault-tolerant, customizable	May face scalability and reliability issues

Applications: e-commerce customer behaviour · moving large data from applications to HDFS quickly · **fraud detection** · **IoT systems** · machine and sensor aggregation · **alerting systems** and **SIEM**.

SQOOP

Sqoop exists because most applications still keep their data in **relational databases like Oracle, MySQL and SQL Server**, and those are important sources of big data.

THE TWO VERBS, LEARN THE DIRECTION

IMPORT move data from RDBMS → Hadoop (HDFS, Hive, HBase)
EXPORT move data from Hadoop → RDBMS

Sqoop works through a command-line interface, CLI.

No sqoop import --connect jdbc:... command lines appear anywhere in these notes. Sqoop is described only conceptually, through the two verbs and the seven architecture steps, so describe the process rather than inventing flags. Across the whole subject the only executable syntax given is MongoDB shell and CRUD, the Spark RDD API tables, and the one PySpark `countByKey()` snippet.

- The client submits an **import or export command**
- Sqoop uses **connectors**, MySQL, Oracle, SQL Server, to connect to the database
- It retrieves data by **dividing it into smaller parts**, to handle large datasets efficiently
- Multiple **map tasks** launch in parallel, each processing a portion, making transfer fast and scalable
- On **import**, data lands in HDFS for storage and **Hive or HBase** for querying and processing
- On **export**, the process reverses, HDFS data is divided and mappers send it back into the RDBMS
- All operations run in a **distributed and parallel** manner

Import: metadata is sent to Sqoop first, Sqoop analyzes the database structure for metadata such as **primary keys**, then launches a **map-only job**, divides the dataset into **splits**, and runs map tasks in parallel into HDFS. **Export**: gather **metadata** by **introspection**, divide the input into **splits**, and transfer with multiple map tasks while **maintaining the data schema**.

Robust	Reliable, easy to use, strong community
Incremental Load	Loads only updated or new data instead of the full table
Full Load	Imports an entire table or database with a single command
Parallel Import/Export	Multiple mappers via YARN, faster transfer with fault tolerance
Compression	Supports compression such as gzip
Connectors	Works with many RDBMS, MySQL, Oracle, SQL Server
Kerberos Security	Secure authentication using Kerberos
Accumulo Support	Can import directly into Accumulo instead of HDFS

Sqoop is a **sub-project of Hadoop**, typically used on **Linux**. **Java must be installed first**, and **Hadoop must be installed and configured** before using Sqoop.

CMP The comparison tables

BATCH VERSUS STREAM

	BATCH	STREAM
Engine	Hadoop MapReduce, stronger in batch mode , for wide volumes	Spark, stronger in streaming mode , for varied workloads
Data state	Static data, stored first, processed later	Consumed on the fly, not stored for later batch processing
Use when	The full dataset is available and completeness beats latency	Data is too fast to store entirely, or an immediate answer is required
Feeder	Sqoop , batch-oriented, not event-driven	Flume , event-driven, real-time streaming
Spark part	Spark Core and Spark SQL	Spark Streaming , micro-batching with D

Big Data Analytics & Visualisation

TEXT ANALYTICS: TURNING UNSTRUCTURED WORDS INTO NUMBERS A MODEL CAN LEARN FROM

The arc of the text half: text is **unstructured**, so no statistical model can touch it directly. You **clean it, count it** (Bag of Words, TF-IDF), then **give the counts meaning** (word embeddings), and only then classify, summarise, compare or generate. Each technique puts back what the previous one threw away: **order** (n-grams), **rarity** (IDF), **meaning** (Word2Vec), **context** (Transformers). Exam Q1 = NLP intro and levels of linguistic processing. Exam Q4 = text pre-processing plus BOW, TF-IDF and word embedding.

Day 4 Language models, transformers, generative AI

Language models exist to answer one question, "what word comes next", and almost every modern NLP product is built on that.

Definition: "a probabilistic model of a natural language that can generate probabilities of a series of words, based on text corpora it was trained on."

Two types: 1 N-gram language model, 2 Neural Language Models. Why predict upcoming words: ASR, "I will be back soonish" versus "I will be bassoon dish"- spelling correction - grammatical error correction - machine translation.

BIGRAM PROBABILITY. NOTE THE START AND END SYMBOLS
P(<S> i want to join dr halizah basiron class today <S>)
= P([<S>i] x P(want|<S>) x P(to|want) x P(join|to) x P(dr|join) x P(halizah|dr) x P(basiron|halizah) x P(class|basiron) x P(today|class) x P(<S>|today)
P(water | I want a glass of) = count(I want a glass of water) / count(I want a glass of)

N-gram advantages: simplicity and low memory usage. Limitations: limited context, a finite window that cannot capture dependencies beyond the previous N-1 words, and sparsity, as N grows the possible N-grams grow exponentially. Why neural beats n-gram: training saw "the cat gets fed" but never "rabbit gets fed", so at test time the n-gram LM cannot predict "fed" while the neural LM uses the similarity of the cat and rabbit embeddings to generalise. Architecture: input layer → embedding layer → hidden layer → output layer softmax.

LARGE LANGUAGE MODELS

AN LLM is a more advanced LM developed on vast text using deep learning, learning **billions of parameters**. Examples: **GPT-4** Generative Pretrained Transformer 4 · **BERT** Bidirectional Encoder Representations from Transformers · **ROBERTa** Robustly Optimized BERT Approach. The **three words:** Large, large training data and computing power · Language, generating human-like text · Models, learning complex patterns from text data.

Top 10 pre-trained NLP language models, in order: 1 BERT · 2 ROBERTa · 3 OpenAI's GPT-3 · 4 ALBERT · 5 XLNet · 6 OpenAI's GPT-2 · 7 StructBERT · 8 T5 · 9 ELECTRA · 10 DeBERTa. Related terms: Transformer, Recurrent Neural Network, Encoder-Decoder, Attention Mechanisms, Transfer Learning, Fine-tuning.

TRANSFORMERS, THE HISTORY TO BE ABLE TO TELL

In 2017 a Google paper proposed a novel neural architecture for sequence modelling. It outperformed RNNs on machine translation in both **translation quality and training cost**. In parallel **ULMFiT** showed that LSTM training on large diverse corpora could produce state-of-the-art text classifiers with **little labelled data**. These were the catalysts for **GPT** and **BERT**: combining the Transformer architecture with unsupervised learning removed the need to train task-specific architectures from scratch.

THE TRANSFORMERS TIMELINE, EXACT ORDER
Transformer → ULMFiT → GPT → BERT → GPT-2 → DistilBERT → GPT-3 → GPT-Neo → GPT-4 → GPT-4o

THE NESTING TO DRAW IN ANY GENAI QUESTION
AI > ML > DL > GenAI
AI simulation of human intelligence in machines, enabling learning, reasoning, problem-solving and autonomous decision-making

ML subset of AI where computers learn from data without explicit programming per task
DL subset of ML using neural networks with many layers
GenAI subset of DL that generates new content, text, images or music, from learned patterns

Six GenAI capabilities: Information Retrieval · Content Generation · Idea Generation · Language Translation · Proofreading and Editing · Visualization.

THE PROMPT FLOW
Prompt → Input → LLM → Output → Response
A prompt is "an instruction or discussion topic a user provides for the ChatGPT AI model to respond to".

Tools. ChatGPT by OpenAI, GPT = Generative Pre-Trained Transformer. Gemini, formerly Google Bard, multimodal, taking text plus images, audio, videos and code. Microsoft Copilot, embedded in Word, Excel, PowerPoint and Outlook, suggests relevant libraries and functions.

Tasks at work with ChatGPT: facts and details · translations · creative and formal writing · summarizing reports and meeting notes · suggestions · brainstorming · comparisons · lists · pros and cons · feedback · role playing · explaining a concept.

Five harms of LLMs, name and define each.
Multitasking, "the generation of outputs that may sound plausible but are either factually incorrect or unrelated to the given context".
Copyright, authors sue OpenAI over novels. The Times sues OpenAI and Microsoft.
Bias, "How Strangers Got My Email Address From ChatGPT's Model".
Privacy and Misinformation.

Plagiarism and Cheating, posing off AI generated content as original, paraphrasing plagiarised content, fabricating data. An AI detector may be used to detect this offence.

Day 3 Word embeddings and vector semantics

Embeddings exist because counts throw away meaning: "squad" and "team" share no vocabulary yet say the same thing.

THE MOTIVATING PAIR FROM THE NOTES
"This squad is ready to win the next Malaysia Cup match."
"This team is prepared to achieve victory in the upcoming MC game."
Same meaning. No shared vocabulary. BOW sees nothing.

RELATION	DEFINITION AND EXAMPLES
Synonymy	Same meaning in some or all contexts. fibert/hazeltut, couch/sofa, big/large, automobile/car, water/H2O. Exceptions: "H2O" in a surfing guide; my big sister is not my large sister
Similarity	Not synonyms but sharing some element of meaning, cat/dog, car/bicycle, cow/horse, man/woman, crocodile/tiger. Simlex-999 (Hill et al. 2015) scores 0 to 10, from near-synonyms (vanish, disappear) to pairs with nothing in common (hole, agreement)
Word Relatedness	coffee and cup share practically no features yet are related, by co-participating in an everyday event. A semantic field covers a semantic domain with structured relations: hospital (surgeon, scaled, nurse, anaesthet), restaurants (waiter, menu, plate, chef)
Antonymy	Opposites in only one feature of meaning, otherwise very similar. dark/light, short/long, fast/slow, rise/fall, hot/cold, up/down. They define a binary opposition or sit at opposite ends of a scale
Connotations	Affective meaning. Positive: copy, replica, reproduction. Negative: fake, knockoff, forgery. Osgood et al. 1957 , three dimensions: valence pleasantness of the stimulus, arousal intensity of emotion provoked, dominance degree of control exerted

Two roots of vector semantics, both 1950s. Wittgenstein: "the meaning of a word is its use in the language", so define meaning by linguistic distribution. Osgood et al. 1957: meaning as a point in multidimensional space. Zellig Harris 1954: if A and B have almost identical environments we call them synonyms, worked on onchloi, inferred to be a leafy green like spinach purely from shared contexts.

Sparse versus dense. BOW and TF-IDF give sparse vectors, 0 and 1 values, no meaning, mostly 0, where words are represented by a function of the counts of **nearby words**. They are the Information Retrieval workhorse and a common baseline. **Word2vec** gives dense vectors, no or few zeros, and they have meaning: the representation is created by **training a classifier to predict whether a word is likely to appear nearby**.

Day 2 The NLP pipeline and pre-processing, in order

Pre-processing exists because raw text carries noise that degrades every downstream algorithm.

STANDARD NLP WORKFLOW, 4 ORDERED STEPS
1 Collect text data, then standard text pre-processing, parsing and basic exploratory data analysis
2 Represent the text using relevant feature engineering
3 Build predictive supervised or unsupervised models, or NLP symbolic models
4 Evaluate against the success criteria with stakeholders, then deploy the final model

Noisy text data is found in informal settings (online chat, SMS, e-mails, message boards) and in text produced through **automated speech recognition** or OCR. Noise degrades **classification, clustering, summarization and information extraction**. Pre-processing converts raw text into well-defined sequences of linguistic components that have **standard structure and notation**. Three common tasks: 1 Word Tokenization, 2 Word Format Normalization, 3 Sentence Segmentation.

THE PRE-PROCESSING ORDER, OUTPUT OF EACH TASK FEEDS THE NEXT
RAW TEXT CORPUS
1 Word tokenization corpus to sentences to words
2 Normalization special chars, noise, accents, HTML
3 Expand contractions mapping list, not! becomes do not
4 Case conversion upper, lower, sentence, proper case
5 Remove stopwords a, the, me
6 Correct words repeated chars, then spelling
7 STEM or LEMMATIZE, then feature engineering

WORD TOKENIZATION

Breaking textual data into smaller meaningful components called **tokens**. Pipeline: a **text corpus to sentences to words**. **Space-based tokenization** segments off a token between instances of spaces. **Three issues.** (a) You cannot blindly remove punctuation: m.p.h., Ph.D., AT&T, cap'n, M'sia, KL. (b) A dtic is a word that does not stand on its own, "are" in "we're", "am" in "I'm". (c) **MultitwoD expressions (MWE)**: New York, rock'n'roll, Kuala Lumpur.

NLTK TOKENIZERS, EXACT NAMES
word_tokenize TreebankWordTokenizer
RegexTokenizer inherited tokenizer from Regexp
Languages without spaces: Chinese, Japanese, Thai. In Chinese it's common to treat each character (zi) as a token, so segmentation is very simple; Thai and Japanese need more complex segmentation. Standard algorithms are **neural sequence models trained by supervised machine learning**.
Subword tokenization, three algorithms: Byte-Pair Encoding (BPE) Senrich et al. 2016 · Unigram language modeling tokenization Kudo 2018 · WordPiece Schuster and Nakajima 2012. All have two parts: a **token learner** that takes a raw training corpus and induces a vocabulary, and a **token segmenter** that tokenizes a raw test sentence according to that vocabulary.

Day 4 Text classification and evaluation

Classification exists to route documents into a set of classes you defined in advance.

FORMAL STATEMENT
"The process of assigning text documents into one or more classes or categories, assuming that we have a predefined set of classes."
Input: a document d, and a fixed set of classes
C = {c1, c2, ..., cq} Output: a class c in C
Supervised adds a training set (d1, c1), ..., (dm, cm) and outputs a classifier gamma: d → c

Rule-Based. Handcrafted linguistic rules; each rule is an **antecedent or pattern plus a predicted category**. Spam: black-list-address OR ("do!lars" AND "you have been selected"). Sports versus Politics: define word lists, count each higher count wins. High accuracy if rules are carefully refined by an expert, and human comprehensible, but it needs **deep domain knowledge**, is slow to build, hard to maintain and it does not scale well.

Machine Learning. Learns classifications from past observations. Supervised, Unsupervised, Reinforcement, Semi-supervised. Usually much more accurate than human-crafted rules on complex NLP tasks, easier to maintain and re-trainable with new tagged data.

Three types. Binary, exactly two classes. **Multi-class or multinomial**, more than two, one predicted per instance. **Multi-label**, each prediction can yield more than one class per data point.

PIPELINE AND BLUEPRINT
Preparation of datasets → Text Pre-Processing → Feature extraction → Model training → Model evaluation & tuning → Model deployment
TRAINING Training Classes + Training Documents
→ Text Normalization → Feature Extraction
→ Training Features → Supervised ML Algorithm
PREDICTION Testing Documents + Testing Classes
→ Testing Features → Classification Model
→ Predicted Classes → Model Prediction Performance Evaluation

Labelling and annotation. Text labelled by human experts is **data annotation**. With more than one annotator an **inter-annotator agreement** is measured, for example **Kappa Statistics**. The notes show worked cases of **100% agreement and 50% agreement**. When the score falls below the threshold, say 80%, the data labelling is redone.

Honest gap on Cohen's kappa. Your notes name **Kappa Statistics** as the inter-annotator agreement measure and give the **80% redpo kappa**, but the kappa formula itself is never written down in either field. Quote the name, the purpose and the threshold. Do not quote a formula you were not given.

Feature extraction models. Traditional: BOW, Bag of N-Gram, TF-IDF. Advanced: Word2Vec, GloVe, FastText, ELMo.
Algorithms named: Naïve Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), Neural Networks (NN), k-Nearest Neighbors (k-NN). Most effective for text: **Multinomial Naïve Bayes (MNB)** and **SVM**. MNB is the special case of Naïve Bayes for more than two classes. Naïve Bayes is supervised and applies Bayes theorem. SVM is supervised, for classification, regression, novelty and anomaly detection; training samples are points in space and labels are separated by a wide gap called a **hyperplane**, and new points are classified by which side they fall on.

Model evaluation tests the validation set against the actual class labels, the ground truth. k-fold Cross Validation divides data into folds, a chunk trains and the remainder validates, then the final evaluation runs on the test data. **Model tuning** is hyperparameter optimisation: hyperparameters cannot be learned from data and must be set before training, by randomized search or grid search.

Day 1 NLP and the levels of linguistic processing

This box is exam Q1: "Intro to NLP and Levels of Linguistic Processing". The student's own flag on the six levels: "Kena tdk each of this please".

NLP exists so humans can address machines in ordinary language instead of a query syntax.

Definition: a specialised field of **Computer Science** under **Artificial Intelligence**, involved with **Computational Linguistics**, enabled through designing and building applications and systems that enable humans to interact with machines using natural languages. Related to **Human-Computer Interaction (HCI)**.
NLP around you. 7: grammar and spelling checker · machine translation · chatbots · sentence paraphraser · email reply suggestions · next word prediction · generative AI tools. **Main applications:** 6: Machine Translations · Speech Recognition Systems · Question Answering Systems · Contextual Recognition and Resolution · Text Summarization · Text Classification. **Philosophy of language**, 4: nature of meaning · use of language · language cognition · relationship between language and reality. **Linguistic acquisition** is the ability of acquiring and producing languages.

ERA	DATES	CONTENT
Symbolic	1950s to early 1990s	1950 Alan Turing "Machine and Intelligence", 1957 Noam Chomsky , Syntactic Structures, Phrase-Structure Grammar, 1964 ELIZA by Joseph Weizenbaum , initiated a psychiatrist by template-matching
Statistical	1990s to 2010s	Statistical inference. SMT built from a corpus of approved previous translations into a statistical model, then a probability-driven match
Deep Learning	2010s to 2020s	Statistical NLP needed elaborate feature engineering. Neural nets. NMT learns codes to sequence transformations directly

THE SIX LEVELS, MEMORISE THE DEFINITIONS

LEVEL	DEFINITION AS GIVEN
1 Phonological	The study of sound patterns as interpreted in the human mind, used to distinguish different phonemes . A phoneme is the smallest individual unit of human speech in a specific language
2 Morphological	The study of word structures and word formation , analysing the individual components of words. A morpheme is the smallest unit of language that has distinctive meaning
3 Syntax	The study of how words are put together to form correct sentences and the structural role each word has. The structure and ordering of components within a sentence
4 Semantic	The study of word meaning in a language and how these meanings combine in sentences to form sentence meanings
5 Discourse	The study of how the immediately preceding sentences affect the interpretation of the next sentence
6 Pragmatic	The study of how sentences are used in different situations and how use affects the interpretation of the sentence

Word classes: Determiner (Det) a, an, the · Pronoun (Pron) he, she, they · Preposition (Prep) to, with, at · Conjunction (Conj). **POS tags:** NN, NNP, NNS. **Five phrase categories**, each round a **head word**: **NP** noun head, subject or object, replaceable by a pronoun · **VP** verb head · **ADJP** adjective head, "too quick" · **ADVP** adverb head, "pretty soon" · **PP** preposition head, acting as adjective or adverb, "going up the stairs". **Clauses** contain a **subject** and a **predicate**, main clauses stand alone, subordinate ones do not.

Five clause types with the given examples. **Declarative**, neutral statement, "Grass is green." · **Imperative**, request, command, rule or advice, "Please do not talk in class." · **Relative**, subordinate and dependent, "he wanted a sofa" · **Interrogative**, questions affirmative or negative, "Did you get my mail?" · **Exclamative**, shock, surprise or compliments, ending with "!", "What an amazing race!"

Seven lexical semantic relations: Lemmas and Wordforms · Homonyms, Homographs and Homophones · Heteronyms and Heterographs · Polysemes · Captonyms · Synonyms and Antonyms · Hyponyms and Hypernyms. Lexical units are **morphemes**; a **lexicon** is the complete vocabulary of those units.

Text corpora are large structured collections of written or spoken text stored electronically, used for linguistic and statistical analysis and to build NLP tools. **Seven popular corpora:** Key Word in Context (KWIC) · Brown Corpus · LOB Corpus · Collins Corpus · CHILDES · WordNet · Penn Treebank.

Challenges in NLP, two families. Data related: low-resource languages, thousands are not catered for; large or multiple documents, cannot process longer context, expensive to obtain. **Understanding related:** Ambiguity same word different meaning · **Synonymy** different word same meaning · **Co-reference resolution** · **Personality, intention, emotion** and styles, irony and sarcasm.

THE 10 NLP LIBRARIES in order: 1 NLTK · 2 spaCy · 3 Gensim · 4 CoreNLP · 5 Pattern · 6 TextBlob · 7 PyNLPI · 8 scikit-learn · 9 Polyglot · 10 Pytorch. **NLP APIs** are machine-learning models for analysing text, for syntax analysis and entity extraction: **Google NLP API**, **IBM Watson API**, **Azure Cognitive Service**, **Aylien API**.

Day 5 Text similarity

Similarity exists because search, plagiarism checks and duplicate detection all reduce to "how close are these two strings".

DEFINITION AND THE METRIC PROPERTIES
"Text similarity process is to determine how 'close' two pieces of text are in both in surface closeness [LEXICAL similarity] and meaning [SEMANTIC similarity]".
Four properties a distance metric d must satisfy:
1 Non-negativity d(x,y) ≥ 0
2 Identity d(x,y) = 0 if and only if x = y
3 Symmetry d(x,y) = d(y,x)
d("sleep", "hungry") = d("hungry", "sleep")
4 Triangle inequality d(x,z) ≤ d(x,y) + d(y,z)

Lexical similarity splits into **term similarity**, individual tokens and words, and **document similarity**, entire documents. **Semantic similarity** determines meaning and context.

METRIC	DEFINITION	READING
Hamming (HD)	Number of positions with different characters or symbols between two strings of equal length . Popular in information theory	Smaller means more similar
Manhattan (HD)	Subtracts the difference between each pair of characters at each position of the two strings, using ASCII codes	Smaller means more similar
Eucledian (ED)	Shortest straight-line distance between two points	Smaller means more similar
Levenshtein Edit (LED)	Minimum number of edits, deletions, insertions, substitutions , to turn the source string into the target	Bigger means LESS similar
Cosine	Cosine of the angle between two non-zero positive vectors. cos θ deg = 1, the two words are equal; cos 90 deg = 0, totally different	Bigger means more similar

Jaccard is not in your notes. The five metrics taught in MAXD 5153 are **Hamming, Manhattan, Eucledian, Levenshtein and Cosine**. Jaccard similarly appears nowhere in either digest, so do not present it as course content.

Applications of term similarity: spelling checker, plagiarism detector. Spelling pairs: calendar/calender, definitely/definitely, tomorrow/tommorrow, noticeable/notifiable. Use cases: Information Retrieval from a corpus · search engines and duplicate-content detection on **Quora and Stackoverflow** · legal, risk mitigation on a new contract by similarity to an existing resilient one · customer services, uniform responses to semantically similar queries.

DOCUMENT SIMILARITY, 3 STEPS
1 Do text pre-processing
2 Vectorize the document, using TF-IDF, BOW or a word embedding
3 Compute similarity between documents, cosine similarity

Word embedding for similarity: dense vector representations capturing semantic meaning, generated by **Word2Vec, GloVe or FastText**. Document similarity is computed by **averaging the word embeddings of the words in each document and computing the cosine similarity between the resulting vectors**.

Document clustering is unsupervised ML: "the process of segregating a whole corpus of documents into various groups based on some distinctive properties, attributes, and features of the documents." **Three techniques:** K-means, **Affinity propagation**, **Ward's Agglomerative Hierarchical**. **Reading:** Text Analytics with Python, Dipanjan Sarker, 1st ed., pages 265 to 317.

Day 5 Topic modelling: LSI and LDA

LSI, LATENT SEMANTIC INDEXING

A statistical technique to correlate semantically linked terms. **Main principle:** "similar terms tend to be used in the same context and hence tend to co-occur more". It uses **Singular Value Decomposition (SVD)**, a matrix factorization method.

LSI, THREE STEPS AND THE FORMULA
1 Creating a document-term matrix
2 Singular Value Decomposition
3 Sentence Selection
 $A = U \cdot S \cdot V^T$ -transpose
A document-term matrix, rows are terms, columns are documents or sentences
U, k singular values, evaluates the "strength" or importance of each topic. Dimensions with the HIGHEST singular values are the most important
V-T-term-topic matrix, values show the strength of association between each term and the derived topic
n = number of documents, m = unique terms, k = number of topics. Values may be TF-IDF, word frequency, binary value, log frequency or root.

Four LSI drawbacks. Word order is completely neglected, causing information loss · unable to account for homonymy or polysemy · difficult to determine the **optimal number of topics** · **lacks interpretability**.

LDA, LATENT DIRICHLET ALLOCATION

A generative probabilistic model. Each topic is a mixture over a set of words, each document a mixture over a set of topic probabilities. Given M documents, N words and prior K topics it outputs psi, the distribution of words for each topic K, and phi, the distribution of topics K for each document i. It converts the document-term matrix into **M1, document-topics (N, K)** and **M2, topic-terms (K, M)**.

Alpha is the Dirichlet prior concentration parameter for **document-topic density**; higher alpha means documents are assumed made of more topics, giving a more specific topic distribution per document. **Beta** is the prior for **topic-word density**; high beta means topics are assumed made of most of the words, giving a more specific word distribution per topic.

THE LDA ALGORITHM, EXACT STEPS
1 Initialize the necessary parameters
2 For each document, randomly initialize each word to one of the K topics
3 Iterate several times. For each document D, word W, topic T: P1 = P(T|D) proportion of words in D currently assigned to topic T
P2 = P(W|T) proportion of assignments to topic T over all documents that have this word W
Reassign word W with topic T with probability P(T|D) x P(W|T)
The assignment is updated with the PRODUCT of P1 and P2.
The model assumes all existing word-topic assignments except the current word are correct. After a number of iterations a steady state is reached, the CONVERGENCE POINT of LDA.

Evaluating a summary. **Human Evaluation:** experts score coverage of points, answering of queries, grammaticality, non-redundancy. **Automatic Evaluation:** ROUGE, BLEU, METEOR, BERTScore. ROUGE = Recall-Oriented Understudy for Gisting Evaluation, compares the sums to human reference summaries, variants **ROUGE-1, ROUGE-2, ROUGE-L**. BLEU = Bilingual Evaluation Understudy, measures **n-gram overlap** and also considers **summary length**, score 0 to 1, closer to 1 is better.

BART appears in your notes **only** as the abstractive notebook name
TextSummarization using BART. Its architecture is described nowhere in either digest, so name it as the abstractive tool you used and leave the internals alone.

Day 2 Text visualisation

Text visualisation exists because you cannot read a corpus, but you can look at one.
DEFINITION, GIVEN TWICE IN THE NOTES
"The technique of using graphs, charts, or word clouds to showcase written data in a visual manner."

Why, four reasons. i To summarize the large amount of text data, get the "gist". ii Grouping, classification or cluster for overview. iii Comparison, compare document collections, or inspect the evolution of a collection over time. iv Correlation, compare patterns in text to those in other data, for example correlate with a social network.

Four purposes. Summarize large amounts of text, automatically highlighting key terms across a series of texts and categorising by topic and sentiment, saving hours of reading time · make text data easy to understand · find insights in qualitative data · discover hidden trends and patterns, analysing insights over time to detect fluctuations and quickly find the root cause.

The eight types: WORD CLOUDS · BAR CHARTS · HEAT MAPS · SCATTER PLOTS · NETWORK GRAPHS · TREE MAPS · SANKEY DIAGRAMS
WORD TREE Plus "Positive, Neutral, Negative", a visualisation of words showing positive words, negative words and neutral function words.

CHALLENGE	WHAT TO DO ABOUT IT
High Dimensionality	Where possible use text to represent text. Which terms are the most descriptive
Context & Semantics	Provide relevant context to help understanding. Show or provide access to the source text
Modeling Abstraction	Determine your analysis task. Understand the abstraction of your language models. Match the analysis task with appropriate tools and models

Day 3 Feature engineering: BOW and TF-IDF

This box and the next are exam Q4, part 2: "Feature engineering: Bag of Words, TF-IDF and Word Embedding."

Feature engineering exists because models understand only numeric representations of features, never raw text.

ML PIPELINE, ORDERED
Data collection → Data pre-processing → Feature extraction → Data training → Data validate & testing

Three reasons text needs it: text is free-flowing words, sentences and paragraphs · ML and DL models only understand **numeric representations of features** as inputs · the lack of structure and noisy nature of text make raw text hard to work with. **Three families. Traditional:** 1 One-hot Encoding, 2 Bag of Words, 3 Bag of N-Gram, 4 TF-IDF. **Stylometric:** writing style analysis. **Advanced:** Word Embedding.

VECTOR SPACE MODEL, ALSO TERM VECTOR MODEL
"A mathematical and algebraic model for representing and representing text documents as numeric vectors of specific terms that form the vector dimensions."
 $VS = \{w_1, w_2, \dots, w_n\}$
 n = number of distinct words across all documents
 $D = \{wD1, wD2, \dots, wDn\}$
 wDn = weight (numeric) for word n in document D
weight may be frequency of words, average frequency of occurrence, embedding weights, or TF-IDF value

BAG OF WORDS

Each document becomes a vector representing the **frequency of all the distinct words** present in the document vector space for that document.

WORKED EXAMPLE, REPRODUCE THIS
D1: John likes to watch movies. Mary likes movies too.
D2: John also likes to watch football games.
Vocabulary = john, likes, like, watch, movies, mary, too, also, football, games
Vector length is ALWAYS the vocabulary size, here = 10
Initialise with zeros: {0,0,0,0,0,0,0,0,0,0}
Compare each vocabulary word, INCREMENT if present.
D1 = {1, 2, 1, 1, 2, 1, 1, 0, 0, 0}
D2 = {1, 1, 1, 1, 0, 0, 0, 1, 1, 1}

Six limitations of BOW. 1 Does not consider the meaning of the word. 2 Vectors are based completely on **absolute frequencies**. 3 Higher-frequency words **overshadow** lower-frequency ones. 4 Vector dimension is **linearly dependent on the number of word types**. 5 It **does not reveal relationships between words**. 6 "Ahmad is smarter than Ali" and "Ali is smarter than Ahmad" give the **same vector representation**.

Bag of N-Grams. A word alone is a **unigram** or **1-gram**. An **N-gram** is "a collection of words taken from a text document such that these tokens are contiguous and occur in a sequence". **Bigram** = 2 words, **Trigram** = 3. It exists because BOW ignores word order: "love mobile" is more informative than the two words apart.

TF-IDF

TF-IDF exists to fix the BOW problem that terms occurring frequently across all documents overshadow the terms that actually identify one document.

THE FORMULAS, MEMORISE
 $tfidf = tf \times idf$
 $tf(w, D) = f_w(D)$
the raw frequency value of word w in document D
 $idf(w) = \log(N / df(w))$
 N = total number of documents in the corpus
 $df(w)$ = number of documents containing t

FOUR WAYS OF CALCULATING TF, IN ORDER
1 raw count, number of times the word appears in a document
2 adjusted for document length:
raw count / number of words in the document
3 logarithmically scaled: $\log(1 + \text{raw count})$
4 Boolean: 1 if the term occurs, 0 if it does not

The interpretation rules, quote these. Words that occur rarely in the corpus have a high IDF. TF-IDF gives larger values for less frequent words. TF-IDF is high when BOTH IDF and TF are high, that is, the word is rare in the whole corpus but frequent in this document. like BOW, TF-IDF only focuses on frequency, neither takes semantic meaning into account.
Two use cases: information retrieval, avoid all search engines, where TF-IDF ranks results by relevance; and text summarization and keyword extraction, where the highest-relevance words become the summary or the tags.

Working readed in the notes: Text 1 gives high values for "but", "hate", "language", "love", "natural", "lyth"; Text 2 for "image" and "like"; in text 3, "and" and "signal" matter more than "image" and "like".

The exercise seen in the notes, do both ways. Corpus of 4 reviews:
1 This mobile is good. 2 This mobile is not good. 3 Mobile is cheap and mobile is best. 4 Mobile is cheap but screen is small.
(a) Extract Bag of Words and Bag of Bigram. (b) Extract TF-IDF from the above review text.

Day 2 Regular expressions and error types

Regex exists because most cleaning tasks are pattern matches, not word lookups. **Definition:** "a formal language for specifying text strings", used in every computer language, word processor and text processing tool, for searching when you have a **pattern** and a **corpus**.

GROUPINGS AND PER-SYMBOL GLOSSES, AS PRINTED
Metacharacters: () [] { } * + ? \ . * + ? \ . * + ?
Quantifiers: ? * + ? * + repetition characters
Anchors: ^ \$
Lookarounds: (?=) positive lookahead
(?) negative lookahead
(?<) positive lookbehind
(?<) negative lookbehind
\b Any word boundary character
\B Anything but a word boundary
\w Any non-word character
\W Any word character (letter, number, underscore)
\d Any digit
\D Any non-digit
\s Any non-whitespace character
\S Any whitespace character

MAXD 5153 • SHEET 2 OF 3

Speech & Image Processing

SOUND AND PICTURES ARE JUST SIGNALS, SO THE WHOLE TOOLKIT IS FILTERING, THRESHOLDING AND DESCRIPTION

REVISION SHEET 2 OF 3

Both halves of this sheet do the same thing to a different signal. **Speech** travels the speech chain from a linguistic idea, through articulators, through air, back to a linguistic idea, and the machine either **makes it (TTS)** or **reads it (ASR)**, then **analyses it** (speech analytics, exam Q2, whose full answer is on Sheet 3). **Images** go through nine stages, acquisition to colour processing, and every one of them is either a **point operation** on one pixel, a **neighbourhood operation** with a kernel, or a **description** of a shape once it has been segmented. Exam Q3 lives in the image half: **HSI conversion, processing stages, and the chain code**.

Day 6

Automatic Speech Recognition

ASR exists to turn the acoustic waveform back into the words that produced it.

DEFINITION

"Speech recognition is the ability of a machine or program to identify words and phrases in spoken language and convert them to a machine-readable format."

Rudimentary software has a limited vocabulary and needs very clear speech. More sophisticated software accepts natural speech.

Honest gap, read this before revising MFCC or WER. In your notes the ASR architecture, the MFCC feature-extraction steps and the Word Error Rate formula exist only as untranscribed slide images. Neither digest contains a written MFCC step list or a written WER formula, so nothing is invented here: copy the feature-extraction pipeline and the WER equation off the Day 6 slides. The same applies to the spectrogram, never defined in writing, and to a numeric sampling rate for speech: the only frequency example written anywhere is 5 cycles in 0.5 seconds = 10 Hz, and the only written sampling and quantisation treatment is the image one on Day 8.

Speech datasets named: Open SLR
https://www.openslr.org/resources.php • Vox Forge
https://www.voxforge.org • Voices
https://github.com/voicelabs/voicelabs • Over 1.5 TB's of Labeled Audio Datasets". Notebook SpeechRecognition.ipynb.

Day 10

Segmentation

Segmentation exists to split an image into the groups of pixels that correspond to actual objects.

DEFINITION

Segmentation "attempts to PARTITION the pixels of an image into groups that strongly correlate with the objects in an image." From the lab: "Segmentation, Partition. Portioning the pixel on which group they belong."

The taxonomy. On the basis of geometry. Discontinuity Detection finds isolated points, lines and edges. On the basis of intensity, Grayvalue Similarity gives thresholding.

Three basic types of gray level discontinuity. Points, isolated points - Lines, horizontal, vertical, oblique at 45 or minus 45 degrees - Edges, the boundaries of objects. "We typically find discontinuities using masks and correlation."

Point detection. An isolated point is "a point whose gray level is significantly different from its background in a homogeneous area", found with a single mask: points are detected at pixels in the filtered image that are above a set threshold. Edge "a set of connected pixels that lie on the boundary between two regions of different intensity or gray value". Edge detection problems: too much detail, for example brickwork, and "one way to overcome this is to smooth images prior to edge detection". The lab adds "Edge detection is very sensitive to the noise" and "1st derivative, where an edge is; 2nd derivative, show the edge direction".

THRESHOLDING

Thresholding is "a simple and effective way of partitioning an image into a foreground and background". It isolates objects by converting grayscale images into binary images, and is most effective in images with high levels of contrast.

Histogram modality. 1 peak is unimodal, for example a uniform forest, one object, so no partitioning is needed. Single value thresholding only works for bimodal histograms. Images with multimodal histograms need more than a single threshold. The lab writes this as "Simple Single Value Thresholding: 1 Peak (Unimodal) becomes 2 Peaks (Bimodal)".

Threshold selection rule, too many pixels from the foreground. Threshold too high, too many pixels from the background.

BASIC GLOBAL THRESHOLDING. THE EXACT 5 STEPS

- 1 Select an initial estimate for T, typically the average gray level in the image
- 2 Segment the image using T to produce two groups of pixels: G1 = pixels with gray levels > T
G2 = pixels with gray levels <= T
- 3 Compute the average gray levels of the pixels in G1 to give mu1, and of G2 to give mu2
- 4 Compute a new threshold value:
T = (mu1 + mu2) / 2
- 5 Repeat steps 2 to 4 until the difference in T in successive iterations is less than a predefined limit

Note on step 4. The Notion source renders this formula garbled as "T=2mu1+mu2", a nangled fraction. The intended formula is the mean of mu1 and mu2, written above. Global thresholding is based on the histogram and partitions it using a single global threshold, so success depends strongly on how well the histogram can be partitioned.

Basic Adaptive Thresholding divides an image into sub-images and thresholds these individually. Because the threshold for each pixel depends on its location, the technique is adaptive. Success is mixed, but troublesome sub-images can be further subdivided until thresholding succeeds. The lab note: "totally depends on location of the pixel."

Honest gap on OTsu and region growing. OTsu's method is never mentioned in either digest, and region growing is never mentioned either. The segmentation methods you were actually taught are discontinuity detection (points, lines, edges) and thresholding (global, the 5-step iterative algorithm above, and adaptive). Do not present OTsu or region growing as course content. Your own flag on this section reads "Steps, Tengok lecture".

Named edge detectors from the lab: Roberts, minus 45 and plus 45, using 0, 1 and 1 only. Prewitt, horizontal and vertical, can contain minus 1 and 1. Sobel, can have any pixel value.

Day 8

History, the eye, sampling and resolution

HISTORY, THE DATES TO NAME

Early 1920s, one of the first applications was in the newspaper industry, the Bartlane cable picture transmission service, with images transferred by submarine cable between London and New York, coded for cable transfer and reconstructed on a telegraph printer. Mid to late 1920s, higher quality from new reproduction processes based on photographic techniques, and an increased number of tones. 1960s, computing improvements plus the space race: in 1964 computers improved the quality of moon images taken by the Ranger 7 probe, and were used in the Apollo landings. 1970s, medical applications; in 1979 Sir Godfrey N. Hounsfield and Prof. Allan M. Cormack shared the Nobel Prize in medicine for the invention of tomography, the technology behind Computerised Axial Tomography (CAT) scans. 1980s to today, image enhancement and restoration, artistic effects, medical visualisation, industrial inspection, law enforcement, human computer interfaces.

THE HUMAN VISUAL SYSTEM

The lens focuses light onto the retina, covered with light receptors: cones, 6 to 7 million, and rods, 75 to 150 million. Cones are concentrated around the fovea and are very sensitive to colour; rods are more spread out and sensitive to low illumination. Muscles change the shape of the lens to focus near or far. Blind-spot experiment: draw a dot and a cross about 6 inches apart, close your right eye, focus on the cross with your left eye, hold the image about 20 inches away and move it slowly towards you; the dot should disappear.

Brightness adaptation and discrimination. The human visual system perceives approximately 10¹⁰ different light intensity levels, but at any one time discriminates a much smaller number, which is brightness adaptation. The perceived intensity of a region is related to the light intensities of surrounding regions, illustrated by Mach bands and by simultaneous contrast. Light is the part of the electromagnetic spectrum sensed by the human eye, split by wavelength. If white light is shone onto a green object most wavelengths are absorbed while green is reflected.

REPRESENTATION, SAMPLING, RESOLUTION

A digital image is composed of M rows and N columns of pixels, most often grey levels in the range 0 to 255, black to white, represented as matrices. Images are generated by illuminating a scene and absorbing the energy reflected by objects; typical notions of illumination may be off, as with X-rays of a skeleton or ultrasound of an unborn baby. In image sensing, incoming energy lands on a sensor material responsive to that energy, generating a voltage, and collections of sensors capture images.

SAMPLING, QUANTISATION, RESOLUTION, BIT DEPTH

A digital sensor can only measure a LIMITED NUMBER OF SAMPLES at a DISCRETE SET OF ENERGY LEVELS.

QUANTISATION "the process of converting a continuous analogue signal into a digital representation of this signal"

SPATIAL RESOLUTION "the smallest discernable detail in an image", determined by how sampling was carried out. Vision specialists talk about PIXEL SIZE, graphic designers about DOTS PER INCH (DPI)

INTENSITY LEVEL RESOLUTION "the number of intensity levels

used to represent the image". The more levels, the finer the detail. Usually given as the number of BITS used to store each intensity level. This is the BIT DEPTH.

Also covered: SATURATION AND NOISE.

How much resolution is enough? It depends on what is in the image and what you want to do with it. The key question is: "Can you see what you need to see within the image?"

Day 6

Phonetics, phonemes, the sound classes

Phonetics exists because a recogniser has to map a continuous wave onto a finite alphabet of sounds.

Five areas of linguistics: Phonology, Morphology, Syntax, Semantics, Pragmatics. Phonology is "the study of linguistic sounds, how they are produced by the articulators of the human vocal tract, how they are realized acoustically, and how this acoustic realization can be digitized and processed". A phoneme is a unit of sound that distinguishes one word from another in a particular language. /d/ for dark, /b/ for dark; English has about 40 phonemes.

Two phonetic alphabet standards: IPA, the International Phonetic Alphabet, an evolving standard originally developed by the International Phonetic Association in 1888 to transcribe the sounds of all human languages; the same utterance can be transcribed in different ways per IPA principles.

ARPAbet, the Advanced Research Project Agency Alphabet, designed specifically for American English using ASCII symbols, a convenient ASCII representation of an American-English subset of the IPA, used where non-ASCII fonts are inconvenient, such as on-line pronunciation dictionaries.

Two main areas of phonetics: Articulatory phonetics, how speech sounds are produced by the articulators in the mouth, and Acoustic phonetics, the acoustic analysis of speech sounds.

Sound production. Air is expelled from the lungs through the windpipe or trachea and out of the mouth and nose, passing the larynx, Adam's apple or voice box. The vocal folds or vocal cords are two small folds of muscle in the larynx, and the glottis is the space between them. Folds close together but not tightly closed vibrate, folds far apart give no vibration.

CLASSIFICATION OF SPEECH SOUNDS, THE FOUR CLASSES

VOICED vocal folds together and vibrating at a particular frequency [b] [d] [g] [v] [z]

VOICELESS made without vocal cord vibration, air flows continuously with high frequency [p] [s]

NASAL air passing through the nose [m] [n] [ŋ]

PLOSIVE complete closure towards the front of the vocal tract [p] [t] [k]

Two phone classes. Consonants restrict or block airflow and may be voiced or voiceless. Vowels have less obstruction, are usually voiced and are generally long-lasting. Semivowels [y] and [w] have properties of both, voiced like vowels but short and less syllabic like consonants.

ACOUSTIC PHONETICS, THE NUMBERS GIVEN

Analyses sound waves through varying FREQUENCIES, AMPLITUDES and DURATIONS, based on sine and cosine functions. Frequency example: 5 cycles in 0.5 seconds = 10 cycles per second = 10 Hz

Waveform axes, utterance "she just had a baby":
X = time in seconds
Y = air pressure above and below normal atmospheric pressure
AMPLITUDE = air pressure at that time
+ compression 0 normal air pressure - rarefaction

Reading phones off a waveform. Vowels [iy], [ay], [ae], [ey] have regular amplitude peaks and are easy to spot because they are voiced. Stop consonants are a closure followed by a release, so silence then slight bursts of emphasis, very clear for the [b] of "baby". Fricatives are made when a narrow channel for airflow causes noise, so the [sh] of "she" gives a hiss, noisy, irregular waveform. Four factors influencing phonetic variation: rate of speech in syllables per second - phonetic, phonological and morphological factors, misinterpretation /t/ as /d/, tap becoming dap - sociolinguistic factor, gender, class, dialect - style, "something" versus "something".

Phonetics resources, with their exact words. Dictionaries: CMU, machine-readable pronunciations for 134,000 English words: CELEX, English, German, Dutch - PROMLEX by the Linguistic Data Consortium (LDC), 90,694 waveform pronunciations. Annotated corpora: TIMIT, a joint project of Texas Instruments, MIT and SRI, 6300 read sentences, 10 each from 630 speakers, lab labelled and automatically aligned with the wave file and the Switchboard Transcription Project. Texts: Praat and Audacity, for wav to mp3, trimming, removing noise, normalising.

Day 10

Point processing and histogram processing

Point processing exists because the cheapest possible enhancement changes each pixel without looking at its neighbours.

Image enhancement is "the process of making images more useful". Three reasons: highlighting interesting detail - removing noise - making images more visually appealing. From the lab, "Image Enhancement, to make it useful", with the named application agriculture industry. Two categories: spatial domain techniques are the direct manipulation of image pixels, frequency domain techniques are the manipulation of the Fourier transform or wavelet transform of an image. The course concentrates on spatial domain techniques.

POINT PROCESSING, THE MASTER EQUATION
s = T(r)

s = processed image pixel value
r = original image pixel value
T is the GREY LEVEL TRANSFORMATION FUNCTION, also called a point processing operation. The simplest spatial domain operation, where the NEIGHBOURHOOD IS SIMPLY THE PIXEL ITSELF.

Negative images, useful for enhancing white or grey detail embedded in dark regions. The tissue is much clearer in the negative image of a mammogram

Thresholding, particularly useful for segmentation, isolating an object of interest from a background

THE THREE BASIC GREY LEVEL TRANSFORMATION FAMILIES

- 1 LINEAR, 2 LOGARITHMIC, 3 POWER LAW
- 1 LINEAR Negative / Identity
s = c - log(1 + r)
inverse log written in the notes as s = log(1 + r)
"We usually set c to 1."
Grey levels must be in the range [0.0, 1.0]
The log transform maps a NARROW range of LOW input grey levels into a WIDER range of output values. The inverse log performs the opposite.
- 3 POWER LAW s = c - r ^ gamma
Maps a narrow range of DARK input values into a wider range of output values, or vice versa.
Varying GAMMA gives a whole family of curves.

Gamma correction: display devices do not respond linearly to different intensities, and this can be corrected using a log transform.

Piecewise linear transformation functions, arbitrary user-defined transforms rather than a well-defined mathematical function, for example a contrast-stretching linear transform. Gray level slicing, highlights a specific range of gray levels, similar to thresholding, and other levels can be suppressed or maintained. Bit plane slicing, isolating particular bits of the pixel values. Higher-order bits usually contain most of the significant visual information, lower-order bits contain subtle details.

HISTOGRAM PROCESSING

An image histogram "shows us the distribution of grey levels in the image. Massively useful in image processing, especially in segmentation."

HISTOGRAM SHAPE RULES FROM THE LAB, MEMORISE

Darker → LEFT distribution Balanced → CENTRE
Brighter → RIGHT distribution

Low grey level = dark pixel = indicates 0
High grey level = brighter pixel = indicates 1
"High Contrast - Image is perfect."

CONTRAST STRETCHING expand the histogram
HISTOGRAM EQUALISATION purpose is to remove a bright area or a dark area, remove the region, remove the crowded

MATLAB: imhist() imadjust() subplot() imshow

Contrast stretching: "We can fix images that have poor contrast by applying a pretty simple contrast specification. The interesting part is how do we decide on this transformation function?" The lab rule for that decision is blunt: "Try and error to get Optimal Solutions."

Honest gap on the histogram equalisation formula. Histogram equalisation is covered in your notes by purpose and before-and-after pictures only. The equalisation equation, and the cumulative distribution function figure that goes with it, exist in the Day 10 material only as slide images, so no formula is written down in either digest. Take it off the slide; nothing is invented here. Your own note on this section reads "Image Enhancement (Tengok Balki)".

Day 6

Text-to-Speech synthesis

TTS exists to go the opposite way from recognition, "to produce speech (acoustic waveforms) from text input".

Five types of TTS. 1 Articulatory, synthesizing speech by controlling the speech articulators, jaw, tongue, lips, by building a model of articulator movement and vocal-tract acoustics. 2 Formant, generating speech by synthesizing specific spectral peaks or formants; a formant is a concentration of acoustic energy around a particular frequency in your voice. 3 Concatenative, concatenating short samples of recorded sound, units or diphones, from a database of stored speech to assemble new utterances. 4 Parametric, uses an HMM model and trains model parameters on speech databases, constructing a model of the acoustic properties of the vocal tract, then analysing speech by determining parameter values. 5 Neural TTS, neural networks directly generate speech waveforms from input text, highly natural but requiring large training data and computational resources.

Honest gap on the histogram equalisation formula. Histogram equalisation is covered in your notes by purpose and before-and-after pictures only. The equalisation equation, and the cumulative distribution function figure that goes with it, exist in the Day 10 material only as slide images, so no formula is written down in either digest. Take it off the slide; nothing is invented here. Your own note on this section reads "Image Enhancement (Tengok Balki)".

DEVELOPMENT OF TTS, 5 STEPS AND THE SUB-STEPS

- 1 TEXT ANALYSIS convert text into an intermediate representation
- 2 WAVEFORM SYNTHESIS convert that representation into a waveform

Text Analysis is text normalization: sentence tokenization, non-standard word normalization, homograph disambiguation Text normalization, 4 ordered steps: Identify tokens in text → Chunk tokens into reasonably sized sections → Map tokens to words → Tag the words

Sentence tokenization challenge: not all sentence boundaries are marked by periods, sometimes by punctuation such as colons, so you need period disambiguation. Non-standard words, numbers, abbreviations and acronyms, must be expanded into word sequences before pronunciation.

Homograph disambiguation handles words with the same spelling but different pronunciation: "It's no use (Yu w s) to ask to use (yu uz) the telephone." "Do you live (li v) near a zoo with five (fay v) animals?" "I prefer bass (b ae s) fishing to playing with bass (b ey s) guitar."

THE 15 MOST COMMON HOMOGRAPHS, LIBERMAN & CHURCH 1992

use, increase, close, record, house, contract, lead, live, lives, protest, survey, project, separate, present, read

Phonetic analysis takes normalized word strings and produces a pronunciation for each word. Its most important component is a large pronunciation dictionary, and the problem is words not in the dictionary. The CMU Pronouncing dictionary holds 127k words, and the Unisyn dictionary is more accurate and covers multiple dialects.

The big problem area is NAMES, proper nouns. Names are 20% of tokens in typical newswire text, and Spiegel 2003 estimates US names at 2 million surnames and 100,000 first names. Personal names cited: McArthur, D'Angelo, Jimenez, Raghavan, Sondhi, Xu, Hsu, Zhang, Nguyen. Brands: Infiniti, Kmart, Cytic, Medamicus, Inforte, Aeon, Index Labs, Bebe.

Grapheme-to-Phoneme (G2P) is "the process of converting a sequence of letters into a sequence of phoneme", and the earliest algorithm is the letter-to-sound or LTS rule. Prosodic analysis is "the study of the intonational and rhythmic aspects of language (intonation, pitch, accent and tune)", outputting a list of phones with prosody, duration and fundamental frequency.

Day 6

Speech processing fundamentals

Speech processing exists to apply signal processing to the one signal humans use most.

DEFINITION

"The application of digital signal processing techniques to the processing and/or analysis of speech signal."

Fundamental purpose of speech = human communication, the transmission of messages between a speaker and a listener.

Four applications. 1 Speech Coding, compressing the digital waveform into a lower bit-rate representation. 2 Text-to-Speech Synthesis, accepts text and converts it to speech. 3 Speech Recognition, transforms speech into corresponding text. 4 Voice Recognition, recognises the voice of a particular person, speaker verification or identification, like thumbprint or iris recognition.

THE SPEECH CHAIN, FIVE STAGES IN ORDER

- 1 Linguistic Level, speaker side. Words organised into sentences, made of syllables built from phonemes or segments, "cat" has three segments. Segments are classified by discrete binary features, so the representation is digital, and word sequences are organised by syntactic constraints.
- 2 Physiological Level, articulatory. Phonemes and features specify instructions for how sounds are produced by the articulators, tongue, lips, larynx, and how they move and coordinate via the muscles. The central problem is the transformation from the discrete linguistic representation to continuous articulator movements.
- 3 Acoustic Level. Articulator movement plus air from the lungs creates sound sources filtered by acoustic cavities shaped by the articulators, drawing on acoustics, source-filter theory and signal processing. The sound radiated from the lips is the most accessible component, and it reaches the listener's ear as mechanical vibration of the eardrum.
- 4 Physiological Level, auditory. Sound becomes mechanical motion, frequency analysis happens in the ear, then electrical activity on the auditory nerve and in higher brain centres, and the listener decodes the intended word sequence.
- 5 Linguistic Level, listener side.

SPEECH PRODUCTION, 5 STEPS, AND PERCEPTION, 3 STEPS

PRODUCTION

- 1 Schematic of production and perception
- 2 The speaker formulates a message in his or her mind
- 3 Converts the message into a LANGUAGE CODE: phoneme sequences plus PROSODY (syntax) markers denoting duration, loudness and pitch
- 4 Executes neuro-muscular commands causing the vocal tract to vibrate and shaping it so the proper sound sequence is created, producing the acoustic signal
- 5 The commands simultaneously control lips, jaw, tongue and velum

PERCEPTION

- 1 The signal is generated and propagated to the listener
- 2 A neural transduction process converts the spectral signal at the output of the BASILAR MEMBRANE into activity signals on the auditory nerve, roughly a feature extraction process
- 3 Neural activity is converted into a language code at higher processing centres in the brain, and message comprehension is achieved

Speech signal: the analog form of speech, an acoustic waveform whose amplitude varies with time and is normalised so the maximum value is 1. Speech is a sequence of articulated sound units, phonemes, and can be segmented into phonemes and non-speech segments.

Day 10

Boundary and regional descriptors

BOUNDARY DESCRIPTORS, THE SEVEN

1 Length, number of pixels along the contour - 2 Xmax, Xmin, Ymax, Ymin, maximum and minimum coordinates of the contour - 3 Xcentroid, Ycentroid, coordinates of the center of the region - 4 Boundary diameter, maximum distance between two points on the boundary - 5 Curvature, rate of change of slope, for example the difference between the slopes of adjacent boundary segments - 6 Shape order or chain count, number of elements in the chain - 7 Shape number, the first difference of smallest magnitude.

REGIONAL DESCRIPTORS AND TOPOLOGY, BOTH FORMULAS

Area

COMPACTNESS = perimeter^2 / area

In the figure, (white area) / (total light area) gives an idea of relative electrical energy consumption.

TOPOLOGY "the study of the properties which are unaffected by any deformation (rubber-sheet distortion)"

H = number of holes
C = number of connected components
EULER NUMBER E = C - H

Worked: Fig b C = 1591, E = 1552, so H = 39
Fig c the connected component with the number of pixels, 8479

Descriptors for video sequences, MPEG-7. The one-liner to memorise: "MPEG-1/2/4 make content available, whereas MPEG-7 allows you to find the content you need." It is a content description standard. For videos and images it covers shape, size, texture, colour, movements and positions; for audio, key mood, tempo, changes and position in sound space. Four applications: Digital Libraries, Multimedia Directory Systems, Broadcast Media Selection, Editing. Two examples: draw an object and find an object with similar characteristics; play a note of music and find a similar type of music.

Day 10

The named kernels, and what your notes really say

KERNEL	WHAT IS WRITTEN DOWN	MATLAB
Mean / average	The simple smoothing filter, f(x,y) = (1/nn) SUM g(i,t). Increasing the filter size increases the blur	fspecial('average',3) used inside filter2(...)/255
Median	An order statistics filter, the midpoint value of the ordered neighbourhood set	medfilt2(img,[3 3])
Gaussian	Appears as the weighted smoothing filter idea, pixels nearer the centre weigh more. The matrix itself is a slide image only	fspecial(...); imnoise(I,'gaussian',0,0.025)
Sobel	Shown as "variants on the simple Laplacian and Sobel Operators". Lab note: "Sobel, can have any pixel value"	imfilter with the kernel
Prewitt	Lab note: "Prewitt, Horizontal and Vertical (can contain -1 and 1)"	imfilter
Roberts	Lab note: "Roberts, minus 45 and plus 45 (0, 1 and only)"	imfilter
Laplacian	"The first sharpening filter", isotropic, one of the simplest. Used on medical images, the fundus example	h=fspecial('laplacian'); imfilter(moon,h)

Honest gap on the kernel matrices. Every actual kernel matrix in this course, the Laplacian variants, the Sobel operators, the averaging and weighted smoothing masks, the point, line and edge detection masks, exists in your notes only as an untranscribed slide image. Neither digest writes out a single 3 by 3 matrix. The table above therefore gives exactly what I written, the verbal descriptions and the MATLAB calls, and no matrix is invented. Copy the matrices off the Day 10 slides.

Combining spatial enhancement methods: successful enhancement is typically not achieved with a single operation; a range of techniques are combined, as worked on a bone scan.

Day 11

Colour image processing, exam Q3A

This box is exam Q3 Part A: "colour image processing, Hue Saturation Intensity". The HSI formula card with three worked examples is on Sheet 3, flagged in your study page as "Yang wajih halal", what must be memorised.

Colour exists in the syllabus because humans perceive thousands of shades of colour but only about two dozen shades of grey, so colour is a powerful descriptor that greatly simplifies object segmentation and identification.

History and physics. In 1666 Sir Isaac Newton discovered that a beam of sunlight passing through a glass prism splits into a spectrum of colours. The colours humans and most animals perceive are determined by the nature of the light reflected from the object. Green objects reflect light with wavelengths primarily in the range 500 to 570 nm while absorbing most of the energy at other wavelengths.

Three characteristics that distinguish one colour from another. Hue, refers to primary and secondary colours, Yellow, Red, Blue, Green. Brightness, the amount of intensity. Saturation, the degree of purity of a colour.

Two major areas. Full-colour processing, images are acquired and processed in full colour. Pseudo-colour processing, images are grayscale by nature and are converted to colour images for visualization purposes.

Day 11

Colour image processing, exam Q3A

THREE QUALITIES OF A CHROMATIC LIGHT SOURCE, WITH UNITS

RADIANCE the total amount of energy that flows from the light source measured in WATTS

LUMINANCE the amount of energy an observer perceives from the light source measured in LUMENS

BRIGHTNESS a subjective, practically unmeasurable notion embodying the intensity of light no unit

Note: we can have high radiation but low luminance. Hue and saturation taken together are called chromaticity, and a colour may be characterized by its brightness and chromaticity.

TRISTIMULUS VALUES AND THE CIE DIAGRAM

TRISTIMULUS VALUES "the amounts of red, green and blue needed to form any colour", denoted X, Y and Z. A colour is specified by its TRICHROMATIC COEFFICIENTS, which means

X + Y + Z = 1

CIE chromaticity diagram

x-axis = the proportion of red
y-axis = written in the notes as "the proportion of red" (a typo in the source; it is GREEN)
proportion of blue: z = 1 - (x + y)
Worked readings: Green 62% green, 25% red, 13% blue
Red 32% green, 67% red, 1% blue

Any colour on the boundary of the chromaticity chart is fully saturated. The point of equal energy has equal amounts of each colour and is the CIE standard for pure white. Any straight line joining two points defines all the colours obtainable by combining those two additively, and this extends to three points, hence the entire colour range cannot be displayed based on any three colours. The triangle shows the typical colour gamut of RGB monitors; the strange shape is the gamut of high quality colour printers.

PRIMARIES AND SECONDARIES. MEMORISE BOTH DIRECTIONS

LIGHT primary red (R), green (G), blue (B)
secondary cyan (green+blue), magenta (red+blue), yellow (red+green)

PIGMENT primary cyan (C), magenta (M), yellow (Y)
secondary red (R), green (G), blue (B)

The models commonly used in digital image processing are RGB, CMY, CMYK and HSI.

RGB, THE 11 FACTS

Each colour appears in its primary spectral components
Based on a CARTESIAN COORDINATE SYSTEM
RGB values are at 3 corners; cyan, magenta and yellow at three other corners
BLACK is at the ORIGIN; WHITE is the corner FURTHEST from it
Different colours are points on or inside the CUBE, represented by RGB vectors
Images consist of THREE COMPONENT IMAGES, one per primary
combined into a composite when fed into a monitor
Bits per pixel = COLOUR DEPTH. A 24-BIT image is a FULL-COLOUR image, allowing (2^8)^3 = 16,777,216 colours
From the lab: [1,0,0]=Red [0,1,0]=Green [0,0,1]=Blue

HSI, and why it exists. "RGB is useful for hardware implementations and is serendipitously related to the way in which the human visual system works. However, RGB is not a particularly intuitive way in which to describe colours." The slogan: "RGB is great for colour generation, but HSI is great for colour description."

Hue, a colour attribute describing a pure colour, pure yellow, orange or red. Saturation, a measure of how much a pure colour is diluted with white light. Intensity, since brightness is nearly impossible to measure because it is subjective, we use intensity, the same achromatic notion as in grey level images. Geometry: take the diagonal on the RGB colour cube running black to white, then stand the cube on the black vertex with the white vertex directly above it.

Day 8

Digital image fundamentals

Image processing starts here because every later operation is arithmetic on a matrix of numbers.

DEFINITION

"A representation of a two-dimensional image as a finite set of digital values, called PICTURE ELEMENTS or PIXELS." Pixel values typically represent gray levels, colours, heights or opacities. DIGITIZATION is performed by digital cameras and scanners, which implies a digital image is an APPROXIMATION of a real scene.

Common image formats: 1 sample per point is black and white or grayscale, 3 samples per point is colour, Red, Green, Blue.

Two major tasks: improvement of pictorial information for human interpretation, and processing of image data for storage, transmission and representation for autonomous machine perception.

LOW, MID AND HIGH LEVEL PROCESSES

LEVEL	INPUT	OUTPUT	EXAMPLES
Low	Image	Image	noise removal, sharpening
Mid	Image	Attributes	object recognition, segmentation
High	Attributes	Understanding	scene understanding, autonomous navigation

KEY STAGES IN DIGITAL IMAGE PROCESSING, THE 9, IN ORDER

- 1 Image Acquisition
- 2 Image Enhancement
- 3 Image Restoration
- 4 Morphological Processing
- 5 Segmentation
- 6 Object Recognition
- 7 Representation and Description
- 8 Image Compression
- 9 Colour Image Processing

The class notes give a shorter four-stage version, Image acquisition, Enhancement, Filtering, Segmentation. Quote the NINE-stage list for the exam.

Day 10

Shape representation and description, exam Q3C

This box is exam Q3 Part C: "Chain code sequence". Memorise both worked strings below.

Shape description exists to compact the raw list of boundary pixels into a few numbers you can compare.

Region representation, two ways. External characteristics, its boundary, focus on shape. Internal characteristics, its internal pixels, focus on colour and textures. Region description, two ways. Boundary descriptors, boundary length, diameter, curvature. Regional descriptors, area, perimeter, compactness, mean value. Both boundary and regional descriptors are often used together.

CHAIN CODES, DEFINITION AND PROCEDURE

"Represent a boundary by a CONNECTED SEQUENCE OF STRAIGHT-LINE SEGMENTS of specified length and direction. The direction of each segment is coded by using a NUMBERING SCHEME."

Types: 4-directional chain codes and 8-directional

Procedure

- 1 A typical chain code is generated by following a boundary in a CLOCKWISE direction and assigning a direction to the segments connected to each pixel
- 2 Before computing the chain code, it may be useful to REAMPLE the data into an ordered set to reduce the code dimension
- 3 The accuracy of the resulting code depends on the SPACING OF THE SAMPLING GRID
- 4 It is advisable to NORMALIZE THE GRID ORIENTATION by aligning the chain code grid to the BASIC RECTANGLE

THE TWO WORKED CODES. THESE ARE THE ANSWER KEYS

4-directional, Figure (c): 0033333322221110101
8-directional, Figure (d): 0766655533321212

Order of a shape is the number of digits

Redefine the starting point by replacing the last number at the first position, which is the circular shift. The lab says the same: "We have put last no and take and in the first position"

First Difference is "obtained by taking two numbers of chain code and calculating the number of transitions required to reach the second number from the first number in counter clockwise direction"

Shape number is the first difference of smallest magnitude", treating the chain code as a circular list

A direction conflict you must handle. The slide says the chain code is generated clockwise; the physical class notes say "counter clock wise direction and count the number of steps". Follow the slide for the exam and state your traversal direction explicitly in any answer. Note that the first difference is separately defined as counter clockwise, which is probably the source of the confusion.

SKELETON VERSUS MAT

Two ways to represent a shape. Reduce it to a graph by obtaining its skeleton via a skeletonization algorithm, or obtain its MAT, medial axis transformation, via a MAT algorithm. "MAT and skeletonization are often used interchangeably, but they have slight difference. The skeleton is simply a binary image showing the simple skeleton. The MAT shows the center of shape with intensity which represents its distance to boundary in the original object."

Day 6

Waveform synthesis and TTS evaluation

WAVEFORM SYNTHESIS, FOUR TYPES

a Diphone synthesis - b Unit selection synthesis - c Hidden Markov Model (HMM) - d Deep NN.

DIPHONE VERSUS UNIT SELECTION

DIPHONE SYNTHESIS generates a waveform from a sequence of phones by selecting and concatenating units from a prerecorded database of DIPHONES. A diphone is an adjacent pair of phones, usually a recording of the TRANSITION between two phones. In [dafaoun]: [da] [af] [fe] [eou] [un] UNIT SELECTION is a modern concatenative algorithm that synthesizes new naturally sounding utterances by selecting appropriate sub-word units from a database of natural speech TWO DIFFERENCES

- 1 unit selection holds MANY COPIES of each diphone, diphone synthesis stores EXACTLY ONE of each
- 2 diphone synthesis MODIFIES THE PROSODY of concatenated units, unit selection applies MINIMAL SIGNAL PROCESSING

BENEFITS a large unit database solves the unseen-word problem, natural data solves database problems, "there's no data like more data", larger units capture wider effects

EVALUATING TTS, FOUR INTELLIGENCE TESTS

1 Diagnostic Rhyme Test (DRT) tests the intelligibility of initial consonants: listeners hear confusing rhyming words differing in a single phonetic feature such as voicing or nasality, so the subject hears "veal" and chooses between "veal" and "feel". Percentage right is the score. 2 Modified Rhyme Test (MRT) uses a set of 6 words differing in either initial consonants (bat, bad, back) or final consonants (went, sent, bent, dent); listeners get a single word and identify it from the closest list of six. 3 Mean Opinion Score (MOS) plays a sentence and asks listeners to rate it from 1 bad to 5 excellent, comparing scores on the same sentences with t-tests. 4 AB Tests, prefer A or prefer B, that is preference tests.

Notebook TTS.ipynb.

Day 9

Image restoration and noise removal

Restoration exists because degradation has a cause, and if you can identify the cause you can reverse it.

DEFINITION

Image restoration "attempts to restore images that have been degraded, identify the degradation process and attempt to reverse it. MORE OBJECTIVE compared to image enhancement."

Noise sources. Noise arises during image acquisition, that is digitization, and during transmission. Images can be affected by environmental conditions such as smoke and haze, and the imaging sensor itself adds noise because of its circuit system, heat and illumination levels.

ARITHMETIC MEAN OR AVERAGE FILTER, THE ONE WRITTEN FORMULA
f(x,y) = (1 / mn) SUM g(s,t)

"This is implemented as the simplest smoothing filter. Blurs the image to remove noise."

Other means named, three: Geometric Mean, Harmonic Mean, Contraharmonic Mean, each with slightly different behaviour. Order Statistics Filters are "spatial filters that are based on ordering the pixel values that make up the neighbourhood operated on by the filter"; the useful ones are the Median filter, Max and min filter, Midpoint filter and Alpha trimmed mean filter.

NOISE TYPE MATCHED TO THE FILTER THAT REMOVES IT

NOISE	WHAT YOUR NOTES ACTUALLY GIVE	FILTER
Salt and pepper impulse	Injected in the lab with imnoise using 'salt & pepper' and 0.02, where 0.02 means 2% of pixels will be noise	Median filter
Gaussian	Injected with imnoise using 'gaussian', 0, 0.025. Lab parameters: Image / gaussian + type of noise / 0 = Mean / 0.025 = intensity, a medium variance. Student caveat: "It is intensity, not density, density is a variance"	Averaging filter, then Wiener
Speckle	Not named anywhere in your notes. Do not claim it was taught	None given
Periodic	Not named anywhere in your notes. The frequency route is mentioned only as "manipulation of Fourier transform or wavelet transform"	None given

THE TWO FILTER CALLS IN FULL, EXACTLY AS THE LAB WROTE THEM

SALT AND PEPPER, median filter
J = imnoise(I,'salt & pepper',0.02);
J_gray = rgb2gray(J);
I = medfilt2(J_gray,[3 3]);
figure, imshow(I)

GAUSSIAN, averaging filter then Wiener
J = imnoise(I,'gaussian',0,0.025);
K = filter2(fspecial('average',3),J)/255;
wiener2 "remove noise back"

Honest gap on noise models. The Day 9 noise models (Gaussian, Rayleigh, Erlang or Gamma, Exponential, Uniform, Impulse) and the degradation and restoration model diagram appear in your notes only as untranscribed slide images, and so do the formulas for the geometric, harmonic, contraharmonic, median, max, min, midpoint and alpha-trimmed filters. There is no written note to filter mapping table and no written noise PDF formula in either digest. The table above is built only from what is written: the arithmetic mean formula and the MATLAB lab work. The standard pairings of these two rows were not stated to you, so read them off the slides rather than guessing.

The one rule that is written: "We can use spatial filters of different kinds to remove different kinds of noise."

Laplacian filtering in the lab is used on medical images, the example being a fundus image, which "detects almost 28 diseases".

THE LAPLACIAN LAB SCRIPT, EXACT

h = fspecial('laplacian');
moon_lap = imfilter(moon,h);
final = moon - moon_lap;
subplot(1,3,1) title('Original')
subplot(1,3,2) title('Laplacian image')
subplot(1,3,3) title('Original-Laplacian')

Day 10

Spatial filtering, convolution and kernels

Neighbourhood operations exist because detail, edges and noise are all defined by how a pixel differs from its neighbours.

Neighbourhood operations work on a neighbourhood of pixels, unlike point operations on a single pixel. The neighbourhood is determined by a kernel, mask, window or filter of any size and shape, but this lecture uses only a square shape.

THE FOUR SIMPLE NEIGHBOURHOOD OPERATIONS

MIN pixel = the minimum value in the neighbourhood
MAX pixel = the maximum value in the neighbourhood
AVERAGE pixel = the average of all pixels in the neighbourhood, also MEAN or SMOOTHING
MEDIAN pixel = the midpoint of the ordered set, from [1, 7, 15, 18, 24] the median is 15

Smoothing spatial filters average all pixels in a neighbourhood around a central value, useful for removing noise and highlighting gross detail. Weighted smoothing filters weight neighbourhood pixels differently because pixels closer to the central pixel are more important, also called weighted averaging.

THE EDGE PROBLEM, FIVE APPROACHES

- 1 Omit missing pixels, only works with some filters, can add extra code and slow processing
- 2 Pad the image, typically all white or all black pixels
- 3 Replicate border pixels
- 4 Truncate the image
- 5 Allow pixels to wrap around the image, can cause strange image artefacts

CORRELATION VERSUS CONVOLUTION, QUOTE THIS

"The filtering we have been talking about so far is referred to as CORRELATION, with the filter itself referred to as the CORRELATION KERNEL. CONVOLUTION is a similar operation, with a ROTATED kernel, mask, window or filter. FOR SYMMETRIC FILTERS IT MAKES NO DIFFERENCE."

SHARPENING AND THE DERIVATIVES

Sharpening spatial filters "seek to highlight fine detail, remove blurring from images, highlight edges". They are based on spatial differentiation, which measures the rate of change of a function.

THE TWO DERIVATIVE FORMULAS, MEMORISE

1st derivative
df/dx = f(x+1) - f(x)
"Just the difference between subsequent values and measures the rate of change of the function"

2nd derivative
d2f/dx2 = f(x+1) + f(x-1) - 2 f(x)
"takes into account the values both before and after the current value"

Why the 2nd derivative is useful for enhancement: a stronger response to fine detail and a simpler implementation.

1ST VERSUS 2ND DERIVATIVE, THE 4-POINT CONCLUSION

1st order derivatives

Visualisation, Delivery & Exam Answers

HOW A CHART IS BUILT, HOW IT IS CHOSEN, HOW IT IS TOLD, AND WHAT THE EXAM ACTUALLY ASKS

This half of the course is about **delivery**. A graphic is not decoration, it is a **grammar**: data, mapped to aesthetics, placed on scales, drawn as geoms, then optionally transformed, positioned, faceted and themed. On top of the grammar sits a **workflow** that insists you **define before you design**, and a **narrative arc** that turns a finished chart into a decision. Maps are the special case where the data carry a location, and each map type buys one insight at the cost of one specific distortion. **Note carefully**: the final test topic list does not include data visualisation, storytelling or mapping, so this sheet is for the other assessment. The exam box on the right is the one you revise from.

Tools
Power BI, Tableau and the named software

Tools appear at this course as a **list to name**, not as a feature set to operate, so answer at this level.

Be honest about how much Power BI you were taught. Microsoft Power BI appears exactly once in the entire course, as the second entry in the Day 2 list of six text-visualisation tools. There is no Power BI feature detail anywhere: no Map, Filled Map, ArcGIS Maps for Power BI, Shape Map or Azure Map visuals, no geocoding behaviour, no plot-point caps, no DAX, no data model. The same is true of Tableau, which is named as a tool and as paid software and nothing more. If asked about Power BI, name it correctly as a data visualisation tool and say what visualisation tools are, for; do not invent a feature walkthrough.

THE SIX TEXT VISUALISATION TOOLS, THE EXACT LIST

Tableau Data Analysis & Data Visualization

Microsoft Power BI

Many Eyes IBM's Free Online Data Visualization Tool

Plotly Express

Dash

ThinkMap

visual thesaurus

ALTERNATIVES NAMED IN LECTURE 12

Other R packages ggvis, shiny, r2d3

Other languages D3, matplotlib, plotly

Paid Software Tableau, Adobe Illustrator

Other Software Excel, SAS, STATA, SPSS

ggplot2 Extensions

Stats and Geoms ggrepel, ggforce

Coordinate Systems ggtm, circulr

Scales scales, colourbrewer, viridis

only on ggplot2-exts.org

The only colour palette in the entire course is that of **ColorBrewer**: and **viridis** appear in that list, and they are listed as **packages**, not as rules. There is no palette theme in your notes. If you must expand, say it is your elaboration: viridis is perceptually uniform and colour-blind-safe, and ColorBrewer separates sequential, diverging and qualitative palettes.

RESOURCES NAMED, USEFUL, FOR A "CITE YOUR SOURCES" MARK

Books and notes: Lecture Notes by **Jeffrey M. Girard**, Carnegie Mellon University - **Wickham's (2010) layered grammar of graphics - R for Data Science**, **ggplot2: Elegant graphics for Data Analysis - The Tufte Art: Data, Charts, and Maps by Cairo - The Visual Display of Quantitative Information**, as printed, which is Tufte's *The Visual Display of Quantitative Information*.

Websites: ggplot2.org - rstudio.com/resources/cheatsheets/ - stackoverflow.com - colorado.edu - geeksforgeeks.org/maths/ - the class notes add **techscience.com**, marked "useful website".

Day 12
Chart selection: intent to chart type

A selection table exists because the chart is chosen by the **task**, never by what looks nice.

THE QUICK DECISION RULE TO MEMORISE

Trend → line

Comparison → bar

Distribution → histogram / line plot / stem-and-leaf / box plot

Composition → pie / donut / treemap

Relationship → scatter / bubble / network

Flow → Sankey (non-spatial) or flow map (spatial)

Geospatial → choose by data type, see the map box

INTENT	CHART TYPE	WHY, IN THE COURSE'S OWN WORDS
TREND over time	Line Graph	"used to show how the value of a particular variable changes with time. We plot this graph by connecting the points at different values of the variable." Class note: "Line graph, trend." "Useful for analyzing the trends in the data and predicting further trends"
COMPARISON between items	Bar Graph	"bars of uniform width are drawn with equal spacing between them on one axis (x-axis usually), depicting the variable. The values of the variables are represented by the height of the bars." Class note: "Bar graph, comparison chart."
DISTRIBUTION of one variable	Histogram	"similar to bar graphs, but it is based on frequency of numerical values rather than their actual values. The data is organized into intervals and the bars represent the frequency of the values in that range." Class note: "numerical data in the form of interval (in a bin), bin = range"
	Line Plot	"a plot that displays data as points and checkmarks above a number line, showing the frequency of the point." Class note: "same as histogram (distributions). Useful for small amount of data"
	Stem and Leaf Plot	"each value is split into a 'leaf' (in most cases, it is the last digit) and 'stem' (the other remaining digits). For example: the number 42 is split into leaf (2) and stem (4)." Class note: "2 digits only. Useful for small dataset"
	Box and Whisker Plot	Comparing whole groups rather than single values, since it shows spread, average and median side by side.
COMPOSITION part to whole	Pie Chart	"a type of graph which represents the data in the form of a circular graph. The circle is divided such that each portion represents a proportion of the whole"
	Donut Chart	Named as an example graphic. Part-to-whole in ring form
	Tree map	A text-visualisation type. Nested part-to-whole with area encoding, so hierarchical composition
RELATIONSHIP between variables	Scatter plot	A text-visualisation type. Two numeric variables, so correlation
	Bubble Chart	Named as an example graphic. Three variables: x, y and size, so relationship plus magnitude
GEOSPATIAL data with a location	Choropleth Map	Rates or ratios on enumeration units. Never with raw counts . See the three-condition gate
	Proportional Symbol Map	Magnitude at points or areas, shown by symbol size
	Dot Density / Point Distribution Map	Density and clustering, where things clump
	Spatial Heat	Magnitude of a spatial phenomenon as colour Map
	Flow Map	Origin, destination and volume, so movement
	Spider Map	Links or associations between two point sets
TEXT a corpus	Word cloud, Heat map, Word tree	Term prominence by frequency; magnitude across a matrix; the branching context of a keyword. See Sheet 1

Where this table comes from. Lecture 12 slide 22 names **seven** "Types of Graphical Representations", each with a definition: Line Graph, Bar Graph, Histogram, Line Plot, Stem and Leaf Plot, Box and Whisker Plot, Pie Chart. The class notes supply the one-word use case for each. The remaining rows come from the example graphics on slides 5 and 6, from the Day 2 text-visualisation list, and from the Lecture 13 map types. Nothing here is invented, but only the seven are "the seven".

Exam
Q4 practice question, quoted in full

THE QUESTION, VERBATIM, WITH ITS RAW TEXT

"Here is an exercise for you to test your understanding of the Text Pre-processing topic.

Given a raw text below:

I'm gonna be home soon and i don't want to talk about this matter anymore tonight, ok? I've cried enough today.... hello!!!!!! how did ur sunday go? I was just texting to see if you'd decide to do anything tomorrow. @.@not that i'm trying to invite myself or anything!

Question:

Identify FOUR (4) pre-processing tasks and apply each task to the given text below. Show the result after each task is performed. Each pre-processing task must be done in sequence, for example, the output of task 1 will become the input for task 2, and so on."

Answer structure to replicate exactly. For each task give **Theory**, then **Purpose** as **three bullets**, then **Result**, then **Explanation**. The model answer in your notes does this four times.

TASK 1. WORD FORMAT NORMALIZATION, LOWERCASING

Purpose: reduce text inconsistency - improve accuracy of text analysis - prevent duplicate representations of the same word.

RESULT

"I'm gonna be home soon and i don't want to talk about this matter anymore tonight, ok? I've cried enough today.... hello!!!!!! how did ur sunday go? I was just texting to see if you'd decide to do anything tomorrow. @.@not that i'm trying to invite myself or anything!"

Explanation: characters such as "!", "!", "??", "!", "" and " @@" have been removed. For example "don't" becomes "dont" and "@.@"not" becomes "not".

TASK 2. REMOVAL OF PUNCTUATION AND SPECIAL CHARACTERS

Purpose: eliminate unnecessary noise - simplify text structure - improve tokenization and feature extraction.

RESULT

"im gonna be home soon and i dont want to talk about this matter anymore tonight ok ive cried enough today hello how did ur sunday go i was just texting to see if youd decide to do anything tomorrow not that im trying to invite myself or anything"

Explanation: common words such as "and", "!", "!", "to", "was", "if" and "or" have been removed.

TASK 3. WORD TOKENIZATION

Purpose: convert unstructured text into structured data - prepare text for machine learning and NLP algorithms - enable word-level analysis.

RESULT

[im, gonna, be, home, soon, and, i, dont, want, to, talk, about, this, matter, anymore, tonight, ok, ive, cried, enough, today, hello, how, did, ur, sunday, go, i, was, just, texting, to, see, if, youd, decide, to, do, any, thing, tomorrow, not, that, im, trying, to, invite, myself, or, anything]

TASK 4. STOP WORD REMOVAL

Purpose: reduce dataset size - improve computational efficiency - highlight important keywords.

RESULT

[gonna, home, soon, dont, want, talk, matter, anymore, tonight, ok, ive, cried, enough, today, hello, ur, sunday, go, texting, see, youd, decide, anything, tomorrow, trying, invite, myself, anything]

Explanation: common words such as "and", "!", "!", "to", "was", "if" and "or" have been removed.

The student's own mnemonic for the four pre-processing tasks: "Walk Periglenese Test SpeedWind". The mnemonics least verbatim in case they are needed: Day 1 "Iman"; Day 3 "Car Upan Tumpal" for BOW, BON, TF-IDF; Day 7 answer structure "Amin Uderstand Sempat 3Organization Cuma 4Benefit" for Part A, "CounterStrike AjakPergi (First Call Resolution) Oman (Average Handling Time) Rabu CameraRoll" for Part B, and "Mari Try Ambl Terung Pergi Kelantan" for Part C.

Day 12
The data visualization workflow

The workflow exists so that design decisions follow from the audience and the message, not from taste.

Two halves, with their exact headers: **"Define before you design"** and **"Design is a process"**

HALF A, DEFINE BEFORE YOU DESIGN, FIVE DEFINES IN ORDER

THE FIVE DEFINES AND THEIR EXACT QUESTIONS

1 **PURPOSE** Why am I 'creating this graphic'? What are my goals? What level of 'polish' is needed? What format will it be displayed in? What constraints on design exist?

2 **AUDIENCE** Who is the graphic intended for? What do they already know? What do they need to know? What will they understand? What will they be expecting?

3 **DATA** What data will be included? Which observations? Which variables? Which groupings to enforce? Will it use raw or summary scores?

4 **MESSAGE** What is the main take-away? What should viewers conclude? How confident should they be? What emotions should they feel? What questions are they left with?

5 **VALUES** What values are to be emphasized? What techniques can achieve them?

THE FIVE VALUES, THE NEAREST THING HERE TO DESIGN PRINCIPLES

- HONESTY
- ACCESSIBILITY
- CLARITY
- BEAUTY
- FEASIBILITY

The slides give no expansion of the five values. They are five words and nothing more. If a question forces you to expand them, say that you are elaborating: honesty means not distorting scales, accessibility means readable by colour-blind and low-vision viewers, clarity means one message per chart, beauty means polish that aids attention, flexibility means reusable across formats. Label it as your own elaboration.

HALF B, DESIGN IS A PROCESS, FIVE STEPS IN ORDER

Consider the grammar → Create some prototypes → Solicit feedback → Update your design → Iterate until satisfied

Cross-link word writing in an answer: Storytelling Step 3 and Step 2 map exactly onto Define your purpose and Define your audience. Storytelling Step 4 maps onto Consider the grammar. Storytelling Step 5 maps onto Iterate until satisfied.

Day 11
object recognition, exam Q3B

This box is exam Q3 Part B: "Object detection, Processing stages of Image (Histogram, morphological, gray scale etc). The nine processing stages, the Histogram, morphology and grayscale material are all on Sheet 2; the detection theory is here.

Object recognition exists to teach a computer to do what comes naturally to humans, gain a level of understanding of what an image contains.

Definition: "a computer vision technique for identifying objects in images or videos", a key output of deep learning and machine learning algorithms. **Applications:** a key technology behind **driverless cars**, letting them recognize a stop sign or distinguish a pedestrian from a lamppost, **plus disease identification in bioimaging, industrial application and robotic vision.**

RECOGNITION VERSUS DETECTION, LIKELY A SHORT-ANSWER QUESTION

"Object DETECTION is the process of FINDING INSTANCES of objects in images. In the case of deep learning, object detection is a SUBSET OF OBJECT RECOGNITION, where the object is not only IDENTIFIED but also LOCATED in an image. This allows for MULTIPLE OBJECTS to be identified and located within the same image."

AI is "a computer system trained to perceive its environment, make decisions, and take action", relying on learning algorithms such as ML and DL plus large sets of sensor data with well-defined representations of objective truth. **Machine learning** means you **manually select features** and then train the model; common techniques are **decision trees, support vector machines and ensemble methods**. **Deep learning** is a subset of ML modeled loosely on the **neural pathways of the human brain**, where "deep" means **multiple layers between input and output** and the algorithm **automatically learns what features are useful**; common techniques are **CNNs, recurrent neural networks such as LSTM, and deep Q networks**.

Two deep learning approaches. 1 **Training a model from scratch**, gather a **very large labeled dataset** and design a network architecture that learns the features. Impressive results, but it needs a lot of training data and you must set up the **layers and weights in the CNN**. 2 **Using a pretrained model**, deep learning applications use **transfer learning**: start with an existing network such as **AlexNet or GoogleLeNet** and feed in new data containing previously unknown classes. **Less time-consuming** and faster, because the model is already trained on thousands or millions of images.

Two machine learning approaches. **Supervised learning** trains on known input and output data to predict future outputs, making predictions in the **presence of uncertainty**, using **classification and regression**. **Unsupervised learning** finds hidden patterns or **intrinsic structures without labeled responses**, and **clustering** is its most common technique, used for **exploratory data analysis**. **Classification** predicts discrete responses, genuine versus spam email, cancerous versus benign tumor, with applications in **medical imaging, speech recognition, credit scoring**. Use **classification if your data can be tagged, categorized, or separated into specific groups or classes**.

THE QUICK DECISION RULES, NAME THEM ALL

CLASSIFICATION support vector machines (SVM), boosted and bagged decision trees, k-nearest neighbor, Naive Bayes, discriminant analysis, logistic regression, neural networks

CLUSTERING k-means clustering, Gaussian mixture models, hidden Markov models, self-organizing maps, fuzzy c-means clustering, subtractive clustering

How to decide which algorithm to use, quote this whole. "There is no best method or one size fits all. Finding the right algorithm is partly just trial and error, even highly experienced data scientists can't tell whether an algorithm will work without trying it out. But algorithm selection also depends on the size and type of data you're working with, the insights you want to get from the data, and how those insights will be used." From the lab: "Deep Learning, CNN, they learn from the environment. LLM is combination of NLP and Deep Learning."

Day 12
The five storytelling steps and the arc

THREE KEY ELEMENTS OF EFFECTIVE DATA STORYTELLING

1 **Reliable data**. "Every compelling story begins with a strong data foundation. If your data is incomplete, inaccurate, or irrelevant, your narrative falls apart." 2 **Narrative structure**. "Rather than dropping stats on a slide, build a narrative arc: **introduce the problem, reveal the insight, and lead your audience toward a solution**. This structure keeps your audience engaged." 3 **Visualizations**. "Visuals turn complex data into something people can instantly act on. Charts, graphs, diagrams, and even short videos can help you clarify patterns, highlight changes, or emphasize contrasts."

HOW TO TELL A COMPELLING STORY, FIVE NAMED STEPS

STEP WHAT THE SLIDE SAYS

1 **Start with a clear purpose** "What story am I trying to tell, and what decision am I trying to inform?" "Data storytelling isn't about pointing to a chart and saying, 'Look, something happened.' It's about communicating insights in a way that leads to action." Examples: explaining a drop in product usage; investigating why customer churn is high

2 **Know your audience** "That shapes everything, from what you say to how technical you can get." "Ensure you want the big picture: What does this mean for the business? What decisions do we need to make?" **Data science team** "care more about your methodology, models, or confidence levels." "Same story, different angles. When you speak your audience's language, your insights stick longer"

3 **Build a narrative arc** "Even the most valuable insights can fall flat if they're dropped into a deck with no structure. So think like a storyteller. Give your data a **beginning, middle, and end**"

4 **Use visuals with intention** "Visuals are one of your best storytelling tools, but only when they're used with purpose. A poorly designed dashboard can bury important trends. But a clear, focused chart can highlight it better." Well-designed visuals are "the moment your audience gets it without needing to decode a spreadsheet"

5 **Edit for clarity** **Blaise Pascal**, "If I had more time, I would have written a shorter letter." "A great data story isn't just about what you include. It's about what you cut. Every sentence, stat, or chart should move the narrative forward. If it doesn't? **Edit ruthlessly**"

what is going wrong **CONFLICT**

CONTEXT the business challenge what you learned, what happens next **RESOLUTION**

The narrative arc of Step 3, with the exact question the slide attaches to each stage.

THE THREE ARC QUESTIONS, QUOTE THEM

Starting with **CONTEXT** "What's the business challenge or opportunity?"

Introducing **CONFLICT** "What's going wrong?"

Offering a **RESOLUTION** "What did you learn, and what needs to happen next?"

Day 11
The HSI formulas, must memorise

RGB TO HSI, "YANG WAJIB HAFAL"

INTENSITY

$$I = (R + G + B) / 3$$

SATURATION

$$S = 1 - [3 / (R + G + B)] \cdot \min(R, G, B)$$

HUE

$$\theta = \arccos \left\{ \frac{0.5 \cdot (R-G) + (R-B)}{\sqrt{(R-G)^2 + (R-G)(B-G) + (B-G)^2}} \right\}$$

then if B <= G, H = theta
if B > G, H = 360 - theta

WORKED EXAMPLE 1, FROM THE STUDY DOCUMENT

R = 255, G = 0, B = 0
I = (255 + 0 + 0) / 3 = 85
S = 1 - (3/255)(0) = 1
H = 0 degrees (pure red)
Final answer: H = 0, S = 1, I = 85

WORKED EXAMPLE 2, FROM THE CLASS NOTES, THE BEST DRILL

R = 20, G = 40, B = 60
I = (20 + 40 + 60) / 3 = 40
S = 1 - 3(20)/120 = 0.5
theta = 150°
B (60) > G (40), so H = 360 - 150 = 210
Final answer: H = 210, S = 0.5, I = 40

WORKED EXAMPLE 3, THE EXAM-SITE VALUE IN THE NOTES

R = 100, G = 150, B = 200
I = 450 / 3 = 150
min = 100, S = 1 - 3(100)/450 = 0.3333
numerator = 0.5(100-150) + (100-200) = 0.5(-150) = -75
denominator = sqrt((-150)^2 + (-100)(-50))
theta = arccos(-0.8660) = 150 degrees
B (200) > G (150), so H = 360 - 150 = 210 degrees

HUE RANGE SECTOR

0 <= H < 120 degrees RG

120 <= H < 240 degrees GB

240 <= H < 360 degrees BR

"Then use the corresponding formula from the slide."

Honest gap on HSI to RGB. The study document gives you the **sector table only** and then explicitly defines: "Then use the corresponding formula from the slide". **The per-sector HSI to RGB formulas are not written anywhere in your notes.** Copy the three sector formulas off the Day 11 slide before the exam, because the exam framing in your study page says the question can come either way round.

Exam framing, quoted from the study page, "Kulau exam keluar, biasanya dia bagi macam ni: Given R = 100, G = 150, B = 200. Calculate H, S and I. And I Given H = 60, S = 0.5, I = 100. Calculate R, G and B."

Day 12
Data representation and what visualisation is

Representation exists because humans process data better in visual form than in any other form.

DATA REPRESENTATION, THE FOUR VERBATIM LINES

DEFINITION "Representing data in a pictured form for easy understanding."

PURPOSE "Simplifies large datasets and reveals hidden patterns."

VISUAL POWER "Transforms abstract numbers into concrete visuals."

"Graphics Representation is a way of representing any data in picturized form. It helps a reader to understand the large set of data very easily as it gives us various data patterns in a visualized form."

The quote to use: "A picture is worth a thousand words", repeated in the class notes. The **justification**, verbatim: "Scientists have found that the restoration and understanding of any information is better when it is available in the form of visuals as Human beings process data better in visual form than any other form." The class-note rule: **"Data Representation, depends upon how 'problem'."**

WHAT DATA VISUALIZATION IS, THE FIVE STATEMENTS

- "the representation of data using graphics such as 1 charts, graphs, maps, and other visual tools"
- "a powerful way for people, to display data so that it can be interpreted easily"
- "It helps tell a story with data, by turning spreadsheets of numbers into stunning graphs and charts"
- "These visualizations allow us to easily understand any patterns, trends, or outliers in a data set"
- "The purpose of data visualization is for DECISION-MAKING"

Statement 5 is a one-mark line. If a question asks why we visualise at all, the answer the course wants is **decision-making**.

Exam
The other three practice questions

Q3 FEATURE ENGINEERING EXERCISE, FROM DAY 3 PART A

VERBATIM, NOT ANSWERED IN YOUR NOTES, SO DO IT YOURSELF

"Given document corpus:

- This mobile is corrupt;
- This mobile is not good.
- Mobile is cheap and mobile is best.
- Mobile is cheap but screen is small.

(a) Extract Bag of Words and Bag of Bigram text."

(b) Extract TF-IDF from the above review text."

Q2 SPEECH ANALYTICS, THE SELF REVIEW ACTIVITY

Answered in full in your notes as "Proposal for Implementing Speech Analytics at Our Customer Care Centre", in three parts: A **what** speech analytics is, B **how** a speech analytics tool can benefit our company, five numbered benefits each with three bullets, C the recommended tool and its features, proposing **CallMiner** with a Feature, Description, Benefit table. The **full answer is the Day 7 box on this sheet**.

Q3 PART A, THE HSI EXAM FRAMING

VERBATIM FROM THE STUDY PAGE

"KALAU exam keluar, biasanya dia bagi macam ni: Given R = 100, G = 150, B = 200. Calculate H, S and I. And I Given H = 60, S = 0.5, I = 100. Calculate R, G and B. Yang wajib hafal ..."

followed by the three RGB to HSI formulas and the HSI to RGB sector table, both in the Day 11 box on this sheet.

Your own priority flags, treat as a to-do list.

"Kena tlgk exact this please", on the six levels of Linguistic Processing. "Computational understanding of word meaning (Bkgg balik yang ni)", that section is empty, go back to the slide.

"Dah serabut. So refer slide word meaning (Start Slide 43)", end of the word-embedding section, the Word2Vec maths is missing.

"Image Enhancement (Tengok Balik)".

"Done, Tengka Tengka", the Unsharping algorithm steps.

"...check this code", the falling sin(x)/x code.

"Done, But Kena Tengok Ulaah", the Q4 practice session must be re-checked on Ulearn.

Day 7
Speech analytics, exam Q2

This box is exam Q2: "Speech analytics: Refer to Ulearn (Week 4 Day 3): Self Review Activity". The self review is a three-part proposal, answered in your notes as "Proposal for Implementing Speech Analytics at Our Customer Care Centre", so structure your answer A, B, C.

Speech analytics exists because the most honest customer feedback a company owns is already sitting in its call recordings.

DEFINITION, GIVEN TWICE IN THE NOTES

"The process of analyzing recorded calls to gather customer information to improve communication and future interaction."

The process is primarily used by CUSTOMER CONTACT CENTRES to extract information from phone conversations between clients and a company.

SPEECH ANALYTICS VERSUS VOICE ANALYTICS, THE FULL TABLE

ASPECT	SPEECH ANALYTICS	VOICE ANALYTICS
Focus	Analyzes what is said	Analyzes how it is said
Data Type	Spoken words and conversation context	Vocal characteristics and audio patterns
Method	Speech-to-text conversion and phonetic indexing	Analyses tone, pitch, tempo, rhythm, stress, emotions
Purpose	Understand customer intent, topics, conversation context	Understand customer emotions, attitudes, behavioural cues
Key Insights	Common words, phrases, trends, customer concerns	Emotions such as anger, frustration, satisfaction, sarcasm
Example	Whether a customer is asking about shipment status, and the reason for the call	Whether a customer saying "amazing" is sincere, sarcastic or frustrated
Benefits	Improve customer experience, service quality, marketing strategies	Improve customer interactions, agent performance, operational quality
Output	Keywords, topics, conversation summaries, trends	Emotional insights, sentiment indicators, behavioural patterns
Keywords	"Highly appreciated", "Very impressed", "Top quality products", "Service was inadequate"	Tone indicating happiness, anger, frustration, sarcasm, satisfaction
Decisions	Strategic decisions from customer feedback and content	Decisions based on emotional responses and sentiment

Two variants. Synchronous Analysis is accomplished **real-time**, continuously monitored and tracked, allowing the service centre superior customer engagement and the ability to handle situations at the right time to mitigate **attrition and escalations**. **Archival Analysis** mines intelligence from **thousands of recorded calls and chats**, pinpointing cost drivers, trends and opportunities, identifying strengths and weaknesses in processes and products, and showing how offerings are perceived by the marketplace.

SIX CORE FUNCTIONS

1 **Real-time Mood Analysis**, analyses the mood or emotions of the customer with **pre-defined time slices**. 2 **Agent Quality Scoring**, score agents on compliance parameters to enhance quality assurance, identifying training and resource re-allocation requirements. 3 **Voice Transcription**, convert voice to text using **statistical machine translation techniques**, plus customer sentiment and meaning. 4 **Topic Detection**, automatically **identify, group and organize** words and phrases spoken during calls into **themes**, revealing rising trends and areas of opportunity or concern. 5 **Performance Analysis of the service centre**, process thousands of pieces of unstructured data rapidly. 6 **Other functions**, four: **Voice activity detection**, identifying segments of a waveform where only speech is present - **Speech enhancement**, filtering and separating noise from the speech segments - **Text to speech**, synthesising natural speech from text, where sounding natural with emotion is the challenge - **Speaker diarisation and recognition**, diarisation segments the signal by speaker, recognition identifies who is speaking at a particular time.

THE SPEECH ANALYTICS PROCESS, 5 ORDERED STAGES

1 Take the **UNSTRUCTURED** data trapped in the audio of recorded calls and turn it into **STRUCTURED** data that can be searched and analyzed. Incorporates conversations from the source system (call recorder, VOIP stream) plus **METADATA**, which agent handled the interaction, what day and time it occurred, who the customer was

2 The audio undergoes the **SPEECH RECOGNITION** process where sounds are turned into text

3 The **ACOUSTIC SIGNALS** such as agitation and silence are extracted, and text transcripts are **NORMALIZED** into a consistent form

4 Analyzes the interactions for certain **LANGUAGE PATTERNS** to categorize or tag contacts, combining the presence of certain language and other key metrics into an **INDEX** that measures agent quality, customer satisfaction, emotion and first contact resolution

5 **DISCOVERY, CATEGORY ANALYSIS and SCORE ANALYSIS** through a web interface that lets users search contacts by any criteria, visualize data and conduct automatic topic analysis. Data is actioned by direct feedback to analysts, supervisors and agents through notifications and reports

BENEFITS TO INDUSTRY, FIVE AREAS

Revenue Generation, reduce customer churn, improve wallet share, collections and payment behaviours, improve effectiveness of cross-sell and up-sell. **Call Handling Behaviour**, improve FCR and reduce defects, reduce AHT and improve agent performance. **Process Improvement**, identifies drivers of long calls, repeat calls and process-induced calls, contact index by customer segment and persona. **Customer Experience**, identify dissatisfaction drivers by drilling down on sentiments, measure and improve the experience across channels. **Compliance**, identify compliance and risk issues, enhance regulatory adherence, effective audit and efficient call quality audits.

General benefits. Reduce Operational Cost by reducing average handling time (AHT), by **call deflection**, re-routing a call, by **first call resolution**, "the ability of contact centers to resolve customer problems, questions or needs the first time they call, with no follow-up required", and by managing **transfers**, relocating a call to another phone. Then **Increase Revenue and Improve Customer Satisfaction and Retention**.

OUT OF BOX

NICHE PRODUCTS	CUSTOM SPEECH ANALYTICS
CallMiner, ClaraBridge, VocTech, Tech	Invoca, Balto Software, Cogito, Voicebase, Gridspace, Nuance

Five challenges, verbatim. 1 SA hasn't gone mainstream yet (in Malaysia?). 2 Analytics-enabled quality assurance (AQAI) is still underused. 3 Real-time SA has a limited number of use cases. 4 Speech is on the rise, despite contrary predictions. 5 Integrating SA tools with predictive analytics is expensive.

THE MODEL ANSWER SKELETON FOR Q2

A. What is Speech Analytics. The process of analyzing recorded customer conversations to extract useful information and insights; it converts spoken conversations into structured data that can be searched and analyzed, allowing organizations to identify trends, customer sentiments and areas for improvement.

B. How it benefits the company, five headings. 1 Improve Customer Satisfaction, identify complaints and dissatisfaction drivers, understand sentiment and expectations, improve experience. 2 **Improve Agent Performance**, quality training and coaching needs, improve **First Call Resolution (FCR)**. 3 **Reduce Operational Costs**, reduce **Average Handling Time (AHT)**, minimize repeated calls and unnecessary transfers. 4 **Increase Revenue**, up-selling and cross-selling, improve retention and reduce churn. 5 **Improve Compliance and Risk Management**, detect compliance issues, support call auditing and QA.

C. Recommended tool: CallMiner, a popular platform widely used in customer contact centres.

CALLMINER, THE FEATURE / DESCRIPTION / BENEFIT TABLE

Real-Time Mood Analysis detects customer emotions during calls, so agents respond appropriately and satisfaction improves

Voice Transcription converts speech into text, so conversations become searchable and easier to analyze

Agent Quality Scoring measures agent performance against predefined criteria, improving service quality and identifying training needs

Topic Detection automatically identifies common topics and trends, so management understands customer concerns and market trends

Performance Analysis analyzes large volumes of customer interactions, giving insight into customer sentiment and operational performance

Keyword Spotting detects important keywords, identifying complaints, service issues and opportunities

Job scope. Responsibilities: input and guidance on new features; interpret speech and non-speech data to measure customer satisfaction, agent performance and capability, operational processes; feedback on optimization opportunities; assist in developing data strategies; process improvement. Requirements: bachelor's degree or equivalent; minimum 2 year's experience with platforms such as CallMiner. **NICE**, Verint, delivery and support experience; ability to define best practices; strong analytical and technical skills.

Day 12
Data storytelling and the narrative arc

Storytelling exists because raw data rarely inspires action on its own.

DEFINITION, VERBATIM

"Data storytelling is the broader skill of communicating insights in a way that resonates with your audience. This technique brings together DATA ANALYTICS, VISUALIZATION, DOMAIN KNOWLEDGE, and THOUGHTFUL PRESENTATION to make insights clear, compelling, and actionable. Instead of just sharing data, you deliver a data story with CLARITY, CONTEXT, and PURPOSE, so your message resonates."

Four ingredients to memorise: data analysis, visualization, domain knowledge, thoughtful presentation. Three qualities: clarity, context, purpose.

WHY IT MATTERS, FOUR REASONS

1 **Makes data understandable**. "Not everyone speaks the language of data. Insights stop being 'just numbers' and start becoming ideas your team can act on." 2 **Builds trust**. "When everyone's working from the same clear story, it creates alignment. That shared understanding builds trust in both the data and the decisions that follow." 3 **Drives better decisions**

Applied Machine Learning

CLASSICAL MACHINE LEARNING: THE WORKFLOW, THE FEATURES, THE EVALUATION, THE NINE CORE ALGORITHMS

REVISION SHEET

The course is one sentence: **right features build right models which achieve the right task**. You **prepare data** so features are trustworthy, **choose a model family** because the task is geometric, probabilistic or logical, **balance bias against variance** so it generalises, and **pick a metric that cannot lie to you**. Nearly every exam mark is one of those four moves applied to a scenario.

L1 Workflow and experiment design

A model is worthless until someone can feed it data and act on the output, so the pipeline comes before the algorithm. AI is the intelligence, ML is the compute methods implementing it, so ML is the enabler for AI.

DATA PREPARATION, 5 STEPS

Collect > **Clean** > **Feature eng.** > **Transform** > **Split**

- Collect and integrate:** databases, APIs, scraping into a warehouse or lake, standardising formats and schemas.
- Clean:** missing values (impute mean or median, or delete), inconsistency ("NYC" vs "New York"), duplicates, outliers (remove, cap, transform).
- Feature engineering:** derived features (birthdate to age, cost per square foot), encoding, dimensionality reduction.
- Transform:** normalization to 0 to 1, standardization to mean 0 sd 1.
- Split:** train 60 to 80%, validation for tuning, test for the final unbiased score. Garbage in, garbage out. Too many features may not be relevant, they only add complexity.

EXPERIMENT DESIGN, 3 STAGES

1 **Choose the model** from the problem, data and class type, categorical means classification and continuous means regression. 2 **Tune parameters** for highest accuracy and lowest error. 3 **Compare** on the test set against other models.

Three errors: training error optimizes the hypothesis parameters, validation error finds the best hyperparameters, test or generalization error estimates real world performance.

Split 70/30 > **k fold CV** > **Accuracy, RMSE** > **Compare** > **GridSearch**

THREE MODEL OUTPUT FAMILIES

Geometric, hyperplanes, linear transformations, distance metrics: SVM, KNN, PCA (the kernel trick changes how the boundary is computed). **Probabilistic**, learning reduces uncertainty: Bayes, likelihood ratio, Naive Bayes. **Logical**, interpretable rules: decision trees.

Course dataset: breast cancer, 681 women, 12 attributes, class 0 malignant 238, 1 benign 443. Mining tasks: Predictive: classification, regression, time series. Descriptive: clustering, association, profiling, anomaly detection. Responsible AI, 6: fairness, reliability and safety, privacy and security, inclusiveness, transparency, accountability.

L1 The four learning types

They exist because the supervision you can afford differs, and that alone fixes the algorithm family.

	SUPERVISED	UNSUPERVISED	SEMI SUPERVISED	REINFORCEMENT
Data	Labeled	Unlabeled	Small labeled plus much unlabeled	No dataset, an environment
Driver	Task driven	Data driven	Label scarcity	Reward driven
Learns	Correlations between instances and labels	Similarities with no names	Initial model refined by unlabeled data	Optimal policy by trial and error
Slogan	"That is a watermelon"	"That is like this other thing"	"Label a few, guess the rest"	"Eat that, it tastes good"
Examples	DT, KNN, NB, spam	K means, hierarchical, association, autoencoders	Tagging with few labels	Q learning, SARSA, Go

Why **semi supervised** exists: building a labelled set is a painstaking process, so a small labelled set builds an initial model which the unlabeled data refines.

Eager vs lazy. Eager builds a generalized model **before** prediction and predicts fast (tree, Naive Bayes). Lazy builds nothing, stores instances, and does more work **at test time** (KNN). **Parametric vs non parametric** is whether the parameter count is fixed; trees and KNN are both non parametric.

L2 Features

A feature is a measurable property of the observed object, formally a mapping $f_i : X \rightarrow F_i$. Features matter more than the algorithm choice. **Six types:** numerical (continuous, discrete), categorical (nominal, ordinal), boolean, text (BoW, TF-IDF), time series, image. **Structured** maps into rows and columns, **unstructured** does not.

ANALYSIS

Dispersion: variance (mean squared deviation), standard deviation (same scale), range, midrange, percentiles, quartiles and IQR (outlier resistant), coefficient of dispersion (unit free). **Shape:** skewness, kurtosis, cumulative distribution, histogram, normality.

TRANSFORMATION, 7 KINDS

1 **Binarization** to 0 / 1 on a condition · 2 **Unordering**, remove a false order with one hot · 3 **Thresholding** at a cutoff · 4 **Discretization** into bins, handling non linearity and reducing noise · 5 **Recursive partitioning**, the basis of trees · 6 **Agglomerative merging** · 7 **Normalization** into 0 to 1.

SCALING

min max normalization: $X_{\text{norm}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$
standardization (z): $x' = (x - \mu) / \sigma$

Which: **standardization for distance based algorithms** (KNN, k means) because it resists outliers and keeps the distribution shape. Min max when a hard 0 to 1 bound is needed. **Encoding: one hot** gives a new binary column per category and is the tool for unordering; **label** gives one integer per category (Male 0, Female 1) but invents an ordering.

SELECTION, FOUR FAMILIES

FAMILY	METHOD	OUTPUT
Filter model Independent	Pearson correlation, Chi square, ANOVA F test, mutual information	Top k features
Wrapper model dependent	Forward selection, backward elimination, RFE	Best subset
Embedded during training	Lasso L1, tree or forest importance, Elastic Net	Nonzero weights
Dim. reduction	PCA, LDA, t-SNE, makes new features	Reduced space

CONSTRUCTION, FIVE TYPES

1 **Mathematical** ranks, differences, sums · 2 **Polynomial** and **interaction**, $x1^2 \cdot x2$ · 3 **Text**, word count, BoW, TF-IDF which fixes the preferential treatment of rare words and drives document search · 4 **Image:** pixel (2D flattened to 1D, RGB is three images, 0 to 255), colour (6 x 3 = 18 features), metadata, **edge detection** Sobel and Canny with edge score = edges / resolution, **HOG** which fails when orientation changes · 5 **Aggregated** group statistics.

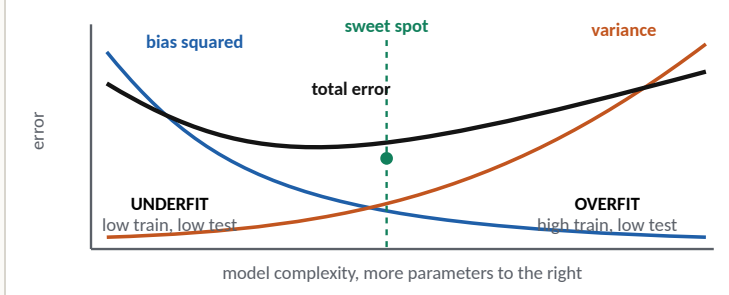
L1b Bias, variance, fitting, CV

Every error splits into two causes and the cure for one worsens the other, so name which you have before fixing anything.

THE IDENTITY

Total expected error = bias + variance

Bias is how far predictions sit from correctness, arising when the algorithm has limited flexibility to learn the true signal. **Variance** is how much predictions vary between iterations, sensitivity to the particular training set.



High bias means too simple, it underfits. High variance means too complex, it memorises noise. The dashed line minimises total error.

TRAIN	TEST	DIAGNOSIS	FIX
High	High	Underfit, high bias	More complex model, add features
Low	High	Overfit, high variance	Simplify, regularize, more data
Low	Low	Good fit	Stop

Six diagnostic moves: separate train and test sets, try another algorithm, fit the parameters, tune the hyperparameters, choose proper metrics, use cross validation. Also more data, better and more features, combine models, inspect misclassified instances.

CROSS VALIDATION

A single split gives a high variance estimate, so average over several splits. **Same data for train and test:** never, it rewards complex models that over fit. **Hold out**, one 70 / 30 split: fast, simple, flexible, but still **high variance** because it depends on the split. **K fold CV**, k = N splits averaged: a much better estimate. **K times slower**. LOOCV, k = N: maximum use of data, computationally expensive. Split table: training 60 to 70%, validation 15 to 20%, test 15 to 20%.

L1b Class imbalance

Accuracy is a fraction of all predictions, so a dominant class lets a model score high while learning nothing about the class you care about.

Worked case: 100 instances, Class 1 = 80, Class 2 = 20, ratio 4 : 1. With 90% in Class 1, always predicting Class 1 gives about **90% accuracy**. Fraud version: non fraud scores 100%, fraud scores 0%, and accuracy is **high because of the non fraud transactions, not because the model is good**.

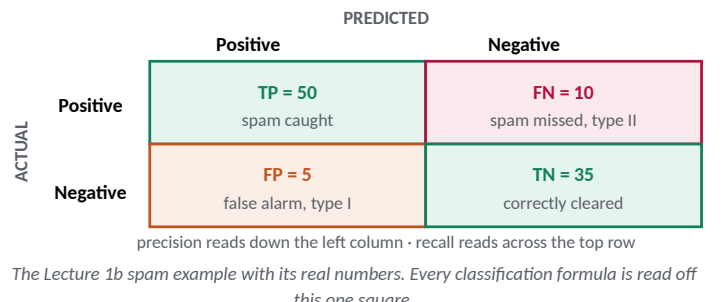
Five options: 1 collect more data · 2 change the metric to precision, recall, F1 or ROC · 3 resample · 4 synthesise samples with SMOTE · 5 try a different algorithm.

Random under sampling drops majority rows: good with millions of rows, but removes potentially valuable information. **Random over sampling** duplicates minority rows: good when data is scarce, but causes overfitting and poor generalization.

SMOTE, 5 steps. 1 Choose a minority instance as the input vector. 2 Find its k nearest neighbours, k being an argument of SMOTE(). 3 Choose one neighbour and place a synthetic point anywhere on the line joining them. 4 Take the vector between them, multiply by a random number in 0 to 1, add it to the current point. 5 Repeat until balanced. **Pro:** beats plain oversampling by interpolating between positives. **Con:** complicates the decision surface and creates many noisy points.

L1b Metrics and the confusion matrix

One accuracy number hides which of the two mistakes you are making, and the two mistakes cost different amounts.



precision reads down the left column · recall reads across the top row

The Lecture 1b spam example with its real numbers. Every classification formula is read off this one square.

CLASSIFICATION, AND THE WORKED NUMBERS ABOVE
accuracy = (TP+TN)/(TP+TN+FP+FN) = 85/100 = 0.85
precision = TP/(TP+FP) = 50/55 = 0.909 positive calls correct
recall = TP/(TP+FN) = 50/60 = 0.833 sensitivity, TPR
specificity = TN/(TN+FP) = 35/40 = 0.875 TNR
F1 = 2*(precision*recall)/(precision+recall)
= 2*(0.909 * 0.833)/(0.909+0.833) = 2*0.758/1.742 = 0.870

ROC AND AUC

ROC plots TPR = TP/(TP+FN) against FPR = FP/(FP+TN) as the decision threshold sweeps from 1 down to 0. AUC = area under that curve. 1.0 perfect, 0.5 random.

REGRESSION

MAE = average(|y_t - y_hat|)
MSE = average((y_t - y_hat)^2)
RMSE = sqrt(MSE) back in the original units
R^2 = 1 - SS_res/SS_tot
= 1 - sum((y_i - y_hat_i)^2) / sum((y_i - y_bar)^2)

Which metric, asked every year. "No positive may be missed" means recall (cancer screening, attack detection). "False alarms are expensive" means precision. Both equally means F1. Imbalanced data means **never accuracy**. Regression: RMSE for original units, MAE when outliers must not dominate. **Class exercise:** y = 5, 7, 10, 4, 8 against predicted 4, 6, 13, 6, 7. Errors 1, 1, -3, -2, 1, so MSE = 16/5 = 3.2, RMSE = 1.789.

L1b Decision trees

Trees exist because people must be able to read the model. Hierarchical divide and conquer, supervised, non parametric, for classification and regression. Internal node = attribute test, branch = value, leaf = label. There can be **more than one tree fitting the same data**, and the goal is the **smallest tree**.

SHANNON ENTROPY, MEASURED 0 TO 1
E(S) = sum over i=1..c of (-p_i * log2(p_i))
all one class -> 0 (pure) half and half -> 1 (max disorder)
100 points, 30 positive: E = -0.3*log2(0.3) - 0.7*log2(0.7) = 0.88

INFORMATION GAIN, TAKE THE LARGEST
Gain(S,A) = Entropy(S)
- sum over v in Values(A) of (|S_v|/|S|) * Entropy(S_v)
slide shorthand: Gain(T,X) = E(T) - E(T,X)

GAIN RATIO, FIXES IG'S BIAS TO MANY VALUED ATTRIBUTES
SplitInfo(S,A) = - sum over v of (|S_v|/|S|) * log2(|S_v|/|S|)
GainRatio(S,A) = Gain(S,A) / SplitInfo(S,A)

GINI INDEX, TAKE THE LOWEST

Gini = 1 - sum over i=1..c of (p_i^2)
how often a random element would be labelled incorrectly

PLAYTENNIS, FULLY WORKED, 14 DAYS, YES 9 NO 5

1 E(PlayTennis) = E(9/14, 5/14) = E(0.64, 0.36)
= -0.64*log2(0.64) - 0.36*log2(0.36) = 0.940
2 E(PlayTennis, Outlook) = (5/14)*E(3/5,2/5)
+ (4/14)*E(4/4,0/4) + (5/14)*E(2/5,3/5)
= (5/14)*0.971 + (4/14)*0 + (5/14)*0.971 = 0.693
3 Gain(PlayTennis, Outlook) = 0.940 - 0.693 = 0.247

ATTRIBUTE	COUNTS YES / NO	GAIN
Outlook	Sunny 3/2, Overcast 4/0, Rain 2/3	0.247 root
Humidity	High 3/4, Normal 6/1	0.152
Wind	Weak 6/2, Strong 3/3	0.048
Temperature	Hot 2/2, Mild 4/4, Cool 3/1	0.029

STEP 5. CONVERT TO IF THEN RULES
Overcast has entropy 0 so it is a leaf. Sunny splits on Wind, Rain splits on Humidity.
R1 Sunny + Wind=Weak -> Yes R2 Sunny +
Wind=Strong -> No
R3 Overcast -> Yes R4 Rain + Humidity=High ->
No
R5 Rain + Humidity=Normal -> Yes

STEPS, SETTINGS, PRUNING

1 Entropy of the target. 2 Split on each attribute, get each branch entropy and the weighted entropy. 3 Gain = target entropy minus weighted entropy, take the largest, divide by its branches, repeat. 4 Entropy 0 makes a leaf, above 0 needs another split. 5 Convert to IF THEN rules.

Hyperparameters: split criterion (entropy, gini, gain ratio), max depth, min samples per split, min information gain, pruning. **Stopping:** one class in the node, no attributes left (majority vote labels the leaf), no samples left, max depth, min gain. **Use it when** you need an interpretable rule set on tabular data.

Pruning reduces depth and removes branches that do not contribute, because a tree classifying training data perfectly may not generalize. **Pre pruning** stops growth when a node's goodness measure is below a threshold, and the stopping point is hard to choose. **Post pruning** grows the full tree then removes nodes lacking evidence, using a validation set to pick the best pruned tree.

Impurity: a node at (0,1) has zero impurity, (0.5,0.5) the highest. Gender (Male 6/4, Female 4/6) is impure; Car type (Family 1/3, Sport 8/0, Luxury 1/7) is pure. **Continuous** variables are discretized into intervals; **feature importance** is impurity reduction summed over all splits.

Strengths: easy to understand, interpret and visualise, inexpensive, tolerates missing values and irrelevant features. **Weaknesses:** complex with many class labels, poor on numeric labels and high dimensional data such as images, overfits easily, sensitive to small data changes.

L1b Ensemble learning

One model carries one set of mistakes, so combining models whose mistakes differ cancels part of the error. Diversity comes from a different hypothesis, technique, seed or population.

METHOD	MAIN IDEA	TRAINING	REDUCES	EXAMPLES
Bagging	One algorithm on random subsets drawn with replacement, then averaged	Parallel	Variance	Random Forest, bagged trees, Extra Trees, MetaCost
Boosting	Train sequentially, reweighting the previous model's errors	Sequential	Bias and variance	AdaBoost, Gradient Boosting, XGBoost
Randomization	Randomness in feature or data selection to force diversity	Parallel	Correlation between learners	Random Forest, Extra Trees
Voting	Combine diverse algorithms: majority vote or average probability	Parallel	Individual error	KNN + DT + SVM
Stacking	A meta model learns the best combination	Parallel plus meta	Both, by learning the models	Stacking Classifier or Regressor

Which suits which: bagging suits **low bias, high variance** models, the complex ones such as deep trees. Boosting suits **low variance, high bias** models, the simple ones such as stumps.

Bagging and AdaBoost steps. 1 Draw multiple random samples of m rows, **same size as the training set, with replacement**, training a model on each. 2 Each model predicts. 3 Combine and average all m results. Four stages: bootstrap sampling, training, forecasting, aggregating. Bagging **unpruned** trees gives good probability estimates because probabilities are averaged rather than voted, and it helps most when data is noisy. **Hyperparameters:** estimators, sample size, base learner.

Boosting and AdaBoost steps. 1 The base learner gives **equal weight** to every observation. 2 Train the first weak learner. 3 **Greater weight goes to wrongly predicted observations** so the next model corrects them. 4 Continue until minimum error or no further improvement. 5 Combine by **weighted voting**. AdaBoost adjusts sample weights each iteration. Gradient Boosting fits new models on the **residuals**, and XGBoost, LightGBM and CatBoost are optimized versions. **Hyperparameters:** estimators, learning rate, base depth. **Caution:** sensitive to noise and outliers.

Randomization, three techniques. 1 **Random subspace**, a random feature subset per learner, decorrelating the models. 2 **Random Forest** = bagging on rows plus a random feature subset at every split. 3 **Extra Trees**, randomized rather than optimal split thresholds, lowering variance at the cost of bias.

Stacking steps. 1 Train several Level 0 base models on the full training set. 2 Train the Level 1 meta model on the base outputs used as features. 3 The meta output is the prediction. The meta learner is usually linear or logistic regression, and it **requires careful cross validation to avoid overfitting**.
WORKED: FIVE STUDENTS RATE A MOVIE 5, 4, 5, 4, 4
Max voting -> majority 4 -> final 4
Averaging -> (5+4+5+4+4)/5 = 4.4 -> final 4.4
Weighted average -> weights .23 .23 .18 .18 .18
(5*.23)+(4*.23)+(5*.18)+(4*.18)+(4*.18) = 4.41
VotingClassifier(estimators=[('lr',m1),('dt',m2)], voting='hard')
Majority voting over 5 emails with classifiers A, B, C:
2:1 Spam, 1:2 Not Spam, 2:1 Spam, 1:2 Not Spam, 3:0 Spam
all five match the truth -> ensemble accuracy 5/5 = 100%

Advantages: higher accuracy, robust to overfitting, mixes model types, handles complex data. **Disadvantages:** expensive, hard to interpret, stacking and boosting often without tuning.

L4 K Nearest Neighbour

The simplest classifier: assume things close together in feature space share a label. Supervised, **instance based, lazy, non parametric**.

Steps. 1 Store the training data, there is no training phase. 2 On a new sample compute its distance to every training sample. 3 Sort ascending. 4 Take the k smallest. 5 Classification: **majority vote**. 6 Regression: **average** of the k values. **Hyperparameters:** k, the distance metric, the scaling applied first. **Use it when** the data is small to moderate, the boundary is irregular, and training cost must be zero.

EVERY DISTANCE FORMULA

Euclidean, the default, translation invariant:
dist(P,Q) = sqrt((q1-p1)^2 + (q2-p2)^2 + ... + (qn-pn)^2)
Manhattan, city block, an approximation to Euclidean:
dist(P,Q) = |q1-p1| + |q2-p2| + ... + |qn-pn|
Minkowski, the general form:
dist(P,Q) = (sum over k=1..m of |x_ik - x_jk|^p)^(1/p)
p = 1 -> Manhattan p = 2 -> Euclidean
One dimensional case: d = |x_query - x_i|

WORKED: P(-2,2) AND Q(2,5), THEN MEDICINE AT K = 3
Euclidean = sqrt((2+2)^2 + (5-2)^2) = sqrt(16+9) = 5
Manhattan = |2+2| + |5-2| = 4 + 3 = 7

A(1,1) Non drug B(2,1) Drug C(4,3) Drug D(5,4) Non drug
new sample X(3,2)
D(X,A) = sqrt((2-1)^2) = 2.2 rank 2
D(X,B) = sqrt(1^2+1^2) = 1.4 rank 1
D(X,C) = sqrt(1^2+1^2) = 1.4 rank 1
D(X,D) = sqrt(2^2+2^2) = 2.8 rank 3
3 nearest B, C, A -> Drug 2/3, Non drug 1/3 -> X is DRUG

WORKED: REGRESSION ON STORE SALES, QUERY 2200, K = 3
S1 1000->200 S2 1500->260 S3 2000->300
S4 2500->340 S5 3000->380
distances 1200, 700, 200, 300, 800 -> nearest S3, S4, S2
prediction = (300 + 340 + 260) / 3 = 300 units

Why **scaling matters most here**, KNN **measures distance**, so a large ranged feature dominates the sum. If feature A runs 0 to 8 and feature B is much larger, the calculation is heavily weighted by B. **Standardization is generally preferred:** less sensitive to outliers, keeps the distribution shape, best for distance based algorithms.

K too small gives a narrow, very flexible model, **low bias and high variance**, sensitive to noise and outliers, which **overfits** and cannot generalize. **K too large** smooths the boundary and averages out noise, **low variance and high bias**, overly broad, which **underfits**, and it costs more to compute.

Tips: choose an **odd k** especially when the class count is even, apply the **elbow method**, and use **cross validation** over several k, which the slides call the best option. Common k: 3, 5, 7. **Slide elbow example:** error is 33.33% for K = 1 to 5, drops to 0% at K = 7, the **elbow**, and returns to 33.33% from K = 9, so optimal K = 7.

Decision boundaries form a **Voronoi diagram**, zones of control around stored points; more examples fragment it and noise makes it nasty, which is overfitting. **Strengths:** any distribution, no linear boundary needed, simple, effective on small to moderate data, no training stage. **Weaknesses:** irrelevant or correlated features must go, ineffective in high dimensions (curse of dimensionality), all cost is at test time, it stores the whole training set, and k is subjective.

L5 K means clustering

Unsupervised learning exists because unlabeled data is far easier to get. A **cluster** is a set of items similar to each other and dissimilar to items in other clusters. **Three unsupervised tasks:** dimensionality reduction, clustering, association rule mining. John Snow plotted cholera deaths in the 1850s and cases clustered around polluted wells.

Steps. 1 Choose k random points as initial centroids, the seeds. 2 Assign each point to the closest centroid. 3 Recompute each centroid, every attribute being the mean of its members, with **majority rules for binary or nominal variables**. 4 If not converged, repeat 2 and 3. **Convergence** means objects no longer move cluster. **Hyperparameters:** k, initial seeds, distance metric, max iterations. **Use it when** clusters are roughly spherical and you can name k.

WCSS, ALSO CALLED SSE, THE COHESION MEASURE
WCSS = sum over clusters k of
sum over x in C_k of ||x - centroid_k||^2
Intra cluster cohesion (compactness) = how near points are to their own centroid, measured by SSE. Inter cluster separation (isolation) = centroids of different clusters must be far apart.

Elbow method: run k means for k = 1, 2, 3 upward and plot WCSS against k. WCSS always falls as k rises so you never take the minimum, you take the k at the **end**, where another cluster stops buying much reduction. Silhouette score confirms the choice.

FULLY WORKED, K = 2, A(1,1) B(2,1) C(4,3) D(5,4)
ITER 0 seeds C1 = A = (1,1), C2 = B = (2,1)
D(C1,.) = 0, 1, 3.61, 5 D(C2,.) = 1, 0, 2.83, 4.24
assign C1 = {A}, C2 = {B,C,D}
new C1 = (1,1), C2 = ((2+4+5)/3, (1+3+4)/3) = (3.67, 2.67)
ITER 1 D(C2,A) = sqrt((1-3.67)^2+(1-2.67)^2) = 3.14
D(C2,B) = 2.36 D(C2,C) = 0.47 D(C2,D) = 1.89
assign C1 = {A,B}, C2 = {C,D}
new C1 = (1.5, 1) C2 = (4.5, 3.5)
ITER 2 D(C1,.) = 0.5, 0.5, 3.2, 4.61
D(C2,.) = 4.3, 3.54, 0.71, 0.71
assignment unchanged, objects no longer move -> CONVERGED

Quality metrics. **Silhouette** s(i) = (b(i)-a(i)) / max(a(i),b(i)): 1 is ideal and well separated, 0 means overlap or wrong k, negative means the point sits in the wrong cluster. **Calinski Harabasz**, between over within dispersion, higher is better. **Davies Bouldin**, average similarity to the most similar cluster, minimum 0, smaller is better.

Strengths: simple, time efficient, best known, finds all clusters at once. **Weaknesses:** needs a defined mean, you must specify k, sensitive to outliers, and it only finds hyperspheres or hyperellipsoids. Mitigation shown: remove points far from every centroid.

L5 Hierarchical clustering

It exists for when you cannot name k in advance: build the whole hierarchy and cut it later. **Agglomerative, bottom up:** every element starts as its own cluster and merges upward. **Divisive, top down:** start with the whole set and split downward. The result is a **dendrogram**.

Steps. 1 Every instance is its own cluster. 2 Compute the distance between every pair using the linkage function. 3 Merge the two closest. 4 Update the distance matrix. 5 Repeat until one cluster remains. **Hyperparameters:** linkage function, distance metric, cut height. **Use it when** k is unknown or the structure itself is the answer.

Single linkage, nearest neighbour, is the distance between the **closest** pair of objects, the shortest link, and it chains clusters along elongated shapes.

Complete linkage, farthest neighbour, is the distance between the **most distant** pair, the longest link, and it produces compact clusters of roughly equal size.

SAME MATRIX, BOTH LINKAGES. D(A,B)=1 D(A,C)=4 D(A,D)=5 D(B,C)=2 D(B,D)=6 D(C,D)=3
SINGLE COMPLETE
d=0 k=4 {A} {B} {C} {D} d=0 k=4 {A} {B} {C} {D}
d=1 k=3 {A,B} {C} {D} d=1 k=3 {A,B} {C} {D}
d=2 k=2 {A,B,C} {D} d=2 k=2 {A,B} {C,D}
d=3 k=1 {A,B,C,D} d=3 k=1 all, joined at height 6
reduced min 2, 5, 3 reduced max 4, 6, then 6

COMPLETE LINKAGE, P1 TO P5
d(P1,P2)=9 d(P1,P3)=3 d(P1,P4)=6 d(P1,P5)=11 d(P2,P3)=7
d(P2,P4)=5 d(P2,P5)=10 d(P3,P4)=9 d(P3,P5)=2 d(P4,P5)=8
1 smallest D(P3,P5) = 2 -> merge P3, P5
2 max update: (P3,P5)-P1 = max(3,11) = 11
(P3,P5)-P2 = max(7,10) = 10
(P3,P5)-P4 = max(9,8) = 9
3 smallest D(P2,P4) = 5 -> merge P2, P4

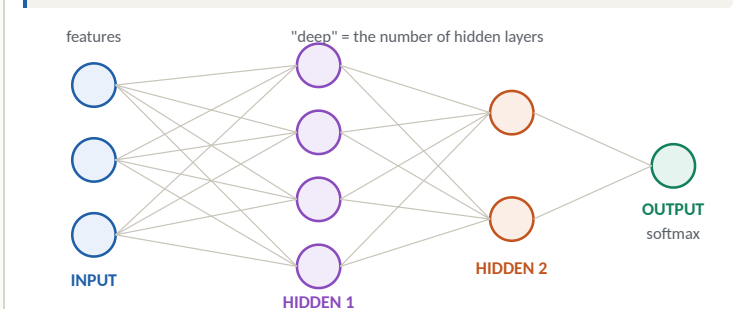
Sheet 1 is the model you can read, this sheet is models you cannot. A deep network replaces hand crafted features with **learned hierarchical representations**, and you pay for that with data, compute and interpretability. So the questions change shape: not 'which algorithm', but **which architecture matches the inputs** (images to CNN, sequences to RNN, goals to reinforcement learning, columns to a plain MLP), **what does each layer do to the tensor**, and **how do you run it when one machine is no longer enough**. The case study answer is nearly always **horizontal scaling**.

L8 The perceptron and the neural network

Neural networks exist because hand designing features stops working once you have raw pixels, audio or text. The network learns the features itself. Networks **mine** the brain: about **100 billion neurons**, each connecting to roughly **100,000 neighbours**. A neural network is built from **perceptrons** that replicate the role of a neuron, a computational unit exchanging messages. Two directions of travel: **feed forward** and **backpropagation**.

ONE NEURON, THE WHOLE OF IT
 $z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$
 $= \text{sum over } i \text{ of } (w_i \times x_i) + b$
 $a = \text{f}(z)$ f is the activation function

w = weights (learned), b = bias (learned), a = output passed on to every neuron in the next layer



Traditional neural networks carry 2 to 3 hidden layers. Deep networks can carry as many as 150. Each layer is a container that receives weighted input, transforms it with mostly non-linear functions, and passes values on.

WHY DEEP, AND WHY NOW

Deep learning stacks many perceptron layers and learns **hierarchical representations**: low level features (lines, dots, curves), then mid level, then high level (eyes, features, wings), then a trainable classifier.

Two reasons it became popular: 1. It requires **large amounts of labeled data** (driverless car development needs millions of images and thousands of hours of video), and 2. It requires **substantial computing power**, high performance CPUs with parallel architecture, which with clusters or cloud cuts training from weeks to hours. Around 2010 deep learning started outperforming other ML techniques, first in speech and vision, then in NLP.

THE CPU NUMBER FROM THE SLIDES
running time without GPU / with GPU = 733 / 27
= 27.1 times faster

WHAT HUMANS STILL DO

ML only automates the optimization of weights on features. Humans still define the task, terminology and evaluation metrics, annotate the training and testing data, supply intuition about useful features, do the error analysis, and define constraints to guide the learner. Features are more important than the choice of learning algorithm.

LIMITS OF DEEP LEARNING

- Needs large data. More powerful models mean more parameters which mean more data.
- Once trained it is **inflexible and cannot multitask**. Even a similar problem needs retraining.
- Anything requiring **reasoning**, long term planning or algorithm like data manipulation is beyond current techniques even with large data.
- Interpretability is the main issue**. You can find which nodes activated, but not what those neurons were supposed to model. Decision trees and logistic regression do not have this problem.

When to use which. Use **traditional ML** for high quality low dimensional data such as database columns. Use **deep learning** when simpler models miss the needed accuracy, when the pattern matching is complex (images, NLP, audio), when dimensionality is high, or when your vectors carry a time dimension. You get poor results from both when data is incomplete or of poor quality. Rule of thumb: you need at least 5,000 labelled training examples to train a neural network.

No Free Lunch Theorem. There is no one model that works for every problem. The assumption that make a good model for one problem may not hold for another, so it is normal practice to try multiple models. **Matching networks to problems**: columnar data to multilayer perceptrons, images to CNNs, time series sequences to RNNs, goal oriented tasks to deep reinforcement learning.

FIVE MAJOR ARCHITECTURES

ARCHITECTURE	WHAT IT IS FOR
UPN unsupervised pretrained networks	Initialises a discriminative net from one trained unsupervised. DBN, deep autoencoder, GANs
CNN	Image classification and anything with spatial structure
RNN	Sequences changing over time: NLP, speech, translation, captioning, conversation, visual Q and A
Recursive	Variable length input. RNNs additionally model hierarchical structure
Deep RL	Agents, environments, states, actions, rewards

By purpose: to generate data use GANs, VAEs or RNNs. To model images use CNNs or DBNs. To model sequences use RNN or LSTM.

Pioneers names: Geoffrey Hinton, Yann LeCun, Andrew Ng, Yoshua Bengio. Ng's line: the rocket engine is the deep learning model and the fuel is the huge amount of data we feed it.

L8 Activation functions

Without a non linear activation, stacking layers collapses into one linear map, so depth would buy nothing. The activation decides whether a neuron fires and is the non linear transformation over the input signal.

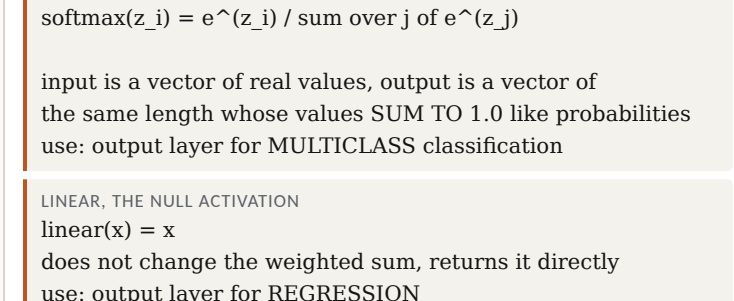
SIGMOID, SQUASHES INTO 0 TO 1
 $\text{sigmoid}(x) = 1 / (1 + e^{-x})$
range 0 to 1, sigmoid(0) = 0.5
use: output layer for BINARY and MULTILABEL classification

TANH, SQUASHES INTO -1 TO 1, ZERO CENTRED
 $\text{tanh}(x) = (e^{-x} - e^{x}) / (e^{-x} + e^{x})$
range -1 to 1, tanh(0) = 0
use: hidden layers of RNNs

RELU, RECTIFIED LINEAR UNIT, THE DEFAULT TODAY
 $\text{ReLU}(x) = \max(0, x)$
any number below 0 becomes 0, any positive number passes through unchanged
use: hidden layers of MLPs and CNNs, ONLY hidden layers
leaky ReLU(x) = $\max(0.01 \times x, x)$ cures dead neurons

SOFTMAX, TURNS A SCORE VECTOR INTO PROBABILITIES
 $\text{softmax}(z_i) = e^{z_i} / (\text{sum over } j \text{ of } e^{z_j})$
input is a vector of real values, SUM is a vector of the same length whose values SUM TO 1.0 like probabilities
use: output layer for MULTICLASS classification

LINEAR, THE NULL ACTIVATION
 $\text{linear}(x) = x$
does not change the weighted sum, returns it directly
use: output layer for REGRESSION



Guidance quoted: there is no rule of thumb. Sigmoid and its combinations generally work better for classifiers, but sigmoid and tanh are sometimes avoided because of the **vanishing gradient problem**. ReLU is the general activation used in most cases today; for dead neurons, leaky ReLU is the best choice. Begin with ReLU and move on only if it does not give optimum results.

CHOOSING AN ACTIVATION

WHERE	USE
Hidden layers, MLP	ReLU
Hidden layers, CNN	ReLU
Hidden layers, RNN	Sigmoid or Tanh
Output, binary classification	Sigmoid
Output, multiclass classification	Softmax
Output, multiclass classification	Sigmoid
Output, regression	Linear

Guidance quoted: there is no rule of thumb. Sigmoid and its combinations generally work better for classifiers, but sigmoid and tanh are sometimes avoided because of the **vanishing gradient problem**. ReLU is the general activation used in most cases today; for dead neurons, leaky ReLU is the best choice. Begin with ReLU and move on only if it does not give optimum results.

FORWARD PASS

layer 1: $z[i] = W[i] * a[i-1] + b[i]$
 $a[i] = \text{f}(z[i])$
 $a[0] = x$, the input, $a[L] = y_{\text{hat}}$, the prediction.

BACKPROPAGATION, THE FOUR STEPS

- Forward propagation generates predictions from the current weights.
- The loss function computes the prediction error against the true label.
- The error is **propagated backwards** layer by layer to determine each neuron's contribution to it, using the chain rule.
- The **gradient of every weight and bias** is computed from that contribution.

BACKPROPAGATION, THE CHAIN RULE AT ITS CORE
 $dL/dW[i] = dL/da[i] * da[i]/dz[i] * dz[i]/dW[i]$

output layer error: $\text{delta}[L] = W[L+1] * f'(z[L])$
propagate backwards: $\text{delta}[i] = (dL/da[i] * \text{delta}[i+1] * f'(z[i]))$
gradient: $dL/dW[i] = \text{delta}[i] * a[i-1]^T$

GRADIENT DESCENT, THE UPDATE RULE

The optimization algorithm updates weights and biases from those gradients. Gradient descent finds the minimum value of a function by stepping downhill.

THE RULE TO WRITE OUT IN THE EXAM
 $w := w - \alpha * (dL / dw)$
 $b := b - \alpha * (dL / db)$

α = learning rate, a HYPERPARAMETER
too large > overshoot and diverge
too small > very slow convergence
The minus sign is there because the gradient points UPHILL.

Forward pass, loss, backward and update repeat over many **epochs** until the loss is minimised.

OPTIMIZERS, ASK DIRECTLY IN THE EXAM

OPTIMIZER	HOW IT WORKS	TRADE OFF
SGD stochastic gradient descent	Updates after one sample or a small batch	Simple and memory efficient, but converges slowly, may fluctuate, uses a fixed learning rate
Adam adaptive moment estimation	Combines momentum with adaptive learning rates, adjusting the rate per parameter	Converges faster and more stably. The usual default
RMSProp root mean square propagation	Adapts the learning rate per parameter using the magnitude of recent gradients	Reduces large oscillations

LOSS FUNCTIONS

REGRESSION LOSSES
 $MSE = (1/n) * \text{sum}((y_i - y_{\text{hat}}_i)^2)$
 $MAE = (1/n) * \text{sum}(|y_i - y_{\text{hat}}_i|)$

BINARY CROSS ENTROPY, SIGMOID OUTPUT
 $L = -(y \log(p) + (1-y) \log(1-p))$
 y is the true label 0 or 1, p is the predicted probability.
If $y = 1$ and $p = 0.9 > L = -\log(0.9) = 0.105$ small
If $y = 1$ and $p = 0.1 < L = -\log(0.1) = 2.303$ large
Confident and wrong is punished hardest.

CATEGORICAL CROSS ENTROPY, SOFTMAX OUTPUT
 $L = -\text{sum over classes } c \text{ of } (y_c * \log(p_c))$
Only the true class has $y_c = 1$, so it reduces to $L = -\log(p \text{ of the correct class })$

AUTOENCODER RECONSTRUCTION LOSS
 $L(x, x_{\text{hat}})$ minimise the reconstruction error plus an added regularizer term

PARAMETERS VS HYPERPARAMETERS

PARAMETER	HYPERPARAMETER
Where it lives	Internal to the model
Where it comes from	Estimated from the data
How it is set	Learned automatically, saved as part of the model
Examples	Weights in an ANN, support vectors in an SVM, coefficients in linear or logistic regression
	Learning rate, C and gamma in SVM, k in KNN, number of layers, batch size, epochs, dropout rate

Whether a model has a fixed or variable number of parameters is what makes it **parametric** or **nonparametric**.

Six architectural principles of deep networks: parameters, hyperparameters, layers, activation functions, loss functions, optimization methods.

TUNING A DEEP LEARNING METHOD, 6 STEPS

- Determine what the input should be, because the input type informs the architecture.
- Determine the intended result, which fixes the output layer type.
- Set up the architecture: model, cost function, number of hidden layers, activation per layer.
- Work with the training data: clean it, visualise it, vectorize and normalize, balance classes, create train, validate and test splits.
- Develop a hyperparameter tuning strategy on a balanced subset, then grow the subset and tweak.
- If the final training set is large, use **Spark** to train on more data faster.

L8 Vanishing gradients and local optima

These two problems are why shallow networks stayed shallow for decades, so both are standard exam material.

Vanishing gradient, quoted: in backpropagation the gradient is **progressively getting more dilute**, and below the top layers the correction signal **della** is minimal. The early layers therefore barely learn.

Why: backprop multiplies derivatives layer by layer. Sigmoid's derivative peaks at 0.25, so ten layers multiply by at most 0.25^10, which is roughly 0.00000005.

Fixes: ReLU instead of sigmoid, tanh, careful weighted initialization, batch normalization, residual or skip connections, and LSTM gates for sequences.

Stuck in local optima: the objective function is usually **not convex**, and random initialization does not guarantee you start anywhere near the global optimum.

Fixes: momentum and Adam, a well chosen learning rate, several random restarts, and mini batch noise which helps escape shallow minima.

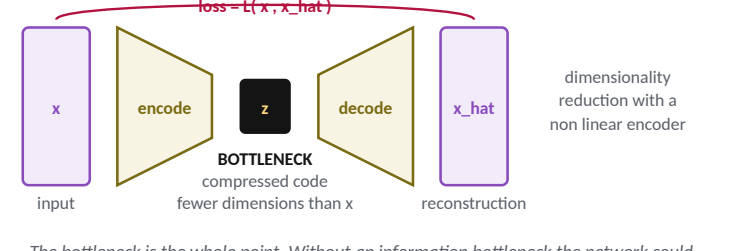
Other limitations of shallow, densely connected networks: high cost, because every interconnected node must be trained, and the network becomes **difficult to train in hidden layers increase**.

Dead neurons. A ReLU unit whose input is always negative outputs 0 forever and its gradient is 0, so it never recovers. Leaky ReLU is the best choice for this.

L8 Autoencoders and the bottleneck

An autoencoder exists to learn a compressed representation with no labels at all, which is representation learning.

It is an unsupervised technique using neural networks, applying backpropagation with the **target values set equal to the inputs**, that is $f(x) = x$. It is trained to copy its input to its output, and trained by minimising the reconstruction error $L(x, x_{\text{hat}})$.



The bottleneck is the whole point. Without an information bottleneck the network could simply memorize the input values by passing them along unchanged.

THE BALANCE AN IDEAL AUTOENCODER STRIVES

- Sensitive enough** to the inputs to accurately build a reconstruction. This is the reconstruction loss term.
- Insensitive enough** that it does not simply memorize or overfit. This is an added regularizer term.

Standard architectures: undercomplete, sparse, denoising, contractive. **Applications**: anomaly detection, data denoising of images and audio, image inpainting, information retrieval, dimensionality reduction.

Hyperparameters: bottleneck size, depth, regularizer strength, noise level for a denoising autoencoder.

L8 Transfer learning and fine tuning

Training from scratch needs data and compute you often do not have, so you reuse a network that already learned generic visual or linguistic structure.

Definition quoted: transfer learning is a technique where knowledge gained during training in one type of problem is used to train in another related task or domain (Pan and Fellow, 2009).

THE THREE TECHNIQUES, IN ORDER

- Truncate the last layer, the softmax layer**, of the pre trained network and replace it with a new softmax relevant to your problem. An ImageNet network has a 1000 category softmax; for a 10 category task the pre trained weights have 10 categories. Then run backpropagation to fine tune the new layers weights, and make sure cross validation is performed.
- Use a smaller learning rate**. Common practice is to make the initial learning rate **10 times smaller** than the one used for training from scratch, because the pre trained weights are already good and you do not want to distort them too quickly.
- Freeze the weights of the first few layers**. They capture universal features such as curves and edges, so freezing them lets the network focus on dataset specific features in the later layers.

Pre trained net > Freeze early layers > Replace softmax > LR / 10 > Backprop > CV

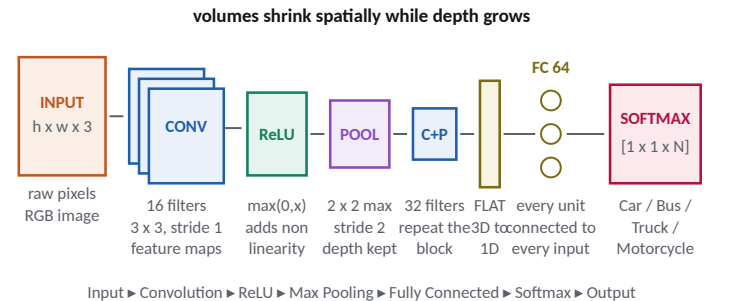
L9 Convolutional neural networks

A fully connected layer on a 224 by 224 RGB image would need over 150,000 weights per neuron and would ignore the fact that neighbouring pixels are related. Convolution exists to exploit locality and reuse the same weights everywhere.

Goal quoted: to learn higher order features in the data via convolutions. Suited to object recognition in images, overlaps with text analysis through OCR, and is also good at analysing sound. The role of the ConvNet is to **reduce the image into a form which is easier to process without losing features critical for a good prediction**.

THREE MAJOR GROUPS OF LAYERS

GROUP	LAYERS IN IT
1 Input	Accepts a 3D input, width by height by depth, depth being the colour channels, generally 3 for RGB. Colour spaces: Grayscale, RGB, HSV, CMYK
2 Feature extraction (learning)	The repeating pattern Convolution plus ReLU , then Pooling , progressively building higher order features
3 Classification	One or more fully connected layers producing class scores, output [b x N], b = mini batch size, N = number of classes



The MNIST demo architecture from the slides, drawn to scale: 16 filters of 3 by 3, then 2 by 2 max pooling stride 2, then one fully connected layer of 64 neurons, then softmax.

THE CONVOLUTION OPERATION

Definition quoted: a mathematical operation describing a rule for how to merge two sets of information. It takes input, applies a convolution kernel, and gives a feature map as output. Convolution is the **feature detector** of a CNN and is interpreted as a filter, so each edge kernel lets through only information from the edge of an image.

THE OPERATION ITSELF, A DOT PRODUCT IN A SLIDING WINDOW
 $\text{FeatureMap}(i,j) = \text{sum over } m, n \text{ of } (\text{Input}(+m, j+n) * \text{Kernel}(m, n) + \text{bias})$

The convolutional layer transforms the input using a patch of locally connecting neurons from the previous layer, computing a dot product between that region and the weights.

OUTPUT SIZE, THE VERSION GIVEN IN THE SLIDES
Image volume (h x w x d), filter (fh x fw x d)
Output volume = $(h - fh + 1) * (w - fw + 1) * 1$
example: a 5 x 5 image with a 3 x 3 filter gives $(5-3+1) * (5-3+1) = 3 * 3$ feature map

OUTPUT SIZE, THE GENERAL FORM WITH PADDING AND STRIDE
 $\text{out} = \text{floor}((h - fh + 2P) / S) + 1$
P = padding pixels added each side, S = stride
check: h=5, fh=3, P=1, S=1 -> (5-3+0)/1 + 1 = 3 correct
h=7, fh=3, P=1, S=2 -> (7-3+2)/2 + 1 = 4

KERNEL, STRIDE, PADDING

TERM	DEFINITION, AS THE SLIDES PUT IT
Kernel / filter	A function with width and height smaller than the input volume, applied across width and height in a sliding window , applied for every depth of the input. Output is the dot product of the filter and the input region.
Filter count	A hyperparameter . It controls how many activation maps are produced and forms the third dimension of the 3D output volume
Stride	The number of pixels the filter shifts over the input matrix. Stride 1 moves one pixel at a time, stride 2 moves two. Larger stride means a smaller output
Padding	What to do when the filter does not fit. Same padding pads the picture with zeros so that it fits and the output keeps the input size. Valid padding drops the part of the image where the filter did not fit, keeping only the valid part
Activation map	The 2D output produced by sliding one filter across the spatial dimensions during the forward pass. When we say the filter "activates" we mean it lets information pass from the input volume into the output volume
Parameter sharing	The same filter weights are reused at every spatial position, which is what keeps the parameter count small

Major components of a convolutional layer: filters, activation maps, parameter sharing, and layer specific hyperparameters. **Gradient descent** trains the parameters so class scores stay consistent with the labels.

POOLING, MAX VS AVERAGE

Pooling exists to reduce the number of parameters and computation, prevents **overfitting**. It downsamples the volume **spatially**, the **depth is preserved**, and it **operates independently on every depth slice**. Also called subsampling or downsampling.

Why it is needed: decrease the computational power required through dimensionality reduction, and extract **dominant features** that are **rotational and positional invariant**.

TYPE	RETURNS	EXTRA EFFECT
Max pooling	The maximum value in the window	Also a noise suppressant . It discards noisy activations altogether, doing de noising as well as dimensionality reduction. Most commonly used
Average pooling	The average of all values covered	Performs dimensionality reduction only, as a milder noise suppressing mechanism
Sum pooling	The sum of all elements in the feature map	Rarely used. L2 norm pooling also exists

WORKED 2 x 2 MAX POOL, STRIDE 2
input 2 1 3 2 9 output max(1,3,5,6) = 6
5 6 4 1 max(2,9,4,1) = 9

7 2 8 4 max(7,2,8,3) = 8
1 3 3 6 max(4,1,3,6) = 6

4 x 4 input, 2 x 2 window, stride 2 -> 2 x 2 output
75% of the values are discarded, the strongest survive.

FLATTEN, FULLY CONNECTED, SOFTMAX

Flatten turns the final 3D feature volume into a 1D vector so a normal dense layer can read it. Neurons in a **fully connected** layer have full connections to all activations in the previous layer, and there may be more than one FC layer at the end, as in AlexNet. The flattened output feeds a feed forward network with backpropagation applied at every training iteration, and over a series of **epochs** the model separates dominating from low level features and classifies with **softmax**.

Fully connected output shape: [b x N] b = mini batch, N = classes
Softmax layer output volume: [1 x 1 x N]

HOW A CONVNET CLASSIFIES AN IMAGE, 6 STEPS

- Provide the input image into the convolution layer.
- Choose parameters, apply filters with strides and padding if required, perform the convolution and apply ReLU to the matrix.
- Perform pooling to reduce the dimensionality.
- Add as many convolutional layers as needed.
- Flatten the output and feed it into a fully connected layer.
- Output the class using an activation function with a cost function, and classify the image.

WELL KNOWN CHNS

NET	WHO AND WHAT
LeNet	Yann LeCun. Recognises handwritten digits. The pioneer CNN
AlexNet	Krizhevsky, Sutskever, Hinton. Won 2012 ImageNet. First CNN using multiple convolution operations
GoogLeNet	Google. Won 2014 ImageNet. Far fewer parameters, so faster and less prone to overfitting
VGGNet	Most popular version VGG16, 16 layers including input, output and hidden
ResNet	Kaiming He. Won 2015 ImageNet. Variants ResNet34 and ResNet50, and the complex ResNeXt

TENSORFLOW SPECIFICS FROM THE SLIDES

Convolutional layer is a 4D tensor:
(Filter X Filter X Input Depth, number of Filters)
 $\text{tf.nn.conv2d}(\text{input}, \text{strides}, \text{padding})$ strides = filter stepping

$\text{tf.nn.max_pool}(\text{value}, \text{ksize}, \text{strides}, \dots)$ ksize = window per dimension
dropout: with probability keep_prob the input element is output

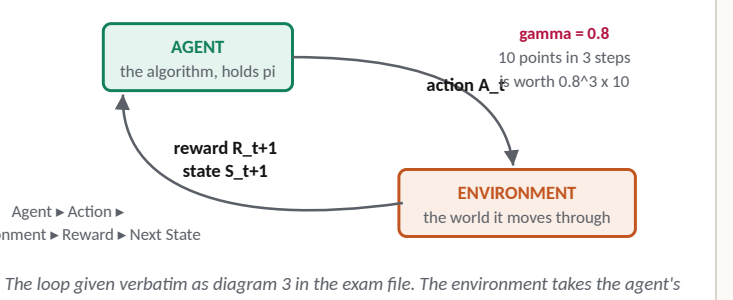
scaled up by 1 / keep_prob, otherwise 0
example X = [1.5, 0.5, 0.75, 1.0, 0.75, 0.6], keep_prob = 0.5
Constant = fixed value. Variable = changes over time.
Placeholder = "a promise to provide a value later",
 $x = \text{tf.placeholder}(\text{tf.float32})$

Tensorboard: Tensorboard --logdir=<dir> then http://localhost:6006

L10 Reinforcement learning

Supervised learning needs a correct answer for every input. Reinforcement learning exists for problems where nobody can supply that answer, only a score after the fact.

Definitions quoted: reinforcement is an occurrence of an event, in the proper relation to a response, that tends to increase the probability that the response will occur again in the same situation. **Reinforcement learning is the science of decision making**. The agent learns by **trial and error** and tries to get the maximum possible reward by performing actions in the environment. It emphasizes learning feedback that **evaluates the learner's performance without providing standards of correctness** as behavioural targets.



The loop given verbatim as diagram 3 in the exam file. The environment takes the agent's current state and action as input, and returns the reward and the next state as output.

EVERY ELEMENT, DEFINED

ELEMENT	DEFINITION FROM THE SLIDES
Agent	An intelligent program that takes actions. The algorithm is the agent . A drone making a delivery, Super Mario navigating a game
Environment	The external condition, the world through which the agent moves. It takes the current state and action as input and returns the reward and next state
Action A	The set of all possible moves the agent can make. Running left or right, jumping, crouching, buying, selling or holding securities
State S	A concrete and immediate situation the agent finds itself in, a specific place and moment, an instantaneous configuration
Reward R	The feedback by which we measure the success or failure of an action. It can be immediate or delayed
Policy pi	The agent's behaviour at a given time, a mapping from states to actions, the strategy determining the next action from the current state. Deterministic (always the same action) or stochastic (a distribution over actions)
Reward function	Defines the goal of the problem. The policy is altered in order to achieve that goal
Value function V	The reward function says what is good immediately, the value function says what is good in the long run . The value of a state is the total reward the agent can expect to accumulate over the future starting from that state
Q value	Like V but with one extra argument, the action. $Q(s,a)$ is the long term return of state s after taking action a under policy pi
Model	Predicts or mimics the behaviour of the environment, used for planning. Splits into Transition (predicts the next state) and Reward (predicts the immediate reward)
Discount gamma	Multiples future rewards to dampen their effect. "a kind of short term hedonism". gamma = 1 makes future rewards worth as much as immediate ones

Three components of an RL agent: policy, value function, model.

EVERY RL FORMULA

DISCOUNTED RETURN, WHAT THE AGENT ACTUALLY MAXIMISES
 $G_t = R_{t+1} + \gamma V_t + \gamma V_{t+1} + \gamma V_{t+2} + \dots$
 $= \text{sum over } k=0, \text{ of } (\gamma^k * R_{t+k+1})$

slide example: gamma = 0.8, a reward of 10 points after 3 time steps has present value $0.8^3 * 10 = 5.12$

BELLMAN EQUATION FOR THE STATE VALUE

$V(s) = E[\$

Flood Segmentation with Cross-Modal Knowledge Distillation

YOUR OWN PROJECT. THIS IS THE ONE YOU WILL BE ASKED ABOUT FOR TWENTY MINUTES

REVISION SHEET

Floods and clouds arrive together. That one sentence is the whole thesis. The satellite that is good at mapping water (optical) is blind exactly when you need it. The satellite that can see through cloud (radar) is the one with the weakest model, because labelled radar is scarce and water in radar is hard. Closing that gap is the problem. The method: train a teacher on optical, distil what it knows into a radar only student, then deploy the student alone.

1 The problem, in the order you should say it

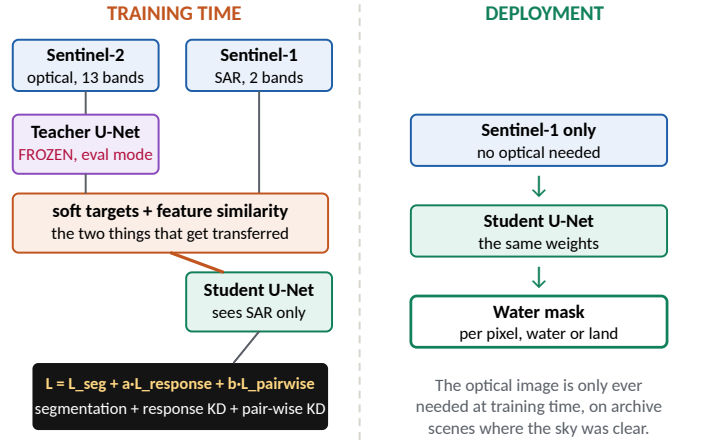
- Sentinel-2 (optical)** maps water accurately. Water is dark and distinctive in the visible and near infrared bands. But optical sensors **see nothing through cloud**, and cloud is exactly the weather a flood brings.
- Sentinel-1 (SAR)** sees through cloud, day or night, because radar is not light. But water in radar is far harder to segment: **speckle, look angle effects**, and **wind roughening the surface** so it stops looking like water.
- Hand labelled SAR is **scarce**, because labelling it is slow and needs expertise most annotators do not have.
- So the sensor that works during the disaster is the one with the weakest model.

THE TWO SENSORS, SIDE BY SIDE

	SENTINEL-2, OPTICAL	SENTINEL-1, SAR
Measures	Reflected sunlight, 13 spectral bands	Backscattered radar energy, 2 polarisations
Through cloud	No	Yes
At night	No	Yes
Water looks like	Dark and distinctive, especially in NIR	Low backscatter, usually
Main nuisance	Cloud and cloud shadow	Speckle, look angle, wind
Labelled data	Plentiful	Scarce

Speckle is **not sensor noise**, it is physics. Radar is coherent, so returns from many scatterers inside one pixel interfere constructively and destructively. That produces the grainy salt and pepper texture, and it **does not average out by taking a better picture**.
Wind: calm water is a mirror for radar and reflects energy away, so it looks dark. Roughen it with wind and it scatters energy back, so **the flood starts to look like land**.

2 The idea, drawn



The deployment claim is structural, not rhetorical. 'forward(s1, s2=None)' returns a SAR only path, and 'predict(s1)' is what you would call in the field.

3 The concepts, with the formulas

SEGMENTATION AND WHY NOT ACCURACY

Segmentation labels **every pixel**, so the output is a mask the same size as the input.

THE TWO METRICS
IoU (Jaccard) = $TP / (TP + FP + FN)$
Dice (F1) = $2 \cdot TP / (2 \cdot TP + FP + FN)$

Both ignore true negatives entirely. That is the point. On a chip that is 95% land, a model that predicts "no water anywhere" scores 95% pixel accuracy. IoU is stricter than Dice for the same prediction, so IoU is the number reported.

KNOWLEDGE DISTILLATION

TEMPERATURE SOFTMAX
 $\text{softmax_T}(z)_i = \exp(z_i / T) / \sum_j \exp(z_j / T)$

T = 1 is the normal softmax. Raising T flattens the distribution and exposes relative confidence.

The KD loss is multiplied by **T squared**, because dividing the logits by T shrinks the gradients by 1 over T squared, and this restores their scale.

Dark knowledge. If the teacher says 0.7 water and 0.3 land on a pixel, it has told the student "this is water, but it is ambiguous". The hard label throws that away. This is cross-modal distillation. It transfers modality, not capacity: teacher and student are the same size.

THE THREE FAMILIES OF DISTILLATION

FAMILY	WHAT IS MATCHED	STRENGTH	WEAKNESS
Response	The output logits	Simple, well understood	Per pixel, ignores spatial structure
Feature	Intermediate activations directly	Richer signal	Needs matching shapes and scales
Relation	The relationships between features, for example a similarity matrix	Transfers structure, shape agnostic	More expensive to compute

Liu et al. 2019 showed per pixel distillation alone is not enough: getting each pixel roughly right does not guarantee a coherent shape. So response KD is the control condition and pair-wise relation KD is the thing under test.

The pair-wise idea in one sentence. Rather than asking "does the student produce the same value as the teacher at this location", ask "does the student think these two regions look alike, as much as the teacher does".

U-NET

The encoder downsamples repeatedly, building what while losing where. The decoder upsamples back to full resolution. Skip connections carry high resolution feature maps from each encoder level straight across to the matching decoder level. Without them the shoreline comes out blurred and blobby. The bottleneck is the deepest, most compressed layer, the most semantic and least spatial, which is why it is the layer pair-wise distillation compares.

4 Data, and the two traps

Sen1Floods11: paired Sentinel-1 and Sentinel-2 chips with hand labels, hosted on Google Cloud Storage. Only the 446 hand labelled chips are wired up so far. The much larger weakly labelled set is for the self supervised stage.

LABEL	MEANING	HANDLING
-1	No data or cloud shadow	Ignored everywhere, loss and metrics
0	Land	Negative class
1	Water	Positive class, and the rare one

NORMALISATION COMES FROM THE SENSORS, NOT THE BATCH
S1_MIN, S1_MAX = -50.0, 1.0 # Sentinel-1 dB range
S2_SCALE = 10000.0 # Sentinel-2 TOA reflectance
IGNORE_INDEX = -1
GCS = 'https://storage.googleapis.com/sen1floods11/v1.1'

Fixed physical constants mean a chip normalises identically alone or in a batch of 64, which matters at inference, when there is no batch.

TRAP 1, THE IGNORE INDEX

Include the -1 pixels in the metric and, depending which way you count, a true IoU of 0.40 can read as 0.25. Wrong, and with no error message. The mask is applied in exactly one place, metrics.py, and losses.py just passes ignore_index=-1.

TRAP 2, DATA LEAKAGE

Chips from a single flood event share terrain, sensor geometry and weather, so they are not independent. A random split lets near identical chips land in both train and test. Fix: split by event, not by chip, using the published split CSVs, and keep Bolivia aside entirely as a held out event.

THE OFFICIAL SPLITS, NEVER RANDOM ONES
flood_train_data.csv flood_valid_data.csv
flood_test_data.csv flood_bolivia_data.csv # held out

CLASS IMBALANCE

Water is the rare class, and plain cross entropy is content to call everything land. Dice loss measures overlap and never counts true negatives. Combining gives L_seg = CE + Dice with a class weight on water: CE keeps the gradients well behaved early, Dice cares about the rare class.

5 Method, architecture, loss

SIX STAGES, ONE ENTRY POINT

RUN.PY
1 explore look at one chip first
2 download --split train,val,test fetch the chips
3 stats --split train water / land / no-data balance
4a train --modality s1 --out-dir runs/base_s1 SAR baseline
4b train --modality s2 --out-dir runs/teacher_s2 optical teacher
5 distill --teacher-ckpt runs/teacher_s2/best.pt
6 eval --ckpt runs/distill/best.pt --split test --modality s1

Order matters. The two baselines come before the framework, because "our method gets 0.62 IoU" means nothing without the number it must beat and the number it chases. The distance between those two is the modality gap, and shrinking it is the whole thesis.

ONE U-NET USED THREE WAYS

2 input channels for SAR, 13 for optical, everything else identical. That identity is what makes the three runs comparable: any difference is the modality, not the architecture.

MODELS.PY
chans = [width * (2 ** i) for i in range(depth + 1)]
= 32, 64, 128, 256, 512
nn.MaxPool2d(2) between encoder blocks
self.bottleneck = conv_block(chans[-2], chans[-1]) # compared by pair-wise KD
self.head = nn.Conv2d(chans[0], num_classes, 1) # num_classes = 2
decoder: bilinear F.interpolate, then torch.cat([x, skip], dim=1)
forward(x, return_features=False) -> (logits, feat)

The teacher is frozen on purpose. requires_grad_(False) plus self.teacher.eval(). That .eval() is not cosmetic: in train mode batch norm keeps updating its running statistics, so a "frozen" teacher would still be drifting and any improvement could be attributed to that rather than to the transfer.

THE LOSS, TERM BY TERM

L = L_seg + ALPHA * L_RESPONSE + BETA * L_PAIRWISE
L_seg CE + Dice, class weighted. DiceLoss zeroes unlabelled pixels with valid = (target != IGNORE_INDEX), uses the SOFT probability not a thresholded mask so it stays differentiable, and eps prevents division by zero on a chip with no water at all.
L_response F.log_softmax(student/t) against F.softmax(teacher/t), F.kl_div, masked, then * (t ** 2).
KL is asymmetric ON PURPOSE, the teacher is the reference.

L_pairwise F.normalize(feat.reshape(b, c, h*w), dim=1) then torch.bmm(f.transpose(1,2), f), MSE between the student and teacher similarity matrices.
F.normalize first makes the dot product a COSINE similarity, direction not magnitude, which is what transfers structure across two different sensors.

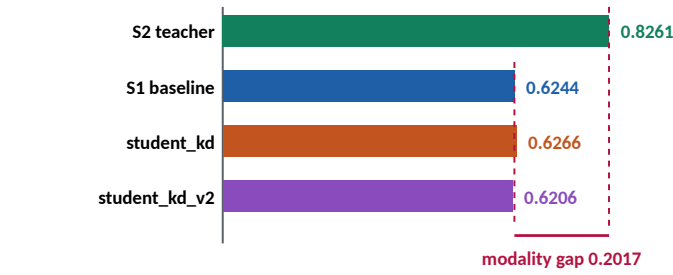
Why pooling is not optional. F.adaptive_avg_pool2d to 16 by 16 gives 256 regions and 65,536 pairs. Unpooled at 512 by 512 it would be 68 billion.

Confidence gating. conf = teacher_logits.softmax(1).max(1).values, then (conf >= threshold). Where the teacher's max softmax falls below the threshold the pixel is marked no data for the distillation term. An unconfident teacher's soft target is just noise wearing a useful shape.

6 Results, stated honestly

CONFIGS/DEFAULT.YAML, THE WHOLE EXPERIMENT SURFACE
modality: s1 # s1 | s2 | both
width: 32 # 16 if GPU memory is tight
depth: 4
epochs: 30
lr: 0.001
dice_weight: 0.5
class_weights: [1.0, 5.0] # water is rare, weight it up
alpha: 1.0 # response KD weight (0 = ablate)
beta: 1.0 # pair-wise KD weight (0 = ablate)
temperature: 4.0
gate_threshold: 0.0 # 0 = off, 0.7 turns gating on
seed: 42

VALIDATION WATER IOU



Read it honestly. The framework, the losses and the plumbing all work. The transfer does not, at these settings. +0.0022 over the baseline is inside run to run noise, and the gated variant is actually worse. The 0.2017 gap to the optical teacher is still entirely open.

THE ABLATION GRID

RUN	COMMAND
No distillation, baseline	train --modality s1
Response KD only	distill --alpha 1 --beta 0
Pair-wise KD only	distill --alpha 0 --beta 1
Both	distill --alpha 1 --beta 1
Both plus gating	distill --alpha 1 --beta 1 --gate-threshold 0.7

Each run writes runs/<name>/log.csv and config.json, so the ablation table can be rebuilt from disk instead of from memory.

7 Viva answers, prepared

DECISIONS YOU CAN DEFEND

- The ignore mask lives in exactly one place. metrics.py masks once, losses.py passes ignore_index=-1, nothing else touches it
- The teacher is frozen, so any improvement is attributable to the transfer
- Inference takes one modality. predict() accepts S1 only
- Seeds fixed, configs saved, and the best checkpoint is chosen on validation water IoU, not on loss, because loss is dominated by the land class
- src/ never prints and never decides. All input, output and control flow live in run.py

THE REAL NOVELTY, AND WHAT IS NOT DONE YET

Self supervised pre-training of the teacher is the actual novelty of the thesis. Labelled SAR is scarce, but unlabelled Sentinel-1 and Sentinel-2 imagery is essentially unlimited. A self supervised teacher could learn from that pool and then distil into the student. The current supervised teacher is the honest baseline it has to beat. The contrastive pretext task belongs in Chapter 3 once the pretext is chosen.

- Data augmentation: the transform hook exists but is unused
- The weakly labelled split is not wired up yet
- Next experiments: the self supervised teacher, augmentation, and a proper sweep of alpha, beta and the gate threshold rather than three hand picked runs

IF THEY ASK "WHY IS IT NOT WORKING YET"

Do not defend the number. Say what is actually true: the transfer is not moving the metric at these hyperparameters, the three runs so far were hand picked rather than swept, the teacher is supervised rather than self supervised, and no augmentation is in play. Then name the next experiment. A candidate who reports +0.0022 as a success is a worse candidate than one who calls it noise.

CROSS LINKS TO YOUR OTHER PROJECTS

- The class imbalance argument is the same as in 03 E-commerce and 02 MVtec: the majority class swamps the average, so pick a metric that cannot be gamed by ignoring the rare class
- The leakage argument is the same as the time based holdout in 04 Airbnb: a random split lets correlated samples appear on both sides and inflates the score

FILE LAYOUT

FLOODKD
run.py explore | download | stats | train | distill | eval
configs/default.yaml every number
src/config.py loads YAML, applies CLI overrides, REJECTS typos
src/data.py src/models.py src/losses.py
src/metrics.py IoU / F1 / precision / recall, mask applied once
src/engine.py docs/original-README.md